

Valus User Guide

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Contents

<u>About This Manual</u>	5
<u>Audience</u>	5
<u>How This Manual Is Organized</u>	5
<u>Conventions Used in This Manual</u>	5

1

<u>Overview</u>	7
<u>Valus - An Overview</u>	8
<u>License Requirements</u>	8
<u>Input Requirements</u>	8
<u>Use Model</u>	8
<u>How to Run Valus</u>	9
<u>Handling TCL Files</u>	9
<u>Tab Auto-Completion of Commands</u>	10

2

<u>TCL Shell Commands</u>	11
<u>add_attribute</u>	13
<u>add_group</u>	15
<u>add_mode_definition</u>	16
<u>add_noise_info</u>	17
<u>add_power_info</u>	20
<u>add_timing_info</u>	22
<u>check_ldb_version</u>	24
<u>copy_attribute</u>	25
<u>define_attribute</u>	26
<u>delete_attribute</u>	27
<u>get_log_file_name</u>	28
<u>get_version</u>	29
<u>help</u>	30

Valus User Guide

<u>read_libs</u>	31
<u>report_library_analysis</u>	32
<u>report_summary</u>	33
<u>set_message</u>	35
<u>set_valus_mode</u>	37
<u>update_library</u>	39
<u>write_ldb</u>	40
<u>write_lib</u>	41

3

<u>Valus Checks</u>	42
<u>LIB_TIMING</u>	43
<u>LIB_NOISE</u>	92
<u>LIB_POWER</u>	108
<u>LIB_MSV</u>	113
<u>LIB_LVF</u>	124
<u>LIB_EM</u>	127
<u>LIB_MERGE</u>	129
<u>LIB_SEMANTIC</u>	131
<u>LIB_MISC</u>	220
<u>LIB_UI</u>	224
<u>Data Integrity Checks</u>	237
<u>Positive Checks</u>	238
<u>Range Checks</u>	241
<u>Monotonic Increase Checks</u>	244
<u>Mandatory Checks</u>	245
<u>Mandatory Ignore Checks</u>	246
<u>Count Checks</u>	247
<u>Enum Value Checks</u>	252

About This Manual

The Valus software is a library validation tool that performs exhaustive checking of the timing libraries. It ensures high-quality library inputs for full Cadence flow.

Audience

This document is intended for library developers, methodology engineers, and designers who need to perform timing library validation checks to detect potential library issues that may cause problems during timing analysis in downstream tools in the design flows.

How This Manual Is Organized

The following sections provide prerequisite information for using the Valus software:

"Overview"

Describes how to install, set up, and run the Valus software.

"TCL Shell Commands"

Describes the Valus commands.

"Valus Checks"

Describes how to interpret and use Valus error messages.

Conventions Used in This Manual

This section describes the typographic and syntax conventions used in this manual.

<code>text</code>	Indicates text that you must type exactly as shown. For example, <code>report_summary -cumulative</code>
-------------------	---

Valus User Guide

About This Manual

<i>text</i>	Indicates information for which you must substitute a name or value. In the following example, you must substitute the message ID and the limit value: <pre>set_message -id TECHLIB-436 -limit 1</pre>
<i>text</i>	Indicates the following: ■ Text found in the graphical user interface (GUI), including form names, button labels, and field names ■ Terms that are new to the manual, are the subject of discussion, or need special emphasis ■ Titles of manuals
[]	Indicates optional arguments. In the following example, you can specify none, one, or both of the bracketed arguments: <pre>command [-arg1] [arg2 value]</pre>
[]	Indicates an optional choice from a mutually exclusive list. In the following example, you can specify any of the arguments or none of the arguments, but you cannot specify more than one: <pre>command [arg1 arg2 arg3 arg4]</pre>
{ }	Indicates a required choice from a mutually exclusive list. In the following example, you must specify one, and only one, of the arguments: <pre>command {arg1 arg2 arg3}</pre>
{ [] [] }	Indicates a required choice of one or more items in a list. In the following example, you must choose one argument from the list, but you can choose more than one: <pre>command {[arg1] [arg2] [arg3]}</pre>
{ }	Indicates curly braces that must be entered with the command syntax. In the following example, you must type the curly braces: <pre>command arg1 {x y}</pre>
...	Indicates that you can repeat the previous argument.

Overview

This chapter contains information about the following topics.

- [Valus - An Overview](#) on page 8
- [License Requirements](#) on page 8
- [Input Requirements](#) on page 8
- [Use Model](#) on page 8
- [How to Run Valus](#) on page 9
- [Handling TCL Files](#) on page 9
- [Tab Auto-Completion of Commands](#) on page 10

Valus - An Overview

The Valus software is a library validation tool that performs exhaustive checking of timing libraries. This ensures high-quality library inputs for a full Cadence flow.

The software supports industry standard timing library formats, such as, NLDM, ECSM, ECSM-Noise, ECSM-Power, CCS, CCS-Noise, CCS-Power, and LVF.

It is recommended that you use Valus to validate the input timing libraries to avoid any undesirable surprises in timing analysis later in the design flow.

License Requirements

The Valus software requires checkout of either Tempus Library Validation (TPS500) or Tempus_Timing_Signoff_XL (TPS200) licenses.

By default, the software will checkout a TPS500 license. If TPS500 is not available, then a TPS200 license will be checked out.

Input Requirements

The Valus software requires timing library files for input information.

The timing library files contain timing information in ASCII format for all standard cells, blocks, and I/O pad cells. Valus reads timing library format files (.lib). You can read the library files using the [read_libs](#) command.

Use Model

The Valus executable file resides in the Tempus installation path. You can set the Tempus release path to the current Tempus build.

To start Valus, you can use the `valus` command.

To view Valus help, type the following command:

```
<tempus_installation_path>/valus -help
```

The `valus` command parameters are described below.

<code>-log logfile</code>	Uses the specified file as the log file.
---------------------------	--

Valus User Guide

Overview

-version	Displays the Valus version.
-help	Displays the help.
-file	Specifies the Tcl file to source at startup.

How to Run Valus

To open the Valus console window, type `valus` in the UNIX window (shell tool, xterm). When a session is active, the Valus console displays the following prompt:

```
valus>
```

To read the timing libraries (.lib), the following command is used:

```
valus> read_libs lib
```

Examples

- The following example shows a script to import a single .lib file:

```
##### Script 1 : "read_lib.tcl" #####
set $rootDir <path of root directory>
read_libs $rootDir/test.lib
report_summary
#####
```

- The following example shows a script to import multiple .lib file:

```
##### Script 2 : "read_libs.tcl" #####
set $rootDir <path of root directory>
read_libs {$rootDir/test.lib $rootDir/test1.lib $rootDir/test2.lib}
report_summary
#####
```

Handling TCL Files

The following command executes the tcl file during startup:

```
valus -file read_lib.tcl
```

Valus validates the .lib file(s) listed in the `read_lib.tcl` script (see [Examples](#)) and displays the checks that are performed during library validation.

If you exit the session by typing `exit`, the `valus>` prompt is no longer displayed and the prompt returns to the Unix shell.

Tab Auto-Completion of Commands

You can use the Tab key at the prompt to complete command names.

After typing the first few letters of a command name, press the Tab key. The software displays the complete command name (if the string is unique to one text command), or shows all the commands that match the text that you typed.

TCL Shell Commands

This chapter contains information about the following Tcl Shell Commands.

- [add_attribute](#) on page 13
- [add_group](#) on page 15
- [add_mode_definition](#) on page 16
- [add_noise_info](#) on page 17
- [add_power_info](#) on page 20
- [add_timing_info](#) on page 22
- [check_ldb_version](#) on page 24
- [copy_attribute](#) on page 25
- [define_attribute](#) on page 26
- [delete_attribute](#) on page 27
- [get_log_file_name](#) on page 28
- [get_version](#) on page 29
- [help](#) on page 30
- [read_libs](#) on page 31
- [report_library_analysis](#) on page 32
- [report_summary](#) on page 33
- [set_message](#) on page 35
- [set_valus_mode](#) on page 37
- [update_library](#) on page 39
- [write_ldb](#) on page 40

- write lib on page 41

add_attribute

```
add_attribute
    [-help]
    -attribute_name <string>
    -group_name {timing_arc pin cell library pg_pin operating_conditions
generated_clock}
    [-name <string>]
    -value <string>
```

Adds the liberty attribute to the ETM/ PTM library.

The supported attributes are:

- Library level:
 - ☐ default_max_fanout
 - ☐ default_max_transition
 - ☐ voltage_map
 - ☐ operating_conditions
 - ☐ default_operating_conditions
- Cell level:
 - ☐ duty_cycle
 - ☐ area
 - ☐ pad_cell
 - ☐ pg_pin attributes
 - ☐ pg_pin group
- Pin level:
 - ☐ related_power_pin
 - ☐ fanout_load
 - ☐ related_ground_pin
 - ☐ prefer_tied
 - ☐ power_down_function
 - ☐ is_analog

- ☐ minimum_pulse_width_low
- ☐ minimum_pulse_width_high
- ☐ min_period
- ☐ max_transition
- ☐ max_capacitance
- ☐ capacitance

■ Operating condition group attributes:

- ☐ process
- ☐ temperature
- ☐ voltage
- ☐ tree_type

Parameters

-help	Prints out the command usage.
-attribute_name <string>	Specifies the attribute name that needs to be added.
-group_name {timing_arc pin cell library pg_pin operating_conditions generated_clock}	Specifies the group name for which attribute needs to be added.
-name <string>	Specifies the name of library, cell or pin for which attribute needs to be added. In case of timing arc, the convention to be followed is toPin_fromPin.
-value <string>	Specifies the attribute value to be added.

Examples

- The following command adds attribute "cell_leakage_power" on cell level with the value 2.2:

```
add_attribute -attribute_name cell_leakage_power -value 2.2 -group_name cell
```

add_group

```
add_group
    [-help]
    -group {pg_pin operating_conditions}
    -group_name <string>
    -parent_group {cell library}
```

Adds liberty groups to the ETM/ PTM library.

Parameters

-help	Prints out the command usage.
-group {pg_pin operating_conditions}	Specifies the group that needs to be added.
-group_name <string>	Specifies the name of the group for which attribute needs to be added.
-parent_group {cell library}	Specifies the parent group for which the group needs to be added.

Examples

- The following command is used to add group named pin_group_1 specified with group pg_pin:

```
add_group -group pg_pin -parent_group cell -group_name pin_group_1
```

add_mode_definition

```
add_mode_definition
    [-help]
    -mode_value <string>
    -name <string>
```

Adds the mode values in the ETM/ PTM library.

Parameters

-help	Prints out the command usage.
-mode_value <string>	Specifies the names of the mode values.
-name <string>	Specifies the name of the group for which mode values are to be set.

add_noise_info

```
add_noise_info
    [-help]
    [-from_pin <string>]
    [-input_ccb <string>]
    [-propagated_noise_height_above_high <float>]
    [-propagated_noise_height_below_low <float>]
    [-propagated_noise_height_high <float>]
    [-propagated_noise_height_low <float>]
    [-propagated_noise_width_above_high <float>]
    [-propagated_noise_width_below_low <float>]
    [-propagated_noise_width_high <float>]
    [-propagated_noise_width_low <float>]
    [-ref_pin <string>]
    [-timing_sense <string>]
    [-timing_type <string>]
    -to_pin <string>
    [-to_ref_pin <string>]
    [-when_condition <string>]
```

Adds the noise data in ETM/PTM library.

Note: It is recommended to use the below command for disabling clipping of noise data while using the `add_noise_info` command:

```
set_valus_mode -ccs_noise_data_disable_clip
```

This command supports the following CCSN-first_stage attributes/groups:

- ccsn_first_stage
- is_inverting
- is_needed
- miller_cap_fall
- miller_cap_rise
- stage_type
- dc_current
- output_voltage_fall
- output_voltage_rise
- propagated_noise_low

Valus User Guide

TCL Shell Commands

- propagated_noise_high
- related_ccb_node

The template corresponding to each sub group will also be copied.

Parameters

-help	Prints out the command usage.
-from_pin <string>	Specifies the from pin.
-input_ccb <string>	Specifies if input_ccb is to be added. Note: This parameter cannot be used without the -ref_pin and -to_pin parameters.
-propagated_noise_height_above_high <float>	
-propagated_noise_height_below_low <float>	
-propagated_noise_height_high <float>	
-propagated_noise_height_low <float>	
-propagated_noise_width_above_high <float>	
-propagated_noise_width_below_low <float>	
-propagated_noise_width_high <float>	
-propagated_noise_width_low <float>	
-ref_pin <string>	Specifies the from pin.
-timing_sense <string>	Specifies the timing sense of the reference arc.
-timing_type <string>	Specifies the timing type of the reference arc.
-to_pin <string>	Specifies the to pin
-to_ref_pin <string>	Specifies the sink pin specified in the reference arc (mandatory for arc-based data).

Valus User Guide

TCL Shell Commands

<code>-when_condition <string></code>	Specifies the when condition of the reference arc
---	---

Examples

- The following commands add noise information to the ETM library in accordance with the specified reference CCS library and write it to a new library:

```
set_valus_mode -ccs_noise_data_disable_clip
read_lib abc.lib xyz.lib #abc.lib is ETM lib and xyz is CCS ref lib
update_library abc.lib #specify the ETM library where noise data needs
to be added
add_noise_info -from_pin A -to_pin Y -propagated_noise_width_high 1 -
propagated_noise_width_low 1
write_lib -lib_name new.lib
```

- The following commands add input_ccb group from reference CCS library xyz cell BUFX2 pin A to pin A of ETM library:

```
add_noise_info -ref_pin xyz/BUFX2/A -to_pin A -input_ccb CCB_D1
write_lib -lib_name new.lib
```

Here, CCB_D1 indicates that the CCB group belongs to pin D and 1 indicates that this is the first input_ccb group of pin D. The pin D can be input/inout pin only.

Important Points

- If propagated_noise_width_low/high is defined, then its corresponding propagated_noise_height_low/high will be mandatory and vice-versa.
- The CCSN data (input_ccb and CCSN first stage) can be defined on input/inout pin only.
- The propagated noise data can be defined on output/ inout pin only.

add_power_info

```
add_power_info
    [-help]
    [-fall_power <string>]
    -group_name {cell pin}
    [-mode <string>]
    [-power <string>]
    -power_type {internal leakage}
    [-related_pg_pin <string>]
    [-related_pin <string>]
    [-rise_power <string>]
    [-to <string>]
    [-when <string>]
```

Adds the power data in ETM/ PTM library.

The supported power types are:

- Internal power
- Leakage_power

Parameters

-help	Prints out the command usage.
-fall_power <string>	Specifies the value of fall_power.
-group_name {cell pin}	Specifies the level where the power arc is to be added
-mode <string>	Specifies the name of mode for which the leakage power arc needs to be added.
-power <string>	Specifies the attribute value.
-power_type {internal leakage}	Specifies the internal_power or leakage_power to be added.
-related_pg_pin <string>	Specifies the 'related_pg_pin' name for the internal/leakage power arc.
-related_pin <string>	Specifies the 'related_pin' name for the internal power arc.
-rise_power <string>	Specifies the value of rise_power.
-to <string>	Specifies the 'to' pin name for which internal power arc needs to be added.

Valus User Guide

TCL Shell Commands

<code>-when <string></code>	Specifies the 'when' condition for the leakage power arc.
-----------------------------------	---

Examples

- The following command adds internal power with value 2.4 to pin x:
`add_power_info -power_type internal -power 2.4 -group_name pin -to x`
- The following command adds leakage power with value 1.2 with a specified when condition at cell level:
`add_power_info -power_type leakage -value 1.2 -related_pg_pin VDD -group_name cell -when !IN1`

add_timing_info

```
add_timing_info
    [-help]
    [-cell_fall <float>]
    [-cell_rise <float>]
    [-fall_constraint <float>]
    [-fall_transition <float>]
    -from <string>
    [-label <string>]
    [-min_delay_arc]
    [-mode <string>]
    [-rise_constraint <float>]
    [-rise_transition <float>]
    [-sdf_cond <string>]
    [-sdf_edge {noedge start_edge end_edge both_edges}]
    [-timing_sense {positive_unate negative_unate non_unate}]
    -timing_type {{<Timing type of Arc >}}
    -to <string>
    [-when <string>]
```

Adds timing data in ETM/ PTM library.

Parameters

-help	Prints out the command usage.
-cell_fall <float>	Specifies the delay value for fall.
-cell_rise <float>	Specifies the delay value for rise.
-fall_constraint <float>	Specifies fall_constraint value for constraint arcs.
-fall_transition <float>	Specifies the transition value for fall.
-from <string>	Specifies the 'from' pin name for which timing arc needs to be added.
-label <string>	Specifies the name of the timing group to identify the timing arc.
-min_delay_arc	Specifies the min_delay_arc value for the combinational/ rising_edge/ falling_edge arc.
-mode <string>	Specifies the name of mode for which timing arc needs to be added.

Valus User Guide

TCL Shell Commands

<code>-rise_constraint <float></code>	Specifies rise_constraint value for constraint arcs.
<code>-rise_transition <float></code>	Specifies the transition value for rise.
<code>-sdf_cond <string></code>	Specifies sdf condition for the corresponding timing arc.
<code>-sdf_edge {noedge start_edge end_edge both_edges}</code>	Specifies the edge-specific information for both the start pins and the end pins. <i>Default:</i> noedge
<code>-timing_sense {positive_unate negative_unate non_unate}</code>	Specifies the unateness of the pin.
<code>-timing_type <string></code>	Specifies the type of timing arc. <i>Default:</i> combinational
<code>-to <string></code>	Specifies the 'to' pin name for which timing arc needs to be added.
<code>-when <string></code>	Specifies the condition to enable the corresponding timing arc.

The following attributes are supported by the `-timing_type` parameter:

```
{combinational combinational_rise combinational_fall three_state_disable
three_state_disable_rise three_state_disable_fall three_state_enable
three_state_enable_rise three_state_enable_fall rising_edge falling_edge
preset clear hold_rising hold_falling setup_rising setup_falling
recovery_rising recovery_falling skew_rising skew_falling removal_rising
removal_falling min_pulse_width minimum_period non_seq_setup_rising
non_seq_setup_falling non_seq_hold_rising non_seq_hold_falling
nochange_high_high nochange_high_low nochange_low_high nochange_low_low}
```

Examples

- The following command can be used to add timing information on w pin having related_pin y with specified delay/transition values:

```
add_timing_info -from y -to w -cell_rise 4.4 -cell_fall 2.2 -
rise_transition 1.6 -fall_transition 7 -timing_sense positive_unate -
timing_type combinational
```

check_ldb_version

```
check_ldb_version
    [-help]
    -ldb string
```

Reports the LDB version, size, and source of the software that is used to generate the library database (LDB) file - which is a Cadence binary library format.

Parameters

-help	Prints out the command usage.
-ldb <i>string</i>	Specifies the LDB file for which version details need to be displayed.

Examples

- The following command reports the LDB version, size, and the source of the software:

```
valus> check_ldb_version -ldb check.ldb
LDB file compiled with tool version: VALUS 17.10-d019_1 (Jan 14 2017
08:35:59)
File Size is 10880
Source Tool: Valus
```


copy_attribute

```
copy_attribute  
    [-help]  
    -attribute_name <string>  
    -group {pin}  
    -reference <string>  
    -target <string>
```

Copies the value from one attribute to another in an ETM/ PTM library. This command works for the user-defined attributes only.

Parameters

-help	Prints out the command usage
-attribute_name <string>	Specifies the attribute name needs to be added.
-group {pin}	Specifies the group name for which attribute needs to be added. Note: the -group argument accepts pin only.
-reference <string>	Specifies where should the attribute be copied from
-target <string>	Specifies where should the attribute be copied to.

Examples

- The following command copies the max_transition attribute from pin a to pin b

```
copy_attribute -attribute_name max_transition -group pin -reference a -  
target b
```

define_attribute

```
define_attribute
    [-help]
    -group {library cell pin timing_arc}
    -name <string>
    -type {string integer float boolean}
```

Adds user-defined attribute to the ETM/ PTM library. You can use the `add_attribute` command after defining attributes using this command.

Parameters

-help	Prints out the command usage.
-group {library cell pin timing_arc}	Specifies the group where attribute should be defined.
-name <string>	Specifies the name for the attribute.
-type {string integer float boolean}	Specifies the type of attribute should be defined.

Examples

- The following commands are used to first define an attribute with name "sampleAttribute" on cell level and then adding it by using `add_attribute` under the same `group_name`.

```
define_attribute -name sampleAttribute -group cell -type string
add_attribute -attribute_name sampleAttribute -value true -group_name
cell
```

delete_attribute

```
delete_attribute  
    [-help]  
    -attribute_name <string>  
    -group {timing_arc pin cell library pg_pin}  
    [-name <string>]
```

Deletes an existing user-defined attribute from the ETM/ PTM library.

Parameters

-help	Prints out the command usage.
-attribute_name <string>	Specifies the attribute name needs to be deleted.
-group {timing_arc pin cell library pg_pin}	Specifies the group name for which attribute needs to be deleted.
-name <string>	Specifies the name of library pin name for which attribute needs to be deleted.

Examples

- The following command deletes the max_transition attribute from pin b to pin a

```
delete_attribute -attribute_name max_transition -group pin -name b
```

get_log_file_name

```
get_log_file_name  
    [-help]
```

Retrieves the current log file name.

Parameters

-help	Prints out the command usage.
-------	-------------------------------

Examples

- The following command retrieves the current log file name:

```
valus> get_log_file_name  
/home/xxx/valus.log
```

get_version

```
get_version  
    [-help]
```

Displays the software version.

Parameters

-help	Prints out the command usage.
-------	-------------------------------

Examples

- The following command displays the software version:

```
valus> get_version  
17.10-d031_1
```

help

```
help
    [-help]
    [-command string]
    [-message string]
```

Displays one of the following types of information based on the specified option:

- Syntax of a command.
- Summary of the message text - based on the specified message ID.

Parameters

<code>-help</code>	Prints out the command usage.
<code>-command <i>string</i></code>	Displays the help of the specified TCL command.
<code>-message <i>string</i></code>	Displays the help of the specified message ID.

Examples

- The following command displays the syntax of the specified command:

```
valus> help -command read_libs
Usage: read_libs [-help] <lib>
-help # Prints out the command usage
<lib> # Specify the library file name or List of Library files in {}
(string, required)
```

- The following command displays a summary of the message text for the specified message ID:

```
valus> help -message TECHLIB-1211
NAME
TECHLIB-1211 (error)
SUMMARY
The %s '%s' is not a pad pin but has '%s' attribute specified on line %d.
DESCRIPTION
The message is issued when pad attributes such as hysteresis,
drive_current etc., have been defined for a pin which is not a pad pin.
Only pad pins are expected to have pad attributes defined on them. To
fix the issue, re-characterize the library with appropriate settings,
such that the correct attributes are defined on the pad pin.
```

read_libs

```
read_libs
    [-help]
    lib
```

Reads one or more library files.

Parameters

<code>-help</code>	Prints out the command usage.
<code>lib</code>	Specifies the name of the library file or a list of library files.

Examples

- The following command reads a library file named “test.lib”:

```
valus> read_libs test.lib
```
- The following command reads the specified multiple libraries sequentially:

```
valus> read_libs {test.lib test1.lib test2.lib}
```

report_library_analysis

```
report_library_analysis  
    [-help]  
    [-ccs_nldm_correlation]  
    -lib_name <string>  
    [-tolerance <float>]
```

Provides correlation report for CCS timing vs NLDM timing.

Parameters

-help	Prints out the command usage.
- ccs_nldm_correlation	Correlation report for CCS timing vs NLDM timing.
-lib_name <string>	Specifies the name of the library file.
-tolerance <float>	Specifies the tolerance (%) value for a check.

report_summary

```
report_summary
    [-help]
    [-cumulative]
    [-file <string>]
    [-severity <string>]
    [-sort {CATEGORY | SEVERITY | COUNT}]]
```

Generates a summary report of all the messages displayed during a session for the specified library.

Parameters

-help	Prints out the command usage.
-cumulative	Generates a cumulative message summary.
-file <i>string</i>	Specifies a timing library file name for which the message summary is to be generated.
-severity <i>string</i>	Specifies the severity list for which a message summary is to be generated.
-sort [CATEGORY SEVERITY COUNT]	Sorts the report summary on the basis of the specified option. You can specify one of the following options: ■ CATEGORY: The messages are categorized on the basis of groups, such as, LIB_TIMING, LIB_NOISE, LIB_POWER, etc, that are defined in the liberty. ■ SEVERITY: The messages are sorted on the basis of the severity level - Error, Warning, or Information. ■ COUNT: Shows the number of occurrences of a message. The message count is displayed in ascending order.

Examples

- The following command generates a cumulative report summary:

```
valus> report_summary -cumulative
```

```
-----  
*** Message summary for timing library test.lib :-  
-----
```

```
Severity ID Count Category Summary  
ERROR TECHLIB-1213 1 LIB_TIMING For '%s' timing group de ...  
ERROR TECHLIB-1223 2 LIB_SEMANTIC The attribute '%s' on li ...  
WARNING TECHLIB-9161 1 LIB_SEMANTIC Empty Group encountered; ...
```

- The following command generates a report summary of the specified file.

```
valus> report_summary -file test.lib
```

```
-----  
*** Message summary for timing library test.lib :-  
-----
```

```
Severity ID Count Category Summary  
ERROR TECHLIB-1213 1 LIB_TIMING For '%s' timing group de ...  
ERROR TECHLIB-1223 2 LIB_SEMANTIC The attribute '%s' on li ...  
WARNING TECHLIB-9161 1 LIB_SEMANTIC Empty Group encountered..
```

- The following command generates a report summary according to the severity level, that is, error.

```
valus> report_summary -severity error
```

```
-----  
*** Message summary for timing library test.lib :-  
-----
```

```
Severity ID Count Category Summary  
ERROR TECHLIB-1213 1 LIB_TIMING For '%s' timing group de ...  
ERROR TECHLIB-1223 2 LIB_SEMANTIC The attribute '%s' on li ...
```

- The following command generates a sorted report summary based on the message count.

```
valus> report_summary -sort count
```

```
-----  
*** Message summary for timing library test.lib :-  
-----
```

```
Severity ID Count Category Summary  
WARNING TECHLIB-9161 1 LIB_SEMANTIC Empty Group encountered;  
ERROR TECHLIB-1213 1 LIB_TIMING For '%s' timing group de ...  
ERROR TECHLIB-1223 2 LIB_SEMANTIC The attribute '%s' on li ...
```

set_message

```
set_message
    [-help]
    [-id list_of_msgIDs]
    [-severity {warn | error | info} | -limit number | -no_limit | -
    suppress | -unsuppress]
```

Controls the type of messages to be displayed. You can specify message severity levels as - information, warning, or error.

Parameters

-help	Prints out the command usage.
-id <i>list_of_msgIDs</i>	Specifies the list of msg IDs to be processed.
-limit <i>number</i>	Sets the message limit for the selected messages. The limit must be greater than zero.
-no_limit	Removes any existing limit on the selected messages. The suppressed messages will not be un-suppressed.
-severity {warn error info}	Specifies the new severity level for the message ID being modified.
-suppress	Suppresses the message.
-unsuppress	Unsuppresses the message.

Examples

- The following command sets the limit of TECHLIB-436 to 1:

```
valus> set_message -id TECHLIB-436 -limit 1
```
- The following command removes any existing limit set for the message TECHLIB-436:

```
valus> set_message -id TECHLIB-436 -no_limit
```
- The following command changes the severity of TECHLIB-436 to error:

```
valus> set_message -id TECHLIB-436 -severity error
```
- The following command suppresses the TECHLIB-436:

```
valus> set_message -id TECHLIB-436 -suppress
```
- The following command unsuppresses the TECHLIB-436:

Valus User Guide

TCL Shell Commands

```
valus> set_message -id TECHLIB-436 -unsuppress
```

set_valus_mode

```
set_valus_mode
    [-help]
    [-capacitance_index_max float]
    [-capacitance_index_min float]
    [-capacitance_value_tolerance float]
    [-ccs_current_peak_tolerance float]
    [-ccs_noise_data_disable_clip]
    [-ecsm_power_peak_current float]
    [-ecsm_power_time float]
    [-merge_library_checks]
    [-setup_hold_checks]
    [-slew_index_max float]
    [-slew_index_min float]
    [-slew_threshold_tolerance float]
    [-slew_value_tolerance float]
    [-timing_value_threshold float]
    [-voltage_waveform_tolerance float]
```

Sets parameters for running Valus.

Parameters

-help	Prints out the command usage.
-capacitance_index_max <i>float</i>	Specifies the upper limit of the capacitance range.
-capacitance_index_min <i>float</i>	Specifies the lower limit of the capacitance range.
-capacitance_value_tolerance <i>float</i>	Specifies the tolerances in which the capacitances values can vary from that of the pins capacitance values. <i>Default: 0.1</i>
-ccs_current_peak_tolerance <i>float</i>	Specifies the percentage tolerance for allowing multiple peaks in the output current waveform. <i>Default: 0.95</i>
-ccs_noise_data_disable_clip	Specifies whether the dc_current data clipping needs to be disabled.

Valus User Guide

TCL Shell Commands

<code>-ecsm_power_peak_current float</code>	Specifies the peak current value against which the ECSM power group current values need to be checked.
<code>-ecsm_power_time float</code>	Specifies the time value against which the ECSM power group time values need to be checked.
<code>-merge_library_checks</code>	Allows the software to performs the merge library check.
<code>-setup_hold_checks</code>	Enables the setup/hold checks.
<code>-slew_index_max float</code>	Specifies the upper limit of the slew range.
<code>-slew_index_min float</code>	Specifies the lower limit of the slew range.
<code>-slew_threshold_tolerance float</code>	Specifies the tolerances in which the sample points can vary near the slew threshold. <i>Default: 0.1</i>
<code>-slew_value_tolerance float</code>	Specifies the tolerances in which the slew value can vary from that of the NLDM slew values. <i>Default: 0.1</i>
<code>-timing_value_threshold float</code>	Specifies the threshold value for ignoring large timing values <i>Default: 100s (seconds)</i>
<code>-voltage_waveform_tolerance float</code>	Specifies the tolerances for voltage waveform to start/end at 0 or 1. <i>Default: 0.1</i>

Examples

- The following command sets the minimum value to 0 and the maximum value to 10 of the capacitance index.

```
valus> set_valus_mode -capacitance_index_min 0  
-capacitance_index_max 10
```

- The following command sets the value of the `-slew_threshold_tolerance` parameter to 0.9.

```
valus> set_valus_mode -slew_threshold_tolerance 0.9
```

update_library

```
update_library  
    [-help]  
    <lib_name>
```

Specifies the ETM/PTM library to be modified when 2 or more libraries are read in a single session.

Parameters

-help	Prints out the command usage
<libName>	Specifies the name of the library to be updated.

Examples

The following command specifies that library the "test2.lib" needs to be modified, when 3 libraries are read.

```
read_libs test1.lib test2.lib test3.lib > read  
update_library test2.lib
```

Note: This command is mandatory when multiple libraries are read.

write_ldb

```
write_ldb
    [-help]
    -library string
    [-opt_level integer]
    -outfile string
```

Converts a text library to the Cadence binary format library called the library database (LDB).

Parameters

-help	Prints out the command usage.
-library <i>string</i>	Specifies the library to be converted to the LDB format.
-opt_level <i>integer</i>	Specifies the optimization level for the library database size. You can set this parameter to 0 or 1. When set to 0, the library database is not optimized. When set to 1, the software reduces the size of the LDB files by excluding the dotlib content, but limits their usage across other tools, such as Conformal. <i>Default: 0</i>
-outfile <i>string</i>	Specifies the output LDB file.

Examples

- The following command will convert the specified library to LDB format:

```
valus> write_ldb -library A.lib -outfile A.ldb
```


write_lib

```
write_lib
    [-help]
    -lib_name <string>
```

Specifies the output library.

Note: This command cannot be used if there are no modifications in the ETM/PTM library.

Parameter

-help	Prints out the command usage.
-lib_name <string>	Specifies the output library name.

Examples

- The following commands first modify the library (in this case copying attribute from one pin to another) and then write the new modified library with the name new.lib:

```
copy_attribute -attribute_name max_transition -group pin -reference a -
target b
write_lib -lib_name new.lib
```

Valus Checks

This chapter contains a category-wise list of checks:

- [LIB_TIMING](#) on page 43
- [LIB_NOISE](#) on page 92
- [LIB_POWER](#) on page 108
- [LIB_MSV](#) on page 113
- [LIB_LVF](#) on page 124
- [LIB_EM](#) on page 127
- [LIB_MERGE](#) on page 129
- [LIB_SEMANTIC](#) on page 131
- [LIB_MISC](#) on page 220
- [LIB_UI](#) on page 224
- [Data Integrity Checks](#) on page 237
 - [Positive Checks](#) on page 238
 - [Range Checks](#) on page 241
 - [Monotonic Increase Checks](#) on page 244
 - [Mandatory Checks](#) on page 245
 - [Mandatory Ignore Checks](#) on page 246
 - [Count Checks](#) on page 247
 - [Enum Value Checks](#) on page 252

LIB_TIMING

The LIB_TIMING checks are performed during the validation of timing attributes/groups of a library. These checks consist of a number of specific checks for NLDM data, ECSM timing data, and CCS timing data.

LIB_TIMING				
Message Prefix	ID	Message	Severity	Description
VALUS	14	The %s capacitances %s table group %s%s%s for %s transition in pin '%s' of cell '%s' is not in the range of min cap value.	Error	For correct calculation of delay value the capacitances value should be more than or equal to min cap value.
VALUS	1008	Identified trigger arcs with timing sense 'non_unate' and having only 'rise' or 'fall' table. These trigger arcs can have only half unateness.	Error	The trigger arcs with only rise or fall table cannot have unateness non_unate. These trigger arcs can have only half unateness i.e positive_unate/negative_unate. This can cause incorrect timing analysis. The generated library should be further looked for correctness.
VALUS	1011	Identified more than 2 parallel arcs with timing type '%s' between '%s'-'>%s' pins of cell '%s'. The number of parallel arcs cannot be more than 2, between pins pair.	Error	Between a pin pair there can be 2 parallel arc possible. One for early and another for late. When more than 2 parallel arcs are present, the generated library should be further checked for correctness.
VALUS	1026	The function specified on the pin '%s' is '%s', but a timing arc is missing between pins '%s' and '%s' of cell '%s'. This missing timing arc can cause issues in timing analysis.	Error	The timing arc needs to be present from all the input pins specified in the function to the output pin. Absence of timing arc can cause inaccuracy during timing analysis.

Valus User Guide

Valus Checks

LIB_TIMING				
Message Prefix	ID	Message	Severity	Description
VALUS	1029	Identified missing tables '%s' and '%s' for the combinational arc in pin '%s' of cell '%s'. This can cause issue during timing analysis.	Error	For a combinational arc all the four tables - (cell_rise and rise_transition) and (cell_fall and fall_transition) should be defined. For combinational_rise, cell_rise and rise_transition should be present and for combinational_fall, cell_fall and fall_transition should be present.
VALUS	36	The timing arc(s) with related pin '%s' timing_type '%s' is closed for the when condition but not for the Input Pin (s) '%s' in pin '%s' for cell '%s'.	Warning	The when condition in a timing arc(s) should be closed for a specific related pin and timing type. Otherwise this can cause problems during timing analysis.
VALUS	1014	Missing ECSM capacitance group on timing arc/input pin. This can cause loss of accuracy during delay calculation.	Warning	It is observed when ECSM capacitance group is present, it can provide accurate delay calculation results.
VALUS	1015	Missing %s on input/inout pin %s of cell %s.	Warning	It is observed when ECSM/ CCS receiver capacitance group is present, it can provide accurate delay calculation results.
VALUS	1028	The %s slew index '%f' is %s than the normalized driver waveform %s defined slew range (%f to %f) in %s table in pin %s of cell %s. This value should be inside the NDW range.	Warning	The boundary timing table arc slew indices should be within the normalized driver waveform slew indices range. Indices beyond this can cause issues during timing analysis.

Valus User Guide

Valus Checks

LIB_TIMING				
Message Prefix	ID	Message	Severity	Description
VALUS	1034	Identified negative values in '%s' table of timing arc. The table values should not have negative values.	Warning	This message is issued when cell_rise/cell_fall tables in timing arc of a non-ETM cell and rise_transition/fall_transition tables in timing arc of any cell have negative values. To fix the issue, re-characterize the library with appropriate settings, such that the delay and transition tables have positive values.
TECHLIB	420	Number of ecsm_waveforms in the '%s' table on pin %s of cell %s does not match the number of transition table axis points specified in the template '%s'. Ignoring waveform data.	Error	The number of ecsm_waveforms must match the number of data points in the transition table for correct waveform lookup. If this is not the case, waveform data for the transition group is ignored.
TECHLIB	421	Invalid ecsm_waveform group name %s specified in pin %s of cell %s. Ignoring waveform data.	Error	Waveform group name must be a digit and the value of the digit must not be greater than or equal to the total number of entries in the output transition table. All the waveforms must start from 0.
TECHLIB	428	Number of index_1 points does not match with the number of values points for the %s waveform group %s specified in pin %s of cell %s Ignoring waveform data.	Error	This message is issued when the total number of index_1 points are not equal to the total number of values specified in pin of cell for ECSM power/PGDC group. To fix the issue, re-characterize the library with appropriate settings, such that the index_1 is correctly defined.

Valus User Guide

Valus Checks

LIB_TIMING				
Message Prefix	ID	Message	Severity	Description
TECHLIB	432	Unknown ecdm_capacitance type defined in pin %s of cell %s. Ignoring ecdm_capacitance defined in the pin.	Error	The ecdm_capacitance type should be either rise or fall. If a value other than rise or fall is encountered, the ecdm_capacitance information will be ignored
TECHLIB	935	No indices specified for the ecdm capacitance in pin %s of cell %s. Ignoring ecdm_capacitance defined in the pin.	Error	The index_1 attribute needs to be defined for ecdm_capacitance. If it is not defined, the ecdm_capacitance information will be ignored
TECHLIB	1006	The normalized_driver_waveform '%s' specified as '%s' for pin '%s' in cell '%s' was not found in the library	Error	This message is issued when the normalized_driver_waveform is specified as driver_waveform/ driver_waveform_rise/ driver_waveform_fall for a pin in a cell but is not found in library. To fix the issue, re-characterize the library with appropriate settings, such that the library does not use undefined normalized_driver_waveform.

Valus User Guide

Valus Checks

LIB_TIMING				
Message Prefix	ID	Message	Severity	Description
TECHLIB	1007	Incorrect axis type specified in the template '%s' for driver waveform. The template and all normalized_driver_waveform groups using this template would be ignored.	Error	This message is issued when the axis type is specified in the normalized_driver_waveform group template is not correct. The valid axes are input_net_transition and normalized_voltage. The template and all normalized_driver_waveform groups using this template would be ignored. To fix the issue, re-characterize the library with appropriate settings, such that the template should be specified with correct axis type.
TECHLIB	1186	The '%s' table found for timing group with timing_type %s.	Error	This message is issued when attributes cell_rise/cell_fall/rise_transition/fall_transition are found for timing group with sequential timing arcs. To fix the issue, re-characterize the library with appropriate settings, such that the correct attributes are specified for sequential timing arcs.
TECHLIB	1213	For '%s' timing group defined on line %d, related_pin %s is not same as the pin %s.	Error	This message is issued when the attribute related_pin name is not same as the parent pin name for 'min_pulse_width' and 'minimum_period' timing groups. To fix the issue, re-characterize the library with appropriate settings, such that the correct related_pin value is set.

Valus User Guide

Valus Checks

LIB_TIMING				
Message Prefix	ID	Message	Severity	Description
TECHLIB	1214	For '%s' timing group defined on line %d, related_pin is same as the parent pin '%s'.	Error	This message is issued when the attribute related_pin name is defined same as parent pin for timing arcs other than min_pulse_width and minimum_period. To fix the issue, re-characterize the library with appropriate settings, such that the correct related_pin value is set.
TECHLIB	1216	The reference time values '%f' and '%f' defined for vector groups on line '%d' and '%d' respectively are different, even though their input_transition_time index is same.	Error	This message is issued when values of the reference_time and input_net_transition attributes are inconsistent inside the output_current_rise or output_current_fall group. To fix the issue, re-characterize the library with appropriate settings, such that there is no such inconsistency.
TECHLIB	1229	The first value %f for '%s' group defined on line %d is the '%s' value.	Error	This message is issued when the first value of the current waveform in CCS library is not correctly specified. The first value in the 'values' attribute of the group 'output_current_rise' cannot be the maximum value and the first value in the 'values' attribute of the group 'output_current_fall' cannot be the minimum value. To fix the issue, re-characterize the library with appropriate settings, such that the correct values are specified.

Valus User Guide

Valus Checks

LIB_TIMING				
Message Prefix	ID	Message	Severity	Description
TECHLIB	1232	The input_transition_time index values '%f' and '%f' defined for vector groups on line '%d' and '%d' respectively are different, even though their respective reference time is same.	Error	This message is issued when values of the reference_time and input_net_transition attributes are inconsistent inside the output_current_rise or output_current_fall group. To fix the issue, re-characterize the library with appropriate settings, such that there is no such inconsistency.
TECHLIB	1233	Number of values in ecsm_waveform_set in the transition table %s on line %d in pin %s of cell %s does not match the number of transition table axis points specified in the template. Ignoring waveform data.	Error	The number of values in ecsm_waveform_set must match the number of data points in the transition table for correct waveform lookup. If this is not the case, waveform data for the transition group is ignored.
TECHLIB	1236	The ecsm_waveform_set group on line %d in the transition table %s in pin %s of cell %s does not have valid index_1 attribute specified. Ignoring waveform data.	Error	Attribute index_1 is mandatory attribute in ecsm_waveform_set group which represents the normalized output voltage sample points which are shared by all voltage waveforms represented by the ecsm_waveform_set group. It can be specified at the group level or at the template level. If it is not specified, waveform data for the transition group is ignored.

Valus User Guide

Valus Checks

LIB_TIMING				
Message Prefix	ID	Message	Severity	Description
TECHLIB	1263	Unknown capacitance type '%s' defined for %s in cell %s.	Error	The ecsm_capacitance/ ecsm_capacitance_set type should be either rise or fall. If a value other than rise or fall is encountered, the ecsm_capacitance information will be ignored
TECHLIB	1296	This library has both CCS and ECSM data. CCS checks will not be thorough for library constructs which have corresponding ECSM data available.	Warning	In case both CCS and ECSM data is present, ECSM data is given priority for timing and power. In this case, CCS data will get ignored and only ECSM data will be used for all the internal tool calculations for the conflicting groups.
TECHLIB	1306	The group %s with name '%s' already defined for %s of cell %s. This will be ignored	Error	The ecsm_waveform group with same name already exists. First will be retained and rest will be ignored
TECHLIB	1313	For cell '%s', in group '%s', voltage swing on pin/bus/ bundle '%s' with load '%f' and slew '%f' is '%.1f%', should at least reach second slew threshold.	Error	This message is issued when the voltage swing defined for cell on pin/bus/bundle in output_current_rise / output_current_fall vector group with specified load and slew is not reaching the second slew threshold. To fix the issue, re-characterize the library with appropriate settings, such that the full voltage swing is reached.

Valus User Guide

Valus Checks

LIB_TIMING				
Message Prefix	ID	Message	Severity	Description
TECHLIB	1336	The first or last value in %s attribute is not equal %d or %d respectively. The attribute will be ignored.	Error	This message is issued when the first/last value of the attribute receiver_capacitance_rise_threshold_pct/ receiver_capacitance_fall_threshold_pct is incorrectly defined. The valid value of first and last point for receiver_capacitance_rise_threshold_pct attribute is [0.0, and 100] and for receiver_capacitance_fall_threshold_pct attribute is [100, and 0.0]. Such attribute will be ignored. This may impact accuracy in delay and glitch analysis. To fix the issue, re-characterize the library with appropriate settings, such that the threshold values are correctly defined at library level.

Valus User Guide

Valus Checks

LIB_TIMING				
Message Prefix	ID	Message	Severity	Description
TECHLIB	1337	The library has '%s' groups defined but the '%s' definition is missing from library group. All '%s' groups will be ignored.	Error	This message is issued when the attribute receiver_capacitance_rise_threshold_pct/ receiver_capacitance_fall_threshold_pct is missing at library level but receiver_capacitance_rise/ receiver_capacitance_fall groups are defined. Complete set of receiver_capacitance groups will be ignored. This may impact accuracy in delay and glitch analysis. To fix the issue, re-characterize the library with appropriate settings, such that the threshold values are not undefined.

Valus User Guide

Valus Checks

LIB_TIMING				
Message Prefix	ID	Message	Severity	Description
TECHLIB	1338	The segment value for '%s' group for %s and cell '%s' has value %d which is not in the expected range [1, %d]. The group will be ignored.	Error	This message is issued when the attribute segment defined for receiver_capacitance_rise/receiver_capacitance_fall group is out of range as per segment definition. The valid range of segment attribute is 1 to N where N is the number of {threshold points -1} defined in the attribute receiver_capacitance_rise_threshold_pct/receiver_capacitance_fall_threshold_pct. For example if seven threshold points are defined then valid range of segment attribute would be {threshold points -1} that is six. Complete set of receiver_capacitance groups will be ignored. To fix the issue, re-characterize the library with appropriate settings, such that the segment values are defined correctly.

Valus User Guide

Valus Checks

LIB_TIMING				
Message Prefix	ID	Message	Severity	Description
TECHLIB	1339	The '%s' groups for all segments in the range [1, %d] for %s and cell '%s' are not defined. The segment '%d' is missing. The '%s' groups will be ignored.	Error	This message is issued when the receiver_capacitance_rise/receiver_capacitance_fall group for all the segments is not defined. The valid range of segment attribute is 1 to N where N is the number of {threshold points -1} defined in the attribute receiver_capacitance_rise_threshold_pct/receiver_capacitance_fall_threshold_pct. Complete set of receiver_capacitance groups will be ignored. To fix the issue, re-characterize the library with appropriate settings, such that the receiver_capacitance groups are defined for all segment values.
TECHLIB	1340	The value %s specified for variable_%d for '%s' group '%s' is invalid. This template group is used in '%s' (line %d). For this group valid variable values are %s only. The group '%s' will be ignored.	Error	This message is issued when invalid variable is defined in the template used for specific groups. For example variable time specified for the receiver_capacitance_rise/receiver_capacitance_fall group is invalid, the valid values are input_net_transition and total_output_net_capacitance. These groups with invalid template will be ignored. To fix the issue, re-characterize the library with appropriate settings, such that the defined variable should be valid.

Valus User Guide

Valus Checks

LIB_TIMING				
Message Prefix	ID	Message	Severity	Description
TECHLIB	1341	The %s index values for '%s' group (line %d) for %s and cell '%s' do not match among all segments. These '%s' groups will be ignored.	Error	This message is issued when there is mismatch in the index values of input_net_transition/total_output_net_capacitance defined for receiver_capacitance_rise/receiver_capacitance_fall group at pin or timing level. Complete set of receiver_capacitance groups will be ignored. To fix the issue, re-characterize the library with appropriate settings, such that the index values should be same.
TECHLIB	1342	The %s group with segment value '%d' is redefined at line %d for %s and cell '%s'. The last definition will be considered.	Error	This message is issued when the receiver_capacitance_rise/receiver_capacitance_fall group with the same segment value is defined more than once at pin or timing level. The multiple definition of same segment value is not allowed. Last definition will be considered and rest will be ignored. To fix the issue, re-characterize the library with appropriate settings, such that the segment value is correctly defined.

Valus User Guide

Valus Checks

LIB_TIMING				
Message Prefix	ID	Message	Severity	Description
TECHLIB	1343	The %s group can have only one or two dimensional tables. All %s groups for %s and cell '%s' will be ignored.	Error	This message is issued when the dimensions of receiver_capacitance_rise/ receiver_capacitance_fall table is more than two. The receiver_capacitance group can have only one or two dimensional tables. Complete set of receiver_capacitance groups will be ignored. To fix the issue, re-characterize the library with appropriate settings, such that the table dimension should be correct.
TECHLIB	1344	The size of %s tables for %s and cell '%s' do not match among all segments. All '%s' groups will be ignored.	Error	This message is issued when there is mismatch in the size of receiver_capacitance_rise/ receiver_capacitance_fall tables among all the segments defined at pin or timing level. Complete set of receiver_capacitance groups will be ignored. To fix the issue, re-characterize the library with appropriate settings, such that the receiver_capacitance tables should be matched.

Valus User Guide

Valus Checks

LIB_TIMING				
Message Prefix	ID	Message	Severity	Description
TECHLIB	1347	Identified mismatch in the number of pins specified in the attributes pin_names and pin_name_map for the cell '%s' for sensitization group '%s'. The sensitization group for that cell will be ignored.	Error	This message is issued when there is mismatch in the number of pins specified in pin_names attribute at library level and pin_name_map attribute at cell/timing level for the sensitization group. The sensitization group for that cell will be ignored. To fix the issue, re-characterize the library with appropriate settings, such that the pins are correctly defined.
TECHLIB	1348	Referred sensitization group '%s' in cell '%s' is not defined at library level. The sensitization group for that cell will be ignored.	Error	This message is issued when the sensitization group name specified in sensitization_master attribute at cell/timing level is not defined at library level. The sensitization group for that cell will be ignored. To fix the issue, re-characterize the library with appropriate settings, such that the undefined sensitization group is not referred.
TECHLIB	1349	The attributes '%s' defined with value ('%d') for cell '%s' is invalid, the value should not be 0.	Error	This message is issued when the attributes wave_rise_sampling_index and wave_fall_sampling_index are defined with '0' value in sensitization group. Such attribute will be ignored. To fix the issue, re-characterize the library with appropriate settings, such that the attributes are defined correctly.

Valus User Guide

Valus Checks

LIB_TIMING				
Message Prefix	ID	Message	Severity	Description
TECHLIB	1350	The attribute '%s' defined with value '%d' for cell '%s' is invalid, the value should not be greater than the number of entries in {wave_rise/wave_fall} -1 ('%d').	Error	This message is issued when the attributes wave_rise_sampling_index and wave_fall_sampling_index are defined with the value which is greater than the number of entries in {wave_rise/wave_fall} -1. Such attribute will be ignored. To fix the issue, re-characterize the library with appropriate settings, such that the attributes are defined correctly.
TECHLIB	1351	The number of entries ('%d') in the attribute '%s' defined for cell '%s' is greater than the number of entries in {wave_rise/wave_fall}-1 ('%d'). The attribute will be ignored.	Error	This message is issued when the number of entries defined in the attributes wave_rise_time_interval and wave_fall_time_interval are greater than the number of entries in {wave_rise/wave_fall} -1. Such attribute will be ignored. To fix the issue, re-characterize the library with appropriate settings, such that the attributes are defined correctly.
TECHLIB	1352	The attribute '%s' has undefined vector ID (%d) for the sensitization group (%s) specified for cell '%s'. The attribute is ignored.	Error	This message is issued when the attributes wave_rise and wave_fall are defined with incorrect vector ID in the sensitization group. Such attribute will be ignored. To fix the issue, re-characterize the library with appropriate settings, such that the attributes are defined correctly.

Valus User Guide

Valus Checks

LIB_TIMING				
Message Prefix	ID	Message	Severity	Description
TECHLIB	1353	The vector ID ('%d') defined is already present in sensitization group '%s'. Last definition of the vector ID will be considered. Vector ID should be unique.	Error	This message is issued when the sensitization group has two or more vectors with the same vector ID. If duplicate vector IDs are present then tool will use the last vector with that ID. To fix the issue, re-characterize the library with appropriate settings, such that the vector ID defined should be unique.
TECHLIB	1354	The sensitization_master attribute is not present in cell '%s' but cell contains sensitization information. Sensitization group information for that cell will be ignored.	Error	This message is issued when the sensitization_master attribute is not defined at cell level but cell contains sensitization information. sensitization_master attribute is required attribute if cell contains sensitization information. Sensitization group information for that cell will be ignored. To fix the issue, re-characterize the library with appropriate settings, such that the attribute is completely specified.

Valus User Guide

Valus Checks

LIB_TIMING				
Message Prefix	ID	Message	Severity	Description
TECHLIB	1355	Pin '%s' specified in pin_name_map attribute at cell/timing level is either not present or refers to bus/bundle in cell '%s'. The sensitization group for that cell will be ignored.	Error	This message is issued when the pin specified for the sensitization in the pin_name_map attribute at cell or timing level either does not exist or refers to bus/bundle in cell. The sensitization group for that cell will be ignored. To fix the issue, re-characterize the library with appropriate settings, such that the pins are correctly specified.
TECHLIB	1356	The number of pins in pin_names ('%d') do not match with the number of pins specified ('%d') in the vector attribute of sensitization group '%s'.	Error	This message is issued when the size of the attribute pin_names does not match with the number of columns of the second argument of vector attribute in the sensitization group. All the attributes defined for the sensitization group for that cell/timing will be ignored. To fix the issue, re-characterize the library with appropriate settings, such that the size of pin_names attribute is defined correctly.
TECHLIB	1357	The attribute vector defined with vector ID ('%d') in sensitization group '%s' is negative. Vector ID should not be negative.	Error	This message is issued when the attribute vector is specified with negative vector ID in the sensitization group. Such vector attribute will be ignored. To fix the issue, re-characterize the library with appropriate settings, such that the vector ID is defined correctly.

Valus User Guide

Valus Checks

LIB_TIMING				
Message Prefix	ID	Message	Severity	Description
TECHLIB	1358	One or more vector values defined in the vector with ID '%d' in sensitization group '%s' are invalid. This vector attribute will be ignored	Error	This message is issued when the attribute vector is specified with invalid value in the sensitization group. The valid values are {0,1,X,Z} . The attribute vector will be ignored. To fix the issue, re-characterize the library with appropriate settings, such that the vector values are defined correctly.
TECHLIB	1398	The '%s' waveform specified in cell '%s' has very long tail with time value reaching '%f'. The last '%d' points in waveform will be ignored and tool will complete the waveform using extrapolation for timing analysis. The last time point considered is '%f' and the first time point ignored is '%f'.	Warning	This message is issued when there are points in CCS timing waveform (output_curent_rise/output_current_fall) which results in very long tail in waveform. These long tails in waveform if used as such may cause issues in delay and glitch calculations. The waveform has very big values in time index. Tool ignores such bad points and completes the waveform using extrapolation for timing analysis. Such waveform indicate issue in characterization and should be reviewed.
TECHLIB	1399	Attributes tied_off and related_pin present in timing group for pin '%s' of cell '%s'. Attribute related_pin cannot be specified on a tied_off output pin.	Error	This message is issued when tied_off output pin consists related_pin attribute in the timing group. To fix the issue, re-characterize the library with appropriate settings, such that the related_pin is not specified for tied_off output pin.

Valus User Guide

Valus Checks

LIB_TIMING				
Message Prefix	ID	Message	Severity	Description
TECHLIB	1401	The timing arc defined on pin '%s' of cell '%s' with timing_sense '%s' and related_pin '%s' has invalid when attribute '%s'. The when condition is not compatible with the pin function '%s'.	Error	This message is issued when timing_arc is defined with invalid when condition for specified timing_sense and related_pin attributes. For example: in case of two input Mux (S0 I0 + !S0 I1) for related pin I0, the valid timing arcs are ((S0 I1) and (S0 !I1)) or (S0), which implies that output will toggle on toggling I0 only if S0=0. If when condition is defined as (!S0 I1), which is an invalid when condition as there would be no transition in this condition at the output. To fix the issue, re-characterize the library with appropriate settings, such that the timing arc is defined with correct when condition.

Valus User Guide

Valus Checks

LIB_TIMING				
Message Prefix	ID	Message	Severity	Description
TECHLIB	1403	The timing arc defined on pin '%s' of cell '%s' with timing_sense '%s' and related_pin '%s' is not defined for all possible 'when' conditions. As per the pin function '%s', the timing arc(s) with when condition '%s' are not defined.	Error	This message is issued when timing arc(s) are missing for reported when condition expression for specified pin function and default arc is also not specified. For example, in case of two input Mux (S0 I0 + !S0 I1) for related pin I0, the valid timing arcs are ((S0 I1) and (S0 !I1)) or (S0), which implies that output will toggle on toggling I0 only if S0=0. If any of the arc ((S0 I1) and (S0 !I1)) is missing, it will be an invalid condition. To fix the issue, re-characterize the library with appropriate settings, such that the timing arc for all valid combinations are specified.

Valus User Guide

Valus Checks

LIB_TIMING				
Message Prefix	ID	Message	Severity	Description
TECHLIB	1404	The timing arc defined on pin '%s' of cell '%s' with related_pin '%s' has incorrect timing sense '%s'. As per the when condition '%s' and pin function '%s', the timing sense should be '%s'.	Error	This message is issued when timing_sense is invalid as per the 'when' condition defined for timing arc. For example, in case of two input Mux (S0 I0 + !S0 I1) for related pin I0, the valid timing arcs are ((S0 I1) and (S0 !I1)) or (S0), which implies that output will toggle on toggling I0 only if S0=0. This implies positive unate relationship between I0 and output. Attribute timing_sense with 'negative_unate' for this condition would be incorrect. To fix the issue, re-characterize the library with appropriate settings, such that the timing arc is specified with correct timing_sense
TECHLIB	1405	The timing arc defined on pin '%s' of cell '%s' with related_pin '%s' has incorrect timing sense '%s'. As per the pin function '%s', the timing sense should be '%s'.	Error	This message is issued when the timing sense defined for timing arc is not consistent with the function of pin. For example, in case of inverter, which has function (!A) then arc should have a negative_unate timing sense. To fix the issue, re-characterize the library with appropriate settings, such that the timing_sense and function are consistent.

Valus User Guide

Valus Checks

LIB_TIMING				
Message Prefix	ID	Message	Severity	Description
TECHLIB	1406	The timing arc defined on pin '%s' of cell '%s' has related_pin '%s'. As per the pin function '%s', the timing arc is not valid.	Error	This message is issued when timing_arc defined for specified 'related_pin' attribute is invalid as per the pin function. To fix the issue, re-characterize the library with appropriate settings, such that the timing arc is correctly defined.
TECHLIB	1407	There is no combinational timing arc defined on pin '%s' of cell '%s' with related pin '%s'. As per the pin function '%s', there should be at least one timing arc defined.	Error	This message is issued when timing_arc is not defined for specified 'related_pin' attribute as per the pin function. To fix the issue, re-characterize the library with appropriate settings, such that the timing arc is defined for all possible valid condition.

Valus User Guide

Valus Checks

LIB_TIMING				
Message Prefix	ID	Message	Severity	Description
TECHLIB	1416	The receiver capacitance %s groups are missing in the timing group%s specified for %s of cell '%s'.	Warning	This message is issued when either the rise group (receiver_capacitance1_rise/ receiver_capacitance2_rise or receiver_capacitance_rise) or the fall group (receiver_capacitance1_fall/ receiver_capacitance2_fall or receiver_capacitance_fall) is missing in the timing arc. Moreover, there is no corresponding pin level receiver_capacitance group with the default or same when condition having the missing information. To fix the issue, re-characterize the library with appropriate settings, such that the receiver_capacitance group is defined correctly

Valus User Guide

Valus Checks

LIB_TIMING				
Message Prefix	ID	Message	Severity	Description
TECHLIB	1467	The 'dynamic_current' waveform specified in cell '%s' has very long tail with long tail ratio '%f' and time value reaching '%f'. This may impact the accuracy of transient power analysis.	Warning	This message is issued when there are points in CCS power waveform resulting in very long tail with long tail ratio exceeding the specified threshold. The long tail ratio is calculated as $(T3-T2)/(T2-T1)$, where T1 is start time, T2 is peak time and T3 is end time. The threshold can be controlled using global 'set timing_library_ccsp_long_tail_threshold' in tempus and using command set_valus_mode - ccsp_long_tail_threshold in Valus. These long tails in waveform if used as such may cause issues in accuracy of power analysis. To fix the issue, review the characterization and modify it with appropriate values.

Valus User Guide

Valus Checks

LIB_TIMING				
Message Prefix	ID	Message	Severity	Description
TECHLIB	1468	The '%s' group on cell '%s'%s, has when condition '%s' which is not mutually exclusive with when condition '%s'.	Warning	This message is issued when the leakage_power or internal_power groups, present on a cell or pin respectively, have when conditions which are not mutually exclusive. For example, if two equivalent internal_power groups have when conditions 'D & !SE' and 'D & SE', they will be mutually exclusive. But if they have when conditions 'D' and 'D & SE', then they will not be mutually exclusive. To fix this issue, you should modify the specified when conditions such that they are mutually exclusive.
TECHLIB	1469	The total net capacitance for slew point %s, in receiver capacitance table for cell '%s', %s, is negative ('%f'). Check capacitance tables in library starting at line '%d' and review the library characterization setup, if needed.	Warning	This message is issued when the total net capacitance across segments for the given slew and load points in the receiver capacitance table on a pin or timing arc is negative. To fix this, check the capacitance tables in the library and review the characterization setup, if needed.

Valus User Guide

Valus Checks

LIB_TIMING				
Message Prefix	ID	Message	Severity	Description
TECHLIB	1482	The '%s' waveform specified in cell '%s' has very long beginning with time values as small as '%f'. The first '%d' points in waveform will be ignored.	Warning	This message is issued when there are points in CCS timing waveform (output_curent_rise/output_current_fall), which results in very slow beginning. These small time points in waveform if used as such may cause issues in delay and glitch calculations. Tempus ignores such bad points. Such waveforms indicate issues in characterization and should be reviewed.
TECHLIB	9067	Missing %s delay value for the path from pin %s to pin %s in the cell %s. Path is not generated. The data in the timing library is incomplete. This can result in a loss of accuracy during delay calculation.	Error	This message is issued when delay and retain values are not specified for the path. Path is not generated if either delay or slew value is missing. To fix the issue, re-characterize the libraries such that delay and retain values are specified correctly.
TECHLIB	9093	Incorrect timing_type specified in timing group of pin '%s' with related pin name '%s' in cell '%s', ignoring arc.	Error	This message is issued when the incorrect timing_type is specified. Such timing arc will be ignored. Arcs from clock_gate_clock_pin and clock_gate_enable_pin to clock_gate_out_pin should be of combinational timing_type. To fix the issue, re-characterize the library with appropriate settings, such that the correct timing_type is defined.

Valus User Guide

Valus Checks

LIB_TIMING				
Message Prefix	ID	Message	Severity	Description
TECHLIB	9119	Missing related pin %s for a timing group of pin %s in cell %s.	Error	This message is issued when the related_pin is specified for a timing group but is not defined in the cell. To fix the issue, re-characterize the library with appropriate settings, such that the related_pin is defined.
TECHLIB	9137	Mismatch in bus width in timing arc with related pin '%s' in pin '%s' of cell '%s'. Ignoring the timing arc.	Error	The bus width of path between source bus pin and target bus pin should match otherwise that timing arc would be ignored.
TECHLIB	9140	No timing sense found for the timing arc at (line no : %d) for cell %s. This is a required attribute for delay timing arcs.	Warning	Timing Sense is a required attribute for delay arcs. If missing can cause issues during timing analysis.
TECHLIB	9141	Identified delay/slew table(s) for timing arc at (line no : %d) with timing type '%s' for cell '%s'. This timing group is check arc and cannot have delay/slew tables.	Error	The cell_rise, cell_fall, rise_transition, fall_transition table are not expected in min_pulse_width arc. These arcs can have only rise_constraint or fall_constraint tables.

Valus User Guide

Valus Checks

LIB_TIMING				
Message Prefix	ID	Message	Severity	Description
TECHLIB	9143	Timing group defined in library has either missing or incorrectly defined lookup tables(s) for cell '%s'. This timing group will be ignored.	Error	Timing groups in library without any delay, transition, or constraint tables are not allowed. The timing group without any indices on lookup table or its referred template is not allowed. All such timing groups in the library will be ignored. To resolve the issue, modify the timing arc definition by adding cell_rise rise_transition/rise_constraint table or cell_fall/fall_transition/fall_constraint table.
TECHLIB	338	Linear timing property '%s' found on pin '%s' of cell '%s' in the non-linear library '%s'	Warning	This message is issued when the non-linear library contains a linear model cell or update a linear model cell to a non-linear library.
TECHLIB	422	The output transition table '%s' for pin '%s' of cell '%s' is not a function of capacitive load. No ECSM data is available for this transition group. Capacitive load indices are required for generating the ECSM data.	Warning	This message is issued when the output transition table is not a function of capacitive load. Such ECSM data will not be available. To fix the issue, re-characterize the library with appropriate settings, such that the capacitive load indices are defined.

Valus User Guide

Valus Checks

LIB_TIMING				
Message Prefix	ID	Message	Severity	Description
TECHLIB	430	Values of the %s waveform group %s should monotonically increase in pin %s of cell %s	Warning	This warning is encountered if the ECSM waveform in the timing library is not monotonically increasing or monotonically decreasing. ECSM waveform models the v-t relationship for a signal. The voltage should monotonically increase (for rise waveforms) and monotonically fall (for fall waveforms). If there is a glitch in the waveform the above message is issued. To avoid this warning, fix the ECSM waveform data in the library so that there are no glitches in the waveform.
TECHLIB	437	The ecsm_waveform_set group in pin '%s' cell '%s' is redefined with different axis points	Warning	Number of index and value points defined in the ecsm_waveform_set group should be same. If the index points are redefined, the number of redefined index points should be same as the number of index points defined in the original template
TECHLIB	458	The %s index of ECSM capacitances table %s %s in pin '%s' of cell '%s' is greater than pin's capacitances value	Warning	For correct calculation of delay value the capacitances value should be less than or equal to pin's cap value.

Valus User Guide

Valus Checks

LIB_TIMING				
Message Prefix	ID	Message	Severity	Description
TECHLIB	800	For the timing arc of pin '%s' of cell '%s', the value '%s' specified for the user defined attribute domino_arc_type is invalid. This attribute is being ignored.	Warning	This message is issued when the value defined for domino_arc_type for the timing arc of pin of cell for user defined group is invalid. The valid values of domino_arc_type are clock_precharge, clock_eval and data_eval. The attribute will be ignored. To fix the issue, re-characterize the library with appropriate settings, such that the valid value is defined.
TECHLIB	809	Time values of the vector group number '%d' specified in '%s' group of the timing arc in pin '%s' of cell '%s', should monotonically increase.	Warning	This message is issued when the time values specified in the vector groups output_current_fall and output_current_rise of the timing arc in pin of cell do not monotonically increase. To fix the issue, re-characterize the library with appropriate settings, such that the time values for output_current_fall and output_current_rise groups vector are monotonic.

Valus User Guide

Valus Checks

LIB_TIMING				
Message Prefix	ID	Message	Severity	Description
TECHLIB	810	Both the '%s' groups must be specified for the timing arc in pin '%s' of cell '%s'. The receiver capacitance data for this timing arc is ignored	Warning	This message is issued when any of the receiver_capacitance1_rise, receiver_capacitance1_fall, receiver_capacitance2_rise, receiver_capacitance2_fall is missing for the timing arc in pin of the cell of receiver_capacitance group. All of them must be defined if any one of them is defined. Such receiver_capacitance data for this timing arc will be ignored. To fix the issue, re-characterize the library with appropriate settings, such that the receiver_capacitance group is defined correctly.
TECHLIB	814	Number of vector groups(%d) specified in '%s' group of timing arc in pin '%s' of cell '%s' do not match unique slew/load(%dx%d) pair value. This output current group will be ignored.	Warning	This message is issued when there is mismatch in the number of vector groups and unique slew/load pair. For this output current group, conversion of the CCS vector groups to ECSM waveform groups is ignored for timing analysis. This may impact delay calculation accuracy. Re-characterize the library such that output current group is specified correctly.

Valus User Guide

Valus Checks

LIB_TIMING				
Message Prefix	ID	Message	Severity	Description
TECHLIB	816	The receiver_capacitance group specified over the output pin '%s' of cell '%s' is ignored.	Warning	This message is issued when the receiver_capacitance group is specified on the output pin of cell. The receiver_capacitance group can be defined on input pin or inout pin at pin level. The receiver_capacitance group will be ignored. To fix the issue, re-characterize the library with appropriate settings, such that the receiver_capacitance group is not specified on output pin of the cell.
TECHLIB	817	The receiver capacitance %s groups are missing in the the receiver capacitance group %s specified for %s of cell '%s'.	Warning	This message is issued when either of the rise group (receiver_capacitance1_rise/ receiver_capacitance2_rise) or fall group (receiver_capacitance1_fall/ receiver_capacitance2_fall) is missing for pin in the receiver_capacitance group for a 'when' condition. If there are duplicate 'when' conditions, this data must be present for at least one of the 'when' condition. In such case, other receiver_capacitance group with same 'when' condition need not have all 4 groups. They can just have either rise or fall groups. To fix the issue, re-characterize the library with appropriate settings, such that the receiver_capacitance group is defined correctly.

Valus User Guide

Valus Checks

LIB_TIMING				
Message Prefix	ID	Message	Severity	Description
TECHLIB	818	Only input_net_transition axis is expected for '%s' group defined within receiver_capacitance group specified on the pin '%s' of cell '%s'. This receiver_capacitance group is ignored.	Warning	This message is issued when the invalid axis is defined for receiver_capacitance1_rise/ receiver_capacitance2_rise/ receiver_capacitance1_fall/ receiver_capacitance2_fall on the pin of the cell for receiver_capacitance group. The valid axis is only input_net_transition. Such receiver_capacitance group will be ignored. To fix the issue, re-characterize the library with appropriate settings, such that the input_net_transition axis should be specified for receiver_capacitance group.
TECHLIB	819	The number of cap values is not equal to the number of time index points for the '%s' group specified in pin '%s' of cell '%s'. This receiver_capacitance group is ignored.	Warning	This message is issued when the number of capacitance value is specified for receiver_capacitance_rise1/ receiver_capacitance_rise2/ receiver_capacitance_fall1/ receiver_capacitance_fall2 are not equal to the number of time index points. Such receiver_capacitance group will be ignored. To fix the issue, re-characterize the library with appropriate settings, such that the number of cap value should be equal to the number of time index points.

Valus User Guide

Valus Checks

LIB_TIMING				
Message Prefix	ID	Message	Severity	Description
TECHLIB	820	Undefined template '%s' for the '%s' group specified for the pin '%s' of cell '%s'. The receiver capacitance data for this pin is ignored.	Warning	This message is issued when an undefined template is used in receiver_capacitance1_rise/ receiver_capacitance2_rise/ receiver_capacitance1_fall/ receiver_capacitance2_fall groups. Such receiver_capacitance data for this pin would be ignored. To fix the issue, re-characterize the library with appropriate settings, such that the template used is defined in library.
TECHLIB	827	The support to threshold construct in ecsm capacitance group has been provided post ECSM VERSION 1.2. The library ECSM VERSION is %s, therefore the threshold construct will be ignored for the ecsm capacitance group in pin '%s' and cell '%s'.	Warning	This message is issued when threshold_pct construct in ecsm_capacitance group is specified in library but the ECSM VERSION < 1.2. Threshold construct will be ignored for the ecsm capacitance group.
TECHLIB	904	Cell '%s' has both edge and combinational arcs between pins '%s' and '%s'. Such arcs can cause problems in the timing analysis flow.	Warning	For correct timing analysis both combinational and edge arcs should not be present between two pins.

Valus User Guide

Valus Checks

LIB_TIMING				
Message Prefix	ID	Message	Severity	Description
TECHLIB	1104	Time values of the %s specified in '%s' group of cell '%s', should monotonically increase.	Warning	This message is issued when the time values specified in the ccs_power vector group of a cell do not monotonically increase. To fix the issue, re-characterize the library with appropriate settings, such that the time value for the ccs_power vector group are monotonic.
TECHLIB	1105	Vector groups specified in '%s' group of cell '%s' are not defined for every slew and load pair value. This may lead to undesirable analysis results.	Warning	This message is issued when vector group is not defined for each load and slew pair. The vector group should be defined for each slew and load pair and number of vector group should match with slew * cap value pair. To fix the issue, re-characterize the library with appropriate settings such that corresponding vector groups are defined for each slew and load pair.
TECHLIB	1108	No normalized driver waveform found in this library. This can cause accuracy issues during delay calculation.	Warning	The normalized driver waveform is required for higher precision during delay calculation.

Valus User Guide

Valus Checks

LIB_TIMING				
Message Prefix	ID	Message	Severity	Description
TECHLIB	1131	The normalized driver waveform '%s' will be extrapolated to span the voltage range 0 to 1.	Warning	This message is issued when the voltage range of normalized_driver_waveform is specified within the range of 0 to 1 but does not span fully from 0 to 1. In such cases, tool will extrapolate to span the voltage from 0 to 1. To fix the issue, re-characterize the library with appropriate settings, such that the normalized_driver_waveform should be defined with appropriate voltage range.
TECHLIB	1148	Found redefined indices in the ecsm_capacitance group of timing arc section of cell '%s'. Indices cannot be redefined in the ecsm_capacitance group of timing arc section as this group would use the indices of transition table. This can cause issues during delay calculation.	Warning	In timing arc section, the ECSM Capacitance group defined is 2 dimensional and would use the indices of transition table. The ECSM capacitance are not expected to have index_1 or index_2 defined in this group of timing arc section.

Valus User Guide

Valus Checks

LIB_TIMING				
Message Prefix	ID	Message	Severity	Description
TECHLIB	1251	For cell '%s', arc '%s', 'CCS '%s' waveform has more than one peak '%s' at line %d. Review the multiple peaks and ensure that these are as per the expectations.	Warning	This message is issued when CCS output current waveform (rise/fall) has distortions and has more than one peak. Review the multiple peaks and ensure that these are as per the expectations. The default threshold value for maximum peak violation is 0.95. To change the default value in valus, use the command line option 'set_valus_mode - ccs_current_peak_tolerance'
TECHLIB	1255	The '%s' table on the timing arc has %s as '%s' on pin '%s' for cell '%s' but it has no 'ecsm_capacitance %s' tables. Ensure that the corresponding ecsm_capacitance/ ecsm_capacitance_set tables are present.	Warning	This message is issued when complete ECSM data is not present for a timing arc. For example, in timing arc, if ecsm waveform data is present but corresponding ecsm capacitance data is missing then software issues this message. This incomplete data can lead to inaccuracy in timing analysis. To fix this issue, re-characterize the library such that ECSM data is complete for the mentioned timing arc.

Valus User Guide

Valus Checks

LIB_TIMING				
Message Prefix	ID	Message	Severity	Description
TECHLIB	1279	For cell '%s', in group '%s', voltage swing on pin/bus/bundle '%s' with load '%f' and slew '%f' is '%.1f%', should be within five percent of '%s'.	Warning	Using current waveform for the cell pin in the library, tool converts it to calculate the charging voltage. If the calculated voltage does not reach a particular threshold (within ~5% of VDD voltage) then this message is issued. This may impact accuracy in delay and glitch analysis. To fix the issue, re-characterize the library with appropriate settings, such that the full voltage swing is reached.

Valus User Guide

Valus Checks

LIB_TIMING				
Message Prefix	ID	Message	Severity	Description
TECHLIB	1280	The group '%s' has not been specified corresponding to the group '%s', but '%s' data has been specified corresponding to other groups on the same arc, on %s for cell '%s'.	Warning	This message is issued when the CCS timing/ECSM/LVF/FIT data is defined for one or more slew tables but missing on other slew tables within the same arc. The valid slew groups are rise_transition, fall_transition, retain_rise_slew, retain_fall_slew, ocv_sigma_rise_transition, ocv_sigma_fall_transition, ocv_sigma_cell_rise, ocv_sigma_cell_fall, ocv_sigma_rise_constraint and ocv_sigma_fall_constraint. To fix the issue, re-characterize the library with appropriate settings, such that the CCS timing/ECSM waveform/LVF/FIT data is defined corresponding to all group within the same arc.

Valus User Guide

Valus Checks

LIB_TIMING				
Message Prefix	ID	Message	Severity	Description
TECHLIB	1281	The timing arc on %s for %s '%s' has CCS timing driver information, but CCS timing receiver information is missing.	Warning	This message is issued when the CCS timing driver information is specified but CCS timing receiver information is missing for the timing arc on pin of the cell. The CCS receiver data on arc is a mandatory if CCS driver model information is present. The valid CCS receiver model pairs are receiver_capacitance1_rise, receiver_capacitance2_rise and receiver_capacitance1_fall, receiver_capacitance2_fall. To fix the issue, re-characterize the library with appropriate settings, such that timing arc should be specified with both driver and receiver information.

Valus User Guide

Valus Checks

LIB_TIMING				
Message Prefix	ID	Message	Severity	Description
TECHLIB	1282	%s data has been defined on one or more %s for %s '%s' but is missing for the %s on %s of the cell.	Warning	This message is issued when the CCS timing/ECSM waveform/LVF/FIT data is specified on one or more timing arc on pin of the cell and missing for another timing arc on pin of the cell. Only those timing arcs will be considered which have slew tables present on them. The valid slew groups are rise_transition, fall_transition, retain_rise_slew, retain_fall_slew, ocv_sigma_rise_transition, ocv_sigma_fall_transition, ocv_sigma_cell_rise, ocv_sigma_cell_fall, ocv_sigma_rise_constraint and ocv_sigma_fall_constraint. To fix the issue, re-characterize the library with appropriate settings, such that CCS timing/ECSM waveform/LVF/FIT data is defined for all timing arcs of all pins of the cell.

Valus User Guide

Valus Checks

LIB_TIMING				
Message Prefix	ID	Message	Severity	Description
TECHLIB	1283	The '%s' timing group is defined, but its corresponding '%s' timing group is missing for %s in cell '%s'.	Warning	The message is issued when the setup_rising/setup_falling timing_type is defined but its corresponding timing_type hold_rising/hold_falling is not defined in the timing group. To fix the issue, re-characterize the library with appropriate settings, such that setup and hold timing arcs are present in pair for fall/rise transitions.
TECHLIB	1284	The '%s' table indices do not match in %s timing group on line %d and %s timing group for %s in cell '%s'	'Warning	This message is issued when there is discrepancy in number of indices of rise_constraint/fall_constraint table for setup_rising and hold_rising timing group. To fix the issue, re-characterize the library with appropriate settings, such that the indices match in constraint tables.

Valus User Guide

Valus Checks

LIB_TIMING				
Message Prefix	ID	Message	Severity	Description
TECHLIB	1285	The sum of values in '%s' table of %s timing group on line %d and corresponding %s timing group, is below %fps for %s in cell '%s'	'Warning	This message is issued when the sum of values of rise_constraint/fall_constraint table for setup_rising/setup_falling and corresponding hold_rising/hold_falling timing group is less than the minimum value. This sum can only be zero or any positive value specified using min_value. To fix the issue, re-characterize the library with appropriate settings, such that the sum of values matches the specified value. Default value of min_value is zero.
TECHLIB	1293	For group '%s', number of vector groups defined is less than four.	Warning	There should be a minimum of four vector groups defined within an output_current_rise or output_current_fall group.
TECHLIB	1294	For group '%s', less than two vector groups are defined having unique values of '%s'.	Warning	There should be a minimum of four vector groups defined within an output_current_rise or output_current_fall group. Current information should be specified for at least two different values of input slew and two different values of output load.

Valus User Guide

Valus Checks

LIB_TIMING				
Message Prefix	ID	Message	Severity	Description
TECHLIB	1345	This library contains both multi-segmented CCS receiver capacitance models and the older two piece receiver capacitance models. The multi-segmented CCS receiver capacitance models will be used in analysis.	Warning	This message is issued when the library contains both multi-segmented CCS receiver capacitance model and the two piece receiver capacitance model. The multi-segmented CCS receiver capacitance models will be used for analysis.
TECHLIB	1360	The voltage values for %s group in '%s' is not %s at point close to %f (%s at '%f'). The value should %s in the tolerance range of %f.	Warning	The voltage waveform for rise/fall transition should start at point close to 0/1 and end at close to 1/0.
TECHLIB	1361	The number of index/values points found is '%d'. The number of sample points(index/values) should be 10 or more than 10 for accurate delay calculation.	Warning	The number of sample time and voltage points should be more than 10 for accurate delay calculation.
TECHLIB	1362	The sample points for ECSM waveform number %d in %s transition are not near the %s threshold.	Warning	The number of sample time and voltage points should be more than 10 for accurate delay calculation.
TECHLIB	1363	The slew value %f for ECSM waveform number %d in %s transition for %s of cell '%s' does not match the corresponding NLDM slew value %f.	Warning	This message is issued when the slew value calculated from ECSM waveform (e.g slew = time values at (upper slew threshold-lower slew threshold)) for rise/fall transition does not match with corresponding NLDM slew value. This can cause delay calculation discrepancies.

Valus User Guide

Valus Checks

LIB_TIMING				
Message Prefix	ID	Message	Severity	Description
TECHLIB	1367	The delay value '%f' calculated for CCS waveform for '%s' vector group on %s of cell '%s' does not match with the corresponding NLDM delay value '%f' at line '%d' and has '%.2g%' error	Warning	This message is issued when there is mismatch in delay value calculated for CCS timing waveform and its corresponding NLDM delay value. The percentage error is calculated with reference to NLDM delay value.
TECHLIB	1368	The slew value '%f' calculated for CCS waveform for '%s' vector group on %s of cell '%s' does not match with the corresponding NLDM slew value '%f' at line '%d' and has '%.2g%' error	Warning	This message is issued when there is mismatch in slew value calculated for CCS timing waveform and its corresponding NLDM slew value. The percentage error is calculated with reference to NLDM slew value.
TECHLIB	1394	Missing %s delay value for the path from pin %s to pin %s in the cell %s. Path is not generated. The data in the timing library is incomplete. This can result in a loss of accuracy during delay calculation.	Warning	This message is issued when delay/retain values are not specified for the path. Path is not generated if either delay or slew value is missing. To fix the issue, re-characterize the libraries such that delay/retain values are specified correctly

Valus User Guide

Valus Checks

LIB_TIMING				
Message Prefix	ID	Message	Severity	Description
TECHLIB	1395	The library level attribute(s) receiver_capacitance_rise_threshold_pct or(and) receiver_capacitance_fall_threshold_pct are defined in library but the library has no 'receiver_capacitance_rise' and 'receiver_capacitance_fall' groups.	Warning	This message is issued when library level attribute(s) receiver_capacitance_rise_threshold_pct or(and) receiver_capacitance_fall_threshold_pct are defined in library but the library has no 'receiver_capacitance_rise' and 'receiver_capacitance_fall' groups. If this is intentional, you can ignore this message. Otherwise, to fix the issue, re-characterize the libraries with appropriate settings.
TECHLIB	1402	The timing arc defined on pin '%s' of cell '%s' with timing_sense '%s' and related_pin '%s' is not defined for all possible 'when' conditions but default arc is present in library. As per the pin function '%s', the timing arc(s) with when condition '%s' are not defined.	Warning	This message is issued when timing arc(s) are missing for reported when condition expression for specified pin function but default arc is present. For example, in case of two input Mux (S0 I0 + !S0 I1) for related pin I0, the valid timing arcs are ((S0 I1) and (S0 !I1)) or (S0), which implies that output will toggle on toggling I0 only if S0=0. If any of the arc ((S0 I1) and (S0 !I1)) is missing, it will be an invalid condition. To fix the issue, re-characterize the library with appropriate settings, such that the timing arc for all valid combinations are specified.

Valus User Guide

Valus Checks

LIB_TIMING				
Message Prefix	ID	Message	Severity	Description
TECHLIB	1463	The retain group attribute(s) '%s' is/are present in the timing arc defined for pin '%s' in cell '%s' but the corresponding timing group attribute(s) '%s' is/are missing. Tool will use retain group information for the corresponding missing timing groups in delay calculation.	Error	This message is issued when one or more retain group attributes (retaining_rise, retaining_fall, retain_rise_slew, retain_fall_slew) are present in a timing arc but the corresponding timing group attributes (cell_rise, cell_fall, rise_transition, fall_transition) are missing. The tool uses retain groups for the corresponding missing timing groups in delay calculation.
TECHLIB	1465	The timing arc defined on pin '%s' of cell '%s' with timing_sense '%s' and related_pin '%s' has invalid when attribute '%s'. The pin '%s' specified in when attribute is not valid for this arc	Error	This message is issued when timing_arc is defined with invalid when condition for specified related_pin attribute. The pin specified in the when condition is not valid for the given timing arc.

Valus User Guide

Valus Checks

LIB_TIMING				
Message Prefix	ID	Message	Severity	Description
TECHLIB	9068	Missing %s slew value for the path from pin %s to pin %s in the cell %s. %s. The data in the timing library is incomplete. This can result in a loss of accuracy during delay calculation.	Warning	This warning is encountered if the delay tables (cell_rise/cell_fall) are specified between a pin pair in the cells and output slew(rise_transition/fall_transition) tables are missing. It is imperative to have both the delay and slew tables to accurately model the slew propagation and hence delay for the cell driven by the cell in question. If the output slew table is missing, the output slew is set to 0 for this cell which would definitely cause wrong delay calculation for the cell driven by the cell in question. To avoid this warning make sure that the timing libraries have complete delay and slew data.
TECHLIB	9124	Missing when or sdf_cond attributes in the min_pulse_width group of pin '%s' in cell '%s'.	Warning	Missing when or sdf_cond attributes in the min_pulse_width group of pin <Pin> in cell <Cell>. Attributes when and sdf_cond define the enabling condition for the timing check. Both attributes are required in the min_pulse_width group.

LIB_NOISE

The LIB_NOISE checks are performed during the validation of ECSM and CCS noise data for a library. These checks include validation of ECSM noise constructs, like `ecsm_vivo` data, `ecsm_internal` transition, noise tolerance attributes, and so on. For CCS noise data, noise vector groups, like CCS first stage, CCS last stage, `dc_current` groups etc, are checked.

LIB_NOISE				
Message Prefix	ID	Message	Severity	Description
VALUS	940	The <code>ecsm_noise</code> version identified is %f. The supported minor version is 1.	Warning	The Cadence tools are backward compatible across all version identifier in a library, but not always forward compatible. Some new constructs might be ignored if <code>ecsm_noise_version</code> is greater than 1.1.
VALUS	941	Identified <code>variable2</code> field in <code>ecsm_itt_template</code> template group. The <code>ecsm_internal_transition</code> group can refer 'ecsm_itt_template' template with <code>variable2</code> defined, only when <code>related_output_pin</code> is defined in <code>ecsm_internal_transition</code> group.	Warning	The <code>ecsm_internal_transition</code> group can use <code>ecsm_itt_template</code> , with <code>variable2</code> defined only when <code>related_output_pin</code> is defined in <code>ecsm_internal_transition</code> group. The characterized library needs to be looked further for correctness.
VALUS	942	Invalid values identified for %s of <code>ecsm_itt_template</code> group. The valid value is %s.	Warning	The valid values for <code>variable_1</code> and <code>variable_2</code> in <code>ecsm_itt_template</code> template group are <code>input_net_transition</code> and <code>related_out_total_output_net_capacitance</code> respectively.

Valus User Guide

Valus Checks

LIB_NOISE				
Message Prefix	ID	Message	Severity	Description
VALUS	943	Non-monotonic increase in index_1 values of ecdsm_itt_template group. This can cause problems during noise analysis.	Warning	The index_1 values of ecdsm_itt_template group should be monotonically increasing for correct noise analysis.
VALUS	944	Missing '%s' attribute found in ecdsm_internal_transition group. This is the required attribute in internal_transition group.	Warning	The following are the required attributes in ecdsm_internal_transition group, ecdsm_transition_type, related_pin, index_1, values. The required attributes when missing, can cause inaccuracy during noise analysis.
VALUS	945	The number of '%s' points in ecdsm_internal_transition group does not match with the number of '%s' points in template referred in ecdsm_internal_transition group.	Warning	The number of index points in ecdsm_internal_transition should match with the number of index point in the template referred by the ecdsm_internal_transition group. Otherwise can cause inaccuracy during noise analysis.
VALUS	946	Identified ecdsm_internal_transition group on output pin '%s'. The groups are present only on input/inout pins.	Warning	The ecdsm_internal_transition groups are present on input pins. If group is present on output pins, the characterized library needs to be checked for correctness.
VALUS	947	Invalid value identified for '%s' in ecdsm_vivo_table_template group. The only valid value for '%s' is '%s'.	Warning	The valid value for variable_1 is input_voltage and variable_2 is output_voltage in ecdsm_vivo_table_template group.

Valus User Guide

Valus Checks

LIB_NOISE				
Message Prefix	ID	Message	Severity	Description
VALUS	948	Missing attribute '%s' found in ecsm_vivo_current_waveform group. This is the required attribute in ecsm_vivo_current_waveform group, if missing can cause problems during noise analysis.	Warning	The following are the required attributes for ecsm_vivo_current_waveform group. 1.) index_1, index_2 and values 2.) ecsm_loadcap 3.) ecsm_millercap 4.) ecsm_arctype 5.) timing_sense 6.) related_pin. The required attributes when missing, can cause in-accuracy during noise analysis.
VALUS	949	Identified non-monotonic increase in '%s' values of ecsm_vivo_current_waveform group. The index_1 values should monotonically increase from 0 to VDD (%f) otherwise can lead to inaccuracy during noise analysis.	Warning	Voltage values in index_1/ index_2 of ecsm_vivo_current_waveform group need to be within 0 and VDD and should be monotonically increasing. Incorrect voltage values may result in inaccuracy during noise analysis. To fix the issue, recharacterize the library such that correct values in index_1 are used.
VALUS	950	The numbers of '%s' points in ecsm_vivo_current_waveform group are less than 5. The minimum number of points should be more than 5 for better accuracy in noise analysis.	Warning	The number of index_1 or index_2 points should be more than 5 in ecsm_vivo_current_waveform group. The more number of sampling points the better the accuracy.

Valus User Guide

Valus Checks

LIB_NOISE				
Message Prefix	ID	Message	Severity	Description
VALUS	951	The '%s' points are increasing in non-equal steps in ecdsm_vivo_current_waveform group.	Warning	The index_1 and index_2 values should increase in equal steps in ecdsm_vivo_current_waveform group.
VALUS	952	The '%d' value point in ecdsm_vivo_current_waveform group found more than 0.1 Amps. The values in ecdsm_vivo_current_waveform should be less than 0.1Amps for better accuracy in noise analysis.	Warning	The values in ecdsm_vivo_current_waveform should be less than 0.1Amps for better accuracy in noise analysis.
VALUS	953	The ecdsm_loadcap value found more than 3 * ecdsm_noise_cap in ecdsm_vivo_current_waveform group. This value should be in the order of ecdsm_noise_cap.	Warning	The ecdsm_loadcap values in ecdsm_vivo_current_waveform group should be in the order of pin.
VALUS	955	Missing ecdsm_vivo_current_waveform group with ecdsm_arctype '%s' on pin '%s'. This loss of information can cause inaccuracy during noise analysis.	Warning	The ecdsm_vivo_current_waveform group should be present for all the following four ecdsm_arctype defined: 'min_rise', 'min_fall', 'max_rise', 'max_fall'.
VALUS	956	Invalid timing_sense in ecdsm_vivo_current_waveform group found. The timing_sense should always be negative_unate in ecdsm_vivo_current_waveform group for correct noise analysis in ETS tool.	Warning	The timing_sense in ecdsm_vivo_current_waveform should always be negative_unate in ETS tool.

Valus User Guide

Valus Checks

LIB_NOISE				
Message Prefix	ID	Message	Severity	Description
VALUS	957	The first row value (s) in ecdsm_vivo_table_template are not positive. This can lead to inaccuracy during noise analysis.	Warning	The first row in ecdsm_vivo_table_template group except the last value in the first row should all be positive.
VALUS	958	The last row value (s) in ecdsm_vivo_table_template are not negative. The last row except the first value should all be negative in ecdsm_vivo_table_template group.	Warning	The last row except the first value should all be negative in ecdsm_vivo_table_template group.
VALUS	959	The '%d %s', '%d' value has increasing current value in ecdsm_vivo_table_template group. For accurate noise analysis, the '%s' should have decreasing values.	Warning	The rows and columns should have decreasing currents values in ecdsm_vivo_table_template group for accurate noise analysis.
VALUS	960	The ecdsm_vdd_level values do not match with '%s' of '%s' pin '%s' for cell '%s'. The ecdsm_vdd_level should match with '%s' for accurate noise analysis.	Warning	If input_signal_level/ output_signal_level/ related_power_pin attributes are present on input/output pins then the ecdsm_vdd_level must match with input_signal_level/ output_signal_level/ related_power_pin.
VALUS	962	The ecdsm_noise_cap value exceeds the 1e-9f. This value should be below 1e-9f.	Warning	The ecdsm_noise_cap should be below 1e-9f.

Valus User Guide

Valus Checks

LIB_NOISE				
Message Prefix	ID	Message	Severity	Description
VALUS	963	The attribute '%s' specified on pin '%s' of cell '%s' has value %fohm which is larger than the derived resistance from characterized timing data.	Warning	This message is issued when the attributes ecdsm_vddres/ vdd_gndres has value larger than the derived resistance from characterized timing data. To fix the issue, re-characterize the library with appropriate settings, such that the ecdsm_vddres/ ecdsm_gndres is correctly defined.
VALUS	964	Missing ecdsm_vivo_current_waveform group on output/inout pin '%s' for cell '%s'. The ecdsm_vivo_current_waveform group is required on output pin/inout pin for accurate noise analysis.	Warning	The ecdsm_vivo_current_waveform group should be present on output/inout pins.
VALUS	966	Missing '%s' template definition at library level. The template '%s' been referred by '%s' group is not defined at library level. This can cause inaccurate noise analysis.	Warning	All the template referred by ecdsm_internal_transition/ ecdsm_vivo_current_waveform group should be defined at library level.
VALUS	967	Identified '%s' attribute on '%s' pin '%s' of cell '%s'. This attribute can be present on '%s' or bidirectional pin.	Warning	The ecdsm_noise_cap, ecdsm_vddres and ecdsm_gndres attributes can be defined only on outpin pin or bidirectional pin. The ecdsm_vhtolerance and ecdsm_vltolerance can be defined only on input pins or bidir pins.

Valus User Guide

Valus Checks

LIB_NOISE				
Message Prefix	ID	Message	Severity	Description
VALUS	1039	Identified invalid '%s' values of ecdm_vivo_current_waveform group. The '%s' values should be from 0 to VDD (%f) otherwise can lead to inaccuracy during noise analysis.	Warning	Voltage values in index_1/ index_2 of ecdm_vivo_current_waveform group need to be within 0 and VDD and should be monotonically increasing. Incorrect voltage values may result in inaccuracy during noise analysis. To fix the issue, recharacterize the library such that correct values in index_1 are used.
TECHLIB	414	Missing %s %s noise propagation specification in pin %s of cell %s in the library. Height and width tables for a particular region (low, high, below_low, above_high) must be specified in pairs.	Error	Height and Width specifications for a particular noise region should always come in a pair. Need to specify both the parameters for propagation.
TECHLIB	708	No pg_pin with name '%s' has been read in the cell '%s'. The attribute '%s' for the dc_current group '%s' is referencing this pg_pin. Ignoring this attribute value.	Error	This message is issued when the pg_pin specified with related_pg_pin/ related_internal_pg_pin does not exist at cell level for dc_current group. Such attribute will be ignored. To fix the issue, re-characterize the library with appropriate settings, such that the pg_pin is defined.

Valus User Guide

Valus Checks

LIB_NOISE				
Message Prefix	ID	Message	Severity	Description
TECHLIB	908	The ccsn_last_stage group is present on the timing arc, but the ccsn_first_stage is not present on %s and cell '%s'. The ccsn_last_stage group can be present on a timing arc only when ccsn_first_stage group is present. The ccsn_last_stage group will be ignored.	Error	The timing group can contain a ccsn_last_group only when ccsn_first_stage group is present. Otherwise it can cause problems during noise analysis.
TECHLIB	914	The vector template '%s' for the vector group number '%d' specified in '%s' group on pin '%s' of cell '%s' is not defined. The CCS Noise data for this %s group is ignored	Error	This message is issued when the vector template of vector group output_voltage_fall and output_voltage_rise is not defined. Therefore CCS noise data for this group will be ignored. To fix the issue, re-characterize the library with appropriate settings, such that the vector template used for output_voltage_fall and output_voltage_rise groups is defined in library.

Valus User Guide

Valus Checks

LIB_NOISE				
Message Prefix	ID	Message	Severity	Description
TECHLIB	915	Invalid axis '%s' specified for the vector group number '%d' specified in '%s' group on pin '%s' of cell '%s'. The CCS Noise data for this %s group is ignored	Error	This message is issued when the axis type specified for the vector group output_voltage_rise/output_voltage_fall is invalid. The valid axes are input_net_transition and total_output_net_capacitance. Such ccs noise data will be ignored for this group. To fix the issue, re-characterize the library with appropriate settings, such that the valid axis type is defined.
TECHLIB	916	The number of time index points does not match with the voltage values of vector group number '%d' specified in '%s' group on pin '%s' of cell '%s'. The CCS Noise data for this %s group is ignored	Error	This message is issued when the voltage value of vector groups propagated_noise_high and propagated_noise_low of a cell do not match with the number of time index points. Therefore CCS noise data for this group will be ignored. To fix the issue, re-characterize the library with appropriate settings, such that the number of time index points should match with the voltage values of vector group propagated_noise_high and propagated_noise_low.

Valus User Guide

Valus Checks

LIB_NOISE				
Message Prefix	ID	Message	Severity	Description
TECHLIB	919	The pin level %s group(s) found on %s %s of cell '%s'. The pin level %s group(s) are expected only on %s or inout pins, therefore the %s group(s) will be ignored.	Error	This message is issued when the ccsn_first_stage group is defined at pin level on output pin or the ccsn_last_stage is defined on input pin of a cell. The ccsn_first_stage group can be defined only on an input pin or inout pin and the ccsn_last_stage can be defined only on an output pin or inout pin. If ccsn group is not defined at proper pin then they will be ignored. To fix the issue, re-characterize the library with appropriate settings.
TECHLIB	921	Invalid number of index_%d points specified for the vector group number '%d', in '%s' group on pin '%s' of cell '%s'. Only single value is expected for this index. The CCS Noise data for this %s group is ignored.	Error	This message is issued when the vector group output_voltage_fall and output_voltage_rise of a cell are not specified with correct number of index points. The axes input_net_transition/total_output_net_capacitance i.e. index_1 and index_2 can have only single value for the vector group output_voltage_fall and output_voltage_rise. To fix the issue, re-characterize the library with appropriate settings, such that the number of index points should be specified correctly for output_voltage_fall and output_voltage_rise groups.

Valus User Guide

Valus Checks

LIB_NOISE				
Message Prefix	ID	Message	Severity	Description
TECHLIB	922	The stage_type of ccsn_*_stage group does not match with the pin's driver type '%s' of cell '%s'. The stage_type of CCSN group must match with the pin's driver type. This ccsn_*_stage group will therefore be ignored.	Error	This message is issued when the ccsn_first_stage and ccsn_last_stage groups are not defined with correct driver type. The valid values are pull_up and pull_down. Therefore ccsn_stage group will be ignored. To fix the issue, re-characterize the library with appropriate settings, such that the correct driver type should be defined.
TECHLIB	1150	The template '%s' for dc_current group on pin '%s' of cell '%s' is not defined. The 'dc_current' group will be ignored.	Error	The dc_current group should have a template defined at library level otherwise group will be ignored. To rectify the problem define the template at library level.
TECHLIB	1333	The %s value defined in index_%d (%s) of group '%s' on %s of cell '%s' is '%f' which is %s '%f'.	Error	This message is issued when the first/last value defined in input_voltage or output_voltage indices on pin of cell is above VSS/below VDD of dc_current group. The first value of input_voltage/ output_voltage indices must be less than or equal to VSS and last value of input_voltage/ output_voltage indices must be greater than or equal to VDD. To fix the issue, re-characterize the library with appropriate settings, such that the input_voltage/ output_voltage indices of dc_current group are correctly defined.

Valus User Guide

Valus Checks

LIB_NOISE				
Message Prefix	ID	Message	Severity	Description
TECHLIB	1334	Identified one or more of the required attributes/groups missing from '%s' group defined %s of cell '%s'. When the stage_type is '%s', the required attributes/groups are %s. This may impact analysis accuracy.	Warning	This message is issued when the required attributes/groups are missing from ccsn_*_stage. The required attributes/groups are when stage_type is both : miller_cap_rise, miller_cap_fall, dc_current, output_voltage_rise, output_voltage_fall, when stage_type is pull_up :miller_cap_rise, dc_current, output_voltage_rise, when stage_type is pull_down : miller_cap_fall, dc_current, output_voltage_fall. This may impact delay and glitch analysis accuracy. To fix the issue, re-characterize the library with appropriate settings, such that the CCSN groups are correctly defined.

Valus User Guide

Valus Checks

LIB_NOISE				
Message Prefix	ID	Message	Severity	Description
TECHLIB	1335	The timing sense and sense of the group '%s' specified on timing arc on %s of cell '%s' is '%s' are inconsistent.	Error	This message is issued when the timing sense is inconsistent with the sense of the ccsn_*_stage group on the arc. The arc sense must match the combination i.e. chaining of the ccsn_*_stage senses; for example, an inverting arc must contain only a single inverting ccsn_first_stage group or an inverting ccsn_first_stage group, and a non-inverting ccsn_last_stage group, or a non-inverting ccsn_first_stage group and an inverting ccsn_last_stage group. To fix the issue, re-characterize the library with appropriate settings, such that the timing sense of the arc should be consistent with sense of the ccsn_*_stage group.
TECHLIB	1412	The '%s' group '%s' referenced in '%s' attribute of '%s' group of %s is not defined in %s of cell '%s'.	Error	This message is issued when input_ccb/output_ccb groups mentioned in attribute, are not defined on pin. To fix the issue, re-characterize the library with appropriate settings, such that the input_ccb/output_ccb groups should be defined.

Valus User Guide

Valus Checks

LIB_NOISE				
Message Prefix	ID	Message	Severity	Description
TECHLIB	1413	The attribute '%s' of '%s' group of %s of cell '%s' does not support timing arcs with three or more inverting stages.	Error	This message is issued when attribute propagating_ccb mentions timing_arcs with three or more inverting stages. To fix the issue, re-characterize the library with appropriate settings.
TECHLIB	1414	In cell '%s', the '%s' vector groups specified on %s are defined for more than one 'total_output_net_capacitance' index value. This group is defined in '%s' group '%s' that does not directly drive an output pin, referenced in '%s' attribute of 'timing' group defined in %s	Error	This message is issued when output_voltage_rise/fall vector groups are defined for more than one load index value. To fix the issue, re-characterize the library with appropriate settings, such that vector groups should be specified for only one load index value as CCB group does not directly drives an output pin.
TECHLIB	1415	In cell '%s', the '%s' vector groups specified on %s are defined for only one 'total_output_net_capacitance' index value. This group is defined in '%s' group '%s' that directly drives an output pin, referenced in '%s' attribute of 'timing' group defined in %s.	Warning	This message is issued when output_voltage_rise/fall vector groups are defined for only one load index value. To fix the issue, re-characterize the library with appropriate settings, such that vector groups should be specified for more than one load index value as CCB group directly drives an output pin.

Valus User Guide

Valus Checks

LIB_NOISE				
Message Prefix	ID	Message	Severity	Description
TECHLIB	405	Template %s should be of size 2D, specified as a function of noise width (input_noise_width) and output load (total_output_net_capacitance). Ignoring this template.	Warning	The Noise Immunity templates must have exactly two axes. The input_noise_width axis specifies the noise width and the output load is specified by the total_output_net_capacitance axis
TECHLIB	410	Invalid combination of axis specified for 'propagation_lut_template' %s. It must be a function of noise width (input_noise_width), noise height (input_noise_height) and output load (total_output_net_capacitance). Ignoring this template.	Warning	The propagation_lut_template must be a 3D table. The axes which need to be specified are - noise width (input_noise_width), noise height (input_noise_height) and output load (total_output_net_capacitance)
TECHLIB	413	Negative data found in table for %s noise immunity in pin %s of cell %s. The data values must be positive.	Warning	Noise immunity data can not be specified as negative.
TECHLIB	707	The attribute '%s' is missing in the dc_current group '%s' in cell '%s'. This may result in incorrect behavior later.	Warning	This message is issued when the attributes related_switch_pin, related_pg_pin and related_internal_pg_pin are not defined for the dc_current group. To fix the issue, re-characterize the library with appropriate settings, such that the attributes should be specified for the dc_current group.

Valus User Guide

Valus Checks

LIB_NOISE				
Message Prefix	ID	Message	Severity	Description
TECHLIB	709	No lu_table_template with name '%s' has been read in the library '%s'. The dc_current group is ignored for cell '%s'.	Warning	This message is issued when lu_table_template is specified in cell but not defined at library level. Such dc_current group will be ignored. To fix the issue, re-characterize the library with appropriate settings, such that the lu_table_template is defined.

LIB_POWER

The LIB_POWER checks are issued during the validation of power attributes/groups of a library - both ECSM and CCS power data.

LIB_POWER				
Message Prefix	ID	Message	Severity	Description
VALUS	21	The internal power arcs on the output/inout pin '%s' of cell '%s' does not have the related pin construct. This can cause problem during power analysis.	Error	For correct calculation of ecsm power tables related pin information should be provided in the internal power tables.
VALUS	22	The '%s' of internal_power group on output/inout pin '%s' of cell '%s' is single dimensional has more than one value.	Error	This message is issued when rise_power/fall_power attribute of internal_power group is one dimensional but has more than one value. It can be two dimensional or one dimensional having one value only. To fix this issue, recharacterize the library such that power arcs are specified correctly.
VALUS	42	The power arc(s) with related pin '%s' is closed for the when condition but not for the input pin(s) '%s' in pin '%s' for cell '%s'.	Warning	The union of when of all power arc(s) that share the same related pin is closed, but not all inputs pins were part of when condition in power arcs. The missing inputs pins are reported in the warning message.

Valus User Guide

Valus Checks

LIB_POWER				
Message Prefix	ID	Message	Severity	Description
VALUS	44	The leakage power arc(s) %s is closed for the when condition but not for the input pin(s) '%s' for cell '%s'.	Warning	The union of when of all power arc(s) that share the same power level/related_pg_pin if present is closed, but not all inputs pins were part of when condition in power arcs. The missing inputs pins are reported in the warning message.
VALUS	1013	Identified less than 3 sample points in ECSM Current waveforms of pin '%s' in cell '%s'. The characterized library needs to be looked further for correctness. This can cause problem during Power Analysis.	Warning	The ECSM Current waveforms should have a minimum of 3 sample points. This is required for correct Power Analysis.
VALUS	1016	Missing ECSM power construct in cell '%s'. This can cause loss of accuracy during power analysis.	Warning	It is observed when ECSM power group is present in cell, it can provide accurate power analysis.
TECHLIB	424	Invalid ECSM_Current/PGDC waveform group name %s specified in pin %s of cell %s. Ignoring waveform data.	Error	Waveform group name must be a digit and the value of the digit must not be greater than or equal to the total number of entries in the output transition table. All the waveforms must start from 0.
TECHLIB	425	Invalid rail connection name %s specified in ECSM_Power/PGDC group for pin %s of cell %s. Ignoring waveform data.	Error	Waveform rail connection name is optional. If provided it must be valid one

Valus User Guide

Valus Checks

LIB_POWER				
Message Prefix	ID	Message	Severity	Description
TECHLIB	427	Empty ECSM_Current/ PGDC waveform group found in cell %s, of library %s	Error	If ECSM_Current/PGDC groups are defined, they must contain waveform groups also. They can not be empty
TECHLIB	1224	The internal_power group is defined multiple times for cell '%s', %s '%s' on lines %d and %d. First definition will be used for analysis	Error	This message is issued for multiple definitions of internal_power groups for a pin with the same related_pg_pin and same 'when' condition attributes. First definition is used for analysis.
TECHLIB	393	Pin '%s' of cell '%s' has power arcs with incomplete when conditions	Warning	This message occurs if pin power arcs has incomplete when conditions Incomplete means that the when conditions for a specific power arcs do not add to 1.
TECHLIB	401	There are multiple power pin defined with the same related_pin and the same when condition on pin %s of cell %s. Only the last data is used.	Warning	Multiple power pin with the same related_pin, same equal_or_opposite_pins, same when conditions, and same transition edges will be merged. i.e. the last data will override the earlier one.
TECHLIB	426	Incompatible ecsm_power_version/ pgdc_version %f in library %s	Warning	The ecsm_power_version/ pgdc_version defined in a library must be equal to 1.0
TECHLIB	460	The internal power arcs on the output/inout pin '%s' of cell '%s' does not have the related pin construct. This can cause problem for some application to generate the ecsm_power tables.	Warning	For correct calculation of ecsm power tables related pin information should be provided in the internal power tables.

Valus User Guide

Valus Checks

LIB_POWER				
Message Prefix	ID	Message	Severity	Description
TECHLIB	461	The %s %s on output/inout pin '%s' of cell '%s' is single dimensional. The power arc should be atleast two dimensional for correct generation of ecm power tables i.e load and slew index point should be defined.	Warning	For correct calculation of ecm power the power arc should have atleast two dimensional template defined.
TECHLIB	1076	The vector template '%s' for the vector group number '%d' of cell '%s' is not defined in pg_current group. The CCS power data in this group is ignored	Warning	This message is issued when the vector template for the vector group of cell is not defined in pg_current group. The CCS power data in this group will be ignored. To fix the issue, re-characterize the library with appropriate settings, such that the template name should be defined for pg_current group.
TECHLIB	1077	Invalid specification for input_net_transition/total_output_net_capacitance axis for the vector group '%s' specified in pg_current group. Only a single slew/cap value is expected. The CCS Power data for this pg_current group is ignored.	Warning	This message is issued when the input_net_transition/total_output_net_capacitance indices for the vector group specified in pg_current group have more than one value. These attributes can have only single value. The CCS Power data for such pg_current group will be ignored. To fix the issue, re-characterize the library with appropriate settings, such that input_net_transition/total_output_net_capacitance in pg_current group have only single value.

Valus User Guide

Valus Checks

LIB_POWER				
Message Prefix	ID	Message	Severity	Description
TECHLIB	1078	The number of current values is not equal to the number of time index points for the vector group number '%d' specified in pg_current. The CCS Power data for this pg_current group is ignored	Warning	This message is issued when the number of current values in CCS power data for pg_current group is not matched with the number of time index points. CCS power data for such pg_current group will be ignored. To fix the issue, re-characterize the library with appropriate settings, such that the number of current values is equal to the number of time index points.
TECHLIB	1154	Identified more than one leakage_power groups with same condition in cell '%s', Last definition will be retained.	Warning	Identified more than one leakage_power groups with same condition in cell <cellName>. The last definition will be retained.
TECHLIB	1446	The related_%s attribute '%s' specified for dynamic_current group of cell '%s' refers to a pg_pin. The value of this attribute should only refer to signal pins(s).	Error	This message is issued when pg_pin is specified in attribute related_inputsor related_outputsdefined in dynamic_current group of a cell. These attributes should only refer to signal pins. To fix this issue, re-characterize the library such that these attributes do not consist pg_pin.

LIB_MSV

The LIB_MSV checks are performed during the validation of multiple power or ground supply voltage attributes or groups of a library. The MSV cells include level-shifters, isolation cells, always_on cells, retention flip-flops, power switch cells, and so on.

LIB_MSV				
Message Prefix	ID	Message	Severity	Description
VALUS	1038	The cell '%s' is %s, but the required attribute '%s' is missing at '%s'.	Error	A cell is a level shifter cell if it has the 'is_level_shifter' attribute present at cell level. For a level shifter cell the following attributes are required: 'level_shifter_type' at cell level, 'input_voltage_range' at either the input pin or at the cell level, 'output_voltage_range' at either the output pin or at the cell level. A cell is a switch cell if it has the attribute 'switch_cell_type' present at cell level. For a switch cell the following attributes are required: 'switch_function' at pin level (pg pin), 'switch_pin' at pin level (input pin), 'power_down_function' at pin level (output pin). A cell is an isolation cell if it has the attribute 'is_isolation_cell' present at cell level. For an isolation cell the attribute 'isolation_cell_enable_pin' should be present at pin level.

Valus User Guide

Valus Checks

LIB_MSV				
Message Prefix	ID	Message	Severity	Description
				A cell is a retention cell if the attribute 'retention_cell' is present at cell level. For a retention cell the attribute 'retention_pin' should be present at pin level.
VALUS	1020	The cell '%s' is %s, but the required attribute '%s' is present '%s'.	Warning	A cell is a level shifter cell if it has the 'is_level_shifter' attribute present at cell level. For a level shifter cell the following attributes are required: 'level_shifter_type' at cell level, 'level_shifter_enable_pin' at pin level, 'input_voltage_range' at either the input pin or at the cell level, 'output_voltage_range' at either the output pin or at the cell level, 'std_cell_main_rail' at pin level (on the pg pin).

Valus User Guide

Valus Checks

LIB_MSV				
Message Prefix	ID	Message	Severity	Description
VALUS	1019	The cell '%s' is %s, but the required attribute '%s' is missing at '%s'.	Info	A cell is a level shifter cell if it has the 'is_level_shifter' attribute present at cell level. For a level shifter cell the following attributes are required: 'level_shifter_type' at cell level 'input_voltage_range' at either the input pin or at the cell level 'output_voltage_range' at either the output pin or at the cell level. A cell is a switch cell if it has the attribute 'switch_cell_type' present at cell level. For a switch cell the following attributes are required: 'switch_function' at pin level (pg pin) 'switch_pin' at pin level (input pin) 'power_down_function' at pin level (output pin). A cell is an isolation cell if it has the attribute 'is_isolation_cell' present at cell level. For an isolation cell the attribute 'isolation_cell_enable_pin' should be present at pin level. A cell is a retention cell if the attribute 'retention_cell' is present at cell level. For a retention cell the attribute 'retention_pin' should be present at pin level.

Valus User Guide

Valus Checks

LIB_MSV				
Message Prefix	ID	Message	Severity	Description
VALUS	1025	The cell '%s' is %s, but the attribute '%s' is missing at '%s'.	Info	A cell is a level shifter cell if it has the 'is_level_shifter' attribute present at cell level. For a level shifter cell the following attributes should be present: 'level_shifter_enable_pin' at pin level, 'std_cell_main_rail' at pin level (on the pg pin).
TECHLIB	433	Cell %s is not an isolation cell, pin %s on line %d is specified isolation %s pin. Ignoring 'isolation_cell_%s_pin' attribute of this pin.	Error	'isolation_cell_enable_pin' attribute specifies the enable pin of an isolation cell and 'isolation_cell_data_pin' attribute specifies the data pin of an isolation cell. The parent cell of this pin is not an isolation cell. To mark the cell as an isolation cell, mention 'is_isolation_cell : true' at the cell level.
TECHLIB	704	The voltage_map attribute with name '%s' referenced by the pg_pin '%s' in the cell '%s', is absent in the library. The required voltage_map group needs to be added. This pg_pin is ignored. Related low power constructs referencing this pg_pin may not be read correctly.	Error	This message is issued when the attribute voltage_map with voltage_name referenced by pg_pin in the cell is not defined in the library. Such pg_pin will be ignored and any low power constructs referencing this pg_pin may not work as expected. To fix the issue, re-characterize the library with appropriate settings, such that the attribute volatge_map referenced by pg_pin is defined at library level.

Valus User Guide

Valus Checks

LIB_MSV				
Message Prefix	ID	Message	Severity	Description
TECHLIB	705	The voltage_name attribute is missing in the pg_pin '%s' in the cell '%s'. This is a mandatory attribute for the pg_pin group. This pg_pin is ignored. Related low power constructs referencing this pg_pin may not be read correctly.	Error	This message is issued when the attribute voltage_name is not defined in the pg_pin group in the cell. The attribute voltage_name is mandatory attribute. Such pg_pin will be ignored. To fix the issue, re-characterize the library with appropriate settings, such that the volatge_map in pg_pin is specified.
TECHLIB	1118	Cell %s is an Isolation cell, pin %s is specified level shifter %s pin. Ignoring '%s' attribute of this pin.	Error	'level_shifter_enable_pin' and level_shifter_data_pin attribute specifies the enable pin and data pin of a level shifter cell respectively. The parent cell of this pin is not a level shifter cell. To mark the cell as a level shifter cell mention 'is_level_shifter : true' at the cell level.
TECHLIB	1144	The pg pin '%s' has also been defined as pin in cell '%s'. The pin will be retained, but the pg_pin will be ignored. Correct the pin names in library cell definition to fix the error.	Error	This error is displayed when both pg_pin and pin of a cell defined in the library, have the same name. The cell definition in the library must have unique pin names for pin and pg_pin for a cell. Due to this error, the software will retain the pin and ignore the pg_pin in the cell definition. The pin name with pg_pin attribute will be ignored. This may lead to issues during design load and analysis, especially if the design includes pg information.

Valus User Guide

Valus Checks

LIB_MSV				
Message Prefix	ID	Message	Severity	Description
TECHLIB	1191	The attribute 'power_down_function' has been defined for %s '%s' of test_cell on line %d.	Error	This message is issued when attribute power_down_function is defined on any pin of test_cell. The test_cell pins cannot have power_down_function attribute. To fix the issue, re-characterize the library with appropriate settings, such that the correct attributes are defined on test_cell pins.
TECHLIB	1227	For cell %s, %s pg pin %s on line %d has related bias pin %s which is not of type %s.	Error	This message is issued when the attribute related_bias_pin has invalid value. For power pins, it can only be nwell/deepnwell and for ground pins, it can only be pwell/deeppwell. To fix the issue, re-characterize the library with appropriate settings, such that the correct related_bias_pin values are set.
TECHLIB	1235	The pg_pin with the name %s specified for %s attribute for %s on line '%d' is not defined for cell '%s'.	Error	This message is issued when the pin name specified for related_pg_pin attribute is not defined at the cell level. To fix the issue, re-characterize the library with appropriate settings, such that the correct pins are specified.

Valus User Guide

Valus Checks

LIB_MSV				
Message Prefix	ID	Message	Severity	Description
TECHLIB	1315	For cell '%s', pg_pin groups %s share the same voltage_name '%s'. These PG pins cannot have same voltage_name. These PG pins definitions will be ignored.	Error	This message is issued when the same voltage_name is specified on the pins among opposite(power/ground) category of pg_type. The power category includes pins with pg_type as primary_power/backup_power/internal_power/nwell/deepnwell and ground category includes pins with pg_type as primary_ground/backup_ground/internal_ground/pwell/deeppwell for the cell for pg_pin group. Such pins will be ignored. To fix the issue, re-characterize the library with appropriate settings, such that the pg_pin(s) of opposite type are not defined with same voltage.

Valus User Guide

Valus Checks

LIB_MSV				
Message Prefix	ID	Message	Severity	Description
TECHLIB	1316	For MSV cell '%s', pg_pin groups %s share the same voltage_name '%s'. These PG pins cannot have same voltage_name. This library will be ignored.	Error	This message is issued when the same voltage_name is specified on the pins among opposite(power/ground) category of pg_type. The power category includes pins with pg_type as primary_power/backup_power/internal_power/nwell/deepnwell and ground category includes pins with pg_type as primary_ground/backup_ground/internal_ground/pwell/deeppwell for MSV cell for pg_pin group. Such library will be ignored. To fix the issue, re-characterize the library with appropriate settings, such that the pg_pin(s) are not defined with same voltage.
TECHLIB	1392	The voltage_name '%s' specified in pg_pin '%s' of pg_type '%s' of cell '%s' is mapped to voltage %gV. The nom_voltage %gV will be used for this pg_pin.	Error	This message is issued when pg_pin of pg_type primary_power/backup_power is mapped with the voltage having 0V or less than 0V. In such case, nominal voltage will be used for power pg_pin. This may impact accuracy in delay and glitch analysis. To fix the issue, re-characterize the libraries with appropriate settings.

Valus User Guide

Valus Checks

LIB_MSV				
Message Prefix	ID	Message	Severity	Description
TECHLIB	434	Cell %s is not a level shifter cell, pin %s is specified level shifter %s pin. No data is ignored.	Warning	'level_shifter_enable_pin' and 'level_shifter_data_pin' attribute specifies the enable pin and data pin of a level shifter cell respectively. The parent cell of this pin is not a level shifter cell. To mark the cell as a level shifter cell mention 'is_level_shifter : true' at the cell level.
TECHLIB	700	Cell %s is not a power switch cell, pin %s is specified switch pin. Ignoring 'switch_pin' attribute of this pin.	Warning	'switch_pin' library pin attribute specifies the enable pin of a power switch cell. This message is issued when the parent cell of this pin is not specified as power switch cell. Tool will ignore 'switch_pin' attribute of this pin. To mark the cell as a power switch, specify 'switch_cell_type : ' attribute at the cell level.

Valus User Guide

Valus Checks

LIB_MSV				
Message Prefix	ID	Message	Severity	Description
TECHLIB	715	Power_rail '%s' specified in rail_connection '%s' of cell '%s' is not present in power_supply group. This rail_connection will be ignored.	Warning	This message is issued when the attribute power_rail is specified in rail_connection of the cell but not specified in the power_supply group. For example, Power_rail 'VDD3' is specified in rail_connection 'PV3' of cell but is not present in power_supply group. Such rail_connection will be ignored. To fix the issue, re-characterize the library with appropriate settings, such that the power_rail should be specified in both i.e. rail_connection as well as power_supply_group.
TECHLIB	1115	Multiple definitions of the power_rail '%s' for the power_supply group at line %d. The last definition would be retained	Warning	This message is issued when power_rail for power_supply group is defined more than once. The same attribute cannot have multiple definition. Last definition will be retained and rest will be ignored.
TECHLIB	1116	Multiple definitions of the voltage '%s' for the voltage_map at line %d. The first definition would be retained	Warning	This message is issued when a supply voltage is defined more than once with different values in voltage_map. The same voltage cannot have multiple definition. First definition will be retained and rest will be ignored.

Valus User Guide

Valus Checks

LIB_MSV				
Message Prefix	ID	Message	Severity	Description
TECHLIB	1217	For the cell '%s', the always-on %s has related_power_pin '%s' which is not of pg_type 'backup_power'.	Warning	This message is issued for an always-on cell, when the cell pin has related_power_pin which is not the pg_type 'backup_power'. To fix the issue, re-characterize the library with appropriate settings, such that the correct pg_type pin is specified.
TECHLIB	1218	For the cell '%s', the always-on %s has related_ground_pin '%s' which is not of pg_type 'backup_ground'.	Warning	This message is issued for an always-on cell, when the cell pin has related_ground_pin which is not the 'backup_ground' pg_type. To fix the issue, re-characterize the library with appropriate settings, such that the correct pg_type pin is specified.
TECHLIB	9133	Identified more than 1 power_gating_pin/ retention_pin attributes for pin %s (Line %d). First definition would be retained.	Warning	Found multiple power_gating_pin/ retention_pin constructs for same pin. Specify only one power_gating_pin/ retention_pin construct for a pin in the library cell.
TECHLIB	9135	%s (Line %d) has both attributes 'power_level' & 'related_pg_pin' defined. power_level would be ignored.	Warning	The power group should be reviewed for correctness.

LIB_LVF

The LIB_LVF checks are performed during the validation of LVF constructs of a library. These include ocv_sigma and derate table.

LIB_LVF				
Message Prefix	ID	Message	Severity	Description
TECHLIB	1393	The %s table is not supported for %s '%s'. This table will be ignored.	Error	This message is issued when a particular attribute is not supported in SOCV side file. For example, RC variation multiplier table cannot have object_type attribute as 'library'. To fix the issue, re-characterize the libraries with appropriate settings.
TECHLIB	1271	No ocv_derate_factors group specified for ocv_derate group %s of cell %s.	Warning	At least one ocv_derate_factors group must be specified within an ocv_derate group.
TECHLIB	1273	'%s' should not be defined in lu_table_template '%s' used in group %s on line %d, in pin '%s' of cell '%s'.)	Warning	The ocv_sigma_cell_rise, ocv_sigma_cell_fall, ocv_sigma_rise_transition, ocv_sigma_fall_transition, ocv_sigma_rise_constraint and ocv_sigma_fall_constraint groups can be 1D, 2D, 3D or scalar only.

Valus User Guide

Valus Checks

LIB_LVF				
Message Prefix	ID	Message	Severity	Description
TECHLIB	1274	The valid value of %s is %s in the lu_table_template '%s' used in group '%s' on line %d, in pin %s of cell %s.	Warning	For the ocv_sigma_cell_rise, ocv_sigma_cell_fall, ocv_sigma_rise_transition, and ocv_sigma_fall_transition groups, the valid values of variable_1, variable_2, and variable_3 are input_net_transition, total_output_net_capacitance, and related_out_total_output_net_capacitance. For the ocv_sigma_rise_constraint and ocv_sigma_fall_constraint groups, the values of variable_1 and variable_2 are related_pin_transition and constrained_pin_transition respectively. The values of variable_3 can be related_out_total_output_net_capacitance, related_out_output_net_length, related_out_net_wire_cap, or related_out_output_net_pin_cap.
TECHLIB	1384	The absolute mean shift value for '%s' group on %s of cell '%s' has value %f which is greater than the value %f. This may impact the delay calculation accuracy.	Warning	This message is issued when the absolute mean shift value is beyond the range. The value should be less than the value calculated as {0.5*std_dev} for accurate delay calculation. To fix the issue, re-characterize the library with appropriate settings, such that the absolute mean shift is within the defined range.

Valus User Guide

Valus Checks

LIB_LVF				
Message Prefix	ID	Message	Severity	Description
TECHLIB	1385	The standard deviation value for '%s' group on %s of cell '%s' has value %f which is greater than the value %f. This may impact the delay calculation accuracy.	Warning	This message is issued when the standard deviation value is beyond the range. The value should be less than the value calculated as $\{0.3 \times \text{input slew}\}$ for accurate delay calculation. To fix the issue, re-characterize the library with appropriate settings, such that the standard deviation is within the defined range.
TECHLIB	1386	The cell '%s' has %u sigma tables out of %u(%g%) which have beyond the range(1/%g and %g) ratios of attribute sigma_type late to early value. This can result in inaccurate analysis. Cadence recommends libraries with LVF moments for such high sigma variation cells.	Warning	This message is issued when the ratio of late and early sigma values is not within the range of (1/x and x). The default value of x is 3. Cadence recommends libraries with LVF moments for such high sigma variation cells.
TECHLIB	1441	The group '%s' in timing arc for %s on cell '%s', is missing for sigma_type '%s'. The table for sigma_type '%s' will be used for both sigma_types 'early' and 'late'.	Error	This message is issued when an ocv_sigma group is not present in a pair on a timing arc. For example 'ocv_sigma_cell_rise' group is present for sigma_type early and missing for sigma_type late. In such cases the same tables would be used for both early and late sigma. To fix the issue, re-characterize the library with appropriate settings, to ensure completeness of ocv_sigma data.

LIB_EM

The LIB_EM checks are performed during the validation of electromigration/max_cap constructs of a library. This includes em_max_toggle_rate look up table.

LIB_EM				
Message Prefix	ID	Message	Severity	Description
TECHLIB	1260	Cell '%s' does not have electromigration/max_cap group on any pin	Error	This message is issued when the cell is not specified with any electromigration/max_cap table on any pin. To fix the issue, re-characterize the library with appropriate settings, such that the electromigration/max_cap table is defined on pin at cell level.
TECHLIB	1288	Incorrect value '%s' specified for variable_1 in 1-D %s '%s' in electromigration group defined on line %d defined in %s and direction '%s'.	Error	For one dimensional template, variable_1 can have only a specific value. For example, in 1-D electromigration template (em_lut_template) variable_1 can only be 'input_transition_time'.
TECHLIB	710	EM group for cell: '%s' and pin: '%s' has negative values in the EM toggle rate group. This group will be ignored.	Warning	This message is issued when the attribute values specified in the em_max_toggle_rate group is defined with negative value. Such EM group will be ignored. To fix the issue, re-characterize the library with appropriate settings, such that the defined value is positive.

Valus User Guide

Valus Checks

LIB_EM				
Message Prefix	ID	Message	Severity	Description
TECHLIB	711	EM group for cell: '%s' and pin: '%s' should have related_pin/related_bus_pin attribute defined for a two dimension EM toggle rate table. This group will be ignored.	Warning	This message is issued when the attributes related_pin and related_bus_pin are not defined for an electromigration group specified with two-dimensional tables. For two-dimensional tables, related_pin and related_bus_pin are mandatory attributes in electromigration group. To fix the issue, re-characterize the library with appropriate settings, such that the related_pin and related_bus_pin are defined.
TECHLIB	1289	em_max_toggle_rate look up table for current_type '%s' is missing in %s for cell '%s'.	Info	This message is issued when any of the three electromigration tables (em_max_toggle_rate look up table) i.e. average/rms/peak is missing, if any one of them is defined.

LIB_MERGE

The LIB_MERGE checks are performed during validation of the merged library format. This includes checking of attributes/groups that are mandatory in a merged library.

LIB_MERGE				
Message Prefix	ID	Message	Severity	Description
VALUS	1040	Necessary CCS noise group (ccsn_first_stage/ccsn_last_stage) is missing on the timing arc with reference pin '%s' and parent pin '%s' for cell '%s' of the merged library. This is a mandatory group for merged library and may lead to inaccuracy in timing analysis.	Error	This message is issued when CCS noise data is missing in the merged library. CCS noise data can be arc-based or pin-based, and is a mandatory requirement in library for accurate analysis.
TECHLIB	1257	Found %s '%s' in the library. This is not a mandatory %s for merged library and may result in increased library size.	Error	For a merged library only the following constructs are mandatory: ecsm_version and 4 ECSM-Timing Constructs - ecsm_waveform_set, ecsm_capacitance_set, ecsm_vtp, ecsm_vtn and 2 CCS-Noise Constructs - ccsn_first_stage, ccsn_last_stage. If any other construct is present, it may cause an increase in size, beyond the expected value for a merge library.

Valus User Guide

Valus Checks

LIB_MERGE				
Message Prefix	ID	Message	Severity	Description
TECHLIB	1258	Necessary %s '%s' is missing from the '%s' table on the timing arc of pin '%s' for cell '%s' of the merged library. This is a mandatory %s for merged library and may lead to inaccuracy in timing analysis.	Error	For a merged library only the following constructs are mandatory: ecdsm_version and 4 ECDSM-Timing Constructs - ecdsm_waveform_set, ecdsm_capacitance_set, ecdsm_vtp, ecdsm_vtn and 2 CCS-Noise Constructs - ccsn_first_stage, ccsn_last_stage If these constructs are not present, it may cause inaccuracy in timing analysis.
TECHLIB	1259	Necessary %s '%s' is %s for the merged library. This is a mandatory %s for merged library and may lead to inaccuracy in timing analysis.	Error	For a merged library only the following constructs are mandatory: ecdsm_version and 4 ECDSM-Timing Constructs - ecdsm_waveform_set, ecdsm_capacitance_set, ecdsm_vtp, ecdsm_vtn and 2 CCS-Noise Constructs - ccsn_first_stage, ccsn_last_stage If these constructs are not present, it may cause inaccuracy in timing analysis. Furthermore ecdsm_vtp and ecdsm_vtn are checked only if the ecdsm_version is greater than 1.2/2.1 as the support for these attributes was added in the aforementioned version.

LIB_SEMANTIC

The LIB_SEMANTIC checks are performed during the validation of library semantics. These checks include data integrity/validity, such as monotonicity, negative value, data range of attributes/groups of a library. This category also includes table or template definition checks.

LIB_SEMANTIC				
Message Prefix	ID	Message	Severity	Description
VALUS	1000	Identified big number in '%s' table for '%s' %s. Big number (s) are not allowed in the library.	Error	Big numbers are not allowed in the checked library. Big number in the library can be because of incorrect generation of library. The library needs to be further looked for correctness.
VALUS	1001	Identified negative values for '%s'. The '%s' value should be a positive number.	Error	The constructs like capacitance, max_capacitance, max_fanout, max_transition, min_period, min_pulse_width_high, min_pulse_width_low and area are positive numbers.
VALUS	1003	The template referred by '%s' table has negative index. The index '%s' has negative value at '%d'. The index cannot have negative values.	Error	The slew and load index of template should have positive values.
VALUS	1004	Incorrect template '%s' with 1*N or N*1 index points identified	Error	The template cannot have 1*N or N*1. Each index values should have atleast 2 values.
VALUS	1005	The template referred by '%s' table has non-monotonic value. The index '%s' has non-monotonic value at '%d'. Index values should monotonically increase.	Error	The index values of a template can monotonically increase.

Valus User Guide

Valus Checks

LIB_SEMANTIC				
Message Prefix	ID	Message	Severity	Description
VALUS	1007	Identified negative values in '%s' table of timing arc. The table values cannot have negative values.	Error	This message is issued when cell_rise/cell_fall tables in timing arc of an ETM library have negative values. To fix the issue, re-characterize the library with appropriate settings, such that the delay tables have positive values.
VALUS	1021	Identified incorrect %s value '%f' at %s for %s '%s'. The value should be positive and less than 100.	Error	The attribute aocv_weight should be a positive value and should be less than 100.
VALUS	1023	Identified incorrect '%s' value '%f' for %s %s of cell '%s'. The acceptable value should be greater than 0.	Error	Negative or 'Zero' values are not permitted for the attribute - 'total_output_net_cap'. Valid values for 'total_output_net_cap' should be greater than 'Zero'. 'Zero' values are not permitted for the attributes - 'minimum_pulse_width' and 'min_period'. Valid values for 'minimum_pulse_width' and 'min_period' should be non-zero.
VALUS	1030	The current value '%f' is greater than the user defined value '%f' in pin '%s' of cell '%s'. This value should be within the user defined value.	Error	The current/time value should be within the user defined value. Indices beyond this can cause issues during power analysis.
VALUS	1031	The time value '%f' is greater than the user defined value '%f' in pin '%s' of cell '%s'. This value should be within the user defined value.	Error	The current/time value should be within the user defined value. Indices beyond this can cause issues during power analysis.

Valus User Guide

Valus Checks

LIB_SEMANTIC				
Message Prefix	ID	Message	Severity	Description
VALUS	1018	'%s' value '%f' is out of range. This value should be in the range '%f' to '%f'.	Warning	In multiple supply libraries, the <code>ecsm_base_rail_*</code> should be in range of $(\text{related_power_pin}/\text{output_signal_level}) > \text{ecsm_base_rail_vdd_} > (\text{related_gnd_pin}/\text{gnd})$. In case of single supply libraries these should be in between $\text{VDD} > \text{ecsm_base_rail_vdd_} > \text{GND}$.
VALUS	1027	The %s %s '%f' is %s than the user defined %s range (%f to %f) in %s table in pin %s of cell %s. This value should be inside the user defined range.	Warning	The boundary slew and capacitance indices have to be within the user defined acceptable range. Indices beyond this can cause issues during timing analysis. To increase the range, use: '-slew_index_max' to specify the upper limit of slew range; '-slew_index_min' to specify the lower limit of slew range; '-capacitance_index_max' to specify the upper limit of cap range; '-capacitance_index_min' to specify the lower limit of cap range.
TECHLIB	104	Incorrect latch/ff/bank combination, cell '%s	Error	This message is issued when the ff/latch bank is specified with negative bit or ff/latch is defined with bit. To fix the issue, re-characterize the library with appropriate settings, such that the correct combination is used.

Valus User Guide

Valus Checks

LIB_SEMANTIC				
Message Prefix	ID	Message	Severity	Description
TECHLIB	106	Could not find any cells in library '%s'	Error	This message is issued when library does not have any cell defined in it.
TECHLIB	107	Could not find pin '%s' on cell '%s'	Error	This message is issued when the undefined related_pin is used in the power arcs/EM arcs/timing arcs of cell. To fix the issue, re-characterize the library with appropriate settings, such that the related_pin specified is defined in cell.
TECHLIB	122	Wireload model %s in selection table not found	Error	This message is issued when the wire_load group is not defined in library but referred in wire_load_table group. To fix the issue, re-characterize the library with appropriate settings, such that the wire_load group used in wire_load_table group is defined.
TECHLIB	123	Undefined template '%s' referred in cell '%s'	Error	This message is issued when a template is referred in cell but is not defined at library level. To fix the issue, re-characterize the library with appropriate settings, such that the template used is defined in library.

Valus User Guide

Valus Checks

LIB_SEMANTIC				
Message Prefix	ID	Message	Severity	Description
TECHLIB	169	Couldn't find pin '%s' specified in the function of cell '%s'	'Error	This message is issued when a pin is used in function statement but is not defined in the cell. Function statement will be ignored. To fix the issue, re-characterize the library with appropriate settings, such that the undefined pins are not used in function statement.
TECHLIB	180	Unknown model for PVT multiplier '%s'	'Error	Currently only constant and linear models are supported for PVT multiplier
TECHLIB	183	The index '%d' of bus '%s' is out of bus range	Error	This message is issued when the scaler bit is specified which is out of range as per the bus bit definition. To fix the issue, re-characterize the library with appropriate settings, such that the correct range is defined.
TECHLIB	190	Unrecognized value '%s' of library attribute '%s'	'Error	This message is issued when the value of the attributes default operating condition/ default wire load selection/ default wireload is undefined in library. To fix the issue, re-characterize the library with appropriate settings, such that no undefined value of these attributes are used in the library.

Valus User Guide

Valus Checks

LIB_SEMANTIC				
Message Prefix	ID	Message	Severity	Description
TECHLIB	191	The value '%f' on cell '%s', pin '%s', '%s' should be non-negative	Error	This message is issued when either the constraint attribute in minimum_period group or the constraint_high/constraint_low attributes in minimum_pulse_width group are defined with negative value. Negative values are not allowed in these attributes. To fix the issue, re-characterize the library with appropriate settings, such that constraints are defined with positive value.
TECHLIB	248	Thresholds cannot have negative values	Error	This message is issued when the slew threshold values are specified as negative. Slew thresholds cannot have negative values. To fix the issue, re-characterize the library with appropriate settings, such that positive slew threshold is defined.
TECHLIB	249	'%s': '%f' should be less than '%s': '%f'	Error	This message is issued when the value of attribute slew_upper_threshold/slew_measured_upper_threshold is less than slew_lower_threshold/slew_measured_lower_threshold. Value of upper threshold should be greater than the lower threshold value. To fix the issue, re-characterize the library with appropriate settings, such that slew threshold values are defined correctly.

Valus User Guide

Valus Checks

LIB_SEMANTIC				
Message Prefix	ID	Message	Severity	Description
TECHLIB	257	%s '%s' has identical lower and upper measured slew thresholds for %s transition	Error	This message is issued when the value of attribute <code>slew_measured_lower_thresh</code> old and <code>slew_measured_upper_thresh</code> old is defined as same for <code>rise_transition</code> or <code>fall_transition</code> in a library. Value of the upper threshold should be greater than the lower threshold value. To fix the issue, re-characterize the library with appropriate settings, such that slew threshold values are defined correctly.
TECHLIB	258	%s '%s' has identical lower and upper slew thresholds for %s transition	Error	This message is issued when the value of attribute <code>slew_lower_threshold</code> and <code>slew_upper_threshold</code> is defined as same for <code>rise_transition</code> or <code>fall_transition</code> in a library. Value of the upper threshold should be greater than the lower threshold value. To fix the issue, re-characterize the library with appropriate settings, such that slew threshold values are defined correctly.

Valus User Guide

Valus Checks

LIB_SEMANTIC				
Message Prefix	ID	Message	Severity	Description
TECHLIB	272	Syntax error in '%s' specification. Expecting a '%s	Error	This message is issued when the attribute input_threshold_fall/ input_threshold_rise is specified with invalid value, for example 'string'. These attributes can accept 'float' values only. To fix the issue, re-characterize the library with appropriate settings, such that valid value is defined for threshold.
TECHLIB	279	The number of input pins for cell '%s' differ between two libraries. First library has %d input pins, whereas second has %d. Aborting update.	Error	To load the library successfully, make sure the information for the cell in both libraries is consistent.
TECHLIB	280	Invalid slew threshold range %.2f-%.2f%. The slew threshold range must be greater than one percent.	Error	This message is issued when the difference between the attributes slew_lower_threshold and slew_upper_threshold is less than 1%. To fix the issue, re-characterize the library with appropriate settings, such that the slew threshold range greater than 1% is defined.
TECHLIB	404	The coefficient value is less than 0 for the hyperbolic %s coefficient for %s in pin %s of Cell %s. Therefore the entire group will be ignored.	Error	Hyperbolic coefficients for area, width and height can not be negative. Even if one of them is specified as negative, the entire hyperbolic noise group is ignored.

Valus User Guide

Valus Checks

LIB_SEMANTIC				
Message Prefix	ID	Message	Severity	Description
TECHLIB	408	The template definition for %s, referenced in pin %s of cell %s is missing in the library. Therefore, the table would be ignored while loading the library.	Error	Specify the missing template at the library level.
TECHLIB	415	Template definition for %s, referenced in pin %s of cell %s missing in the library. Table would be ignored while library loading.	Error	The template used must be defined at the library level.
TECHLIB	419	Number of index point in axis redefinition of template %s in pin %s in cell %s is not equal to that in the original template.	Error	The number of points in axis redefinition should be exactly equal to the number of axis points in the template.
TECHLIB	429	Waveform group count does not match with the template %s in pin %s of cell %s	Error	The total number of waveform groups must be equal to the total number of values described by the corresponding template
TECHLIB	431	Mismatch in the number values redefined for Index_%d of template %s in pin %s of cell %s	Error	Index redefinitions should contain exactly the same number of values as defined in the original template

Valus User Guide

Valus Checks

LIB_SEMANTIC				
Message Prefix	ID	Message	Severity	Description
TECHLIB	702	No pg_pin with name '%s' has been read in the cell '%s'. The attribute '%s' specified for the pin '%s' is being ignored.	Error	This message is issued when the pg_pin specified with related_power_pin/ related_ground_pin/ related_bias_pin does not exist at cell level. These attributes will be ignored. To fix the issue, re-characterize the library with appropriate settings, such that the undefined pg_pin is not used.
TECHLIB	1005	The normalized_driver_waveform '%s' specified as '%s' for cell '%s' was not found in the library	Error	This message is issued when the normalized_driver_waveform is specified as driver_waveform/ driver_waveform_rise/ driver_waveform_fall for a cell but is not defined in library. To fix the issue, re-characterize the library with appropriate settings, such that the library does not use undefined normalized_driver_waveform.
TECHLIB	1123	Identified multiple definitions of the related_pin for the '%s' group defined on pin '%s' of cell '%s'. The first definition would be retained and rest will be ignored.	Error	This message is issued when related_pin is specified more than once for max_trans and max_cap groups defined at pin level. The first definition would be retained and rest will be ignored. To fix the issue, re-characterize the library with appropriate settings, such that the related_pin is defined correctly.

Valus User Guide

Valus Checks

LIB_SEMANTIC				
Message Prefix	ID	Message	Severity	Description
TECHLIB	1143	The pin '%s' in the statetable/ input_map is not defined in the cell '%s'. The statetable will be ignored. Fix incorrect pin names or any special characters in the statetable to rectify the error.	Error	The pin names defined in the input_map or statetable should be the same as defined in the library cell definition. Also, per liberty statetable syntax, pin names are identified based on white space delimiter. Any special characters, if any in the statetable, separated by white spaces could be also be identified as pins. To rectify the Error, correct the incorrect pin names or remove special characters from the statetable, if any.
TECHLIB	1151	Invalid axis '%s' specified for '%s' group referred by template name '%s' on pin '%s' of cell '%s'. This group is ignored.	Error	The valid index name needs to be specified for group template. To rectify the problem change the variable_1 or variable_2 or variable_3 to these valid values in the template group.
TECHLIB	1170	Name of '%s' group is missing on line %d. To fix this issue, define the missing group name in the library.	Error	The message is issued when a group such as cell, pin etc., is defined in the library without any name. For example, the group pin is defined as: pin(). To rectify the issue, correct the library to define the missing name of the group in the library.

Valus User Guide

Valus Checks

LIB_SEMANTIC				
Message Prefix	ID	Message	Severity	Description
TECHLIB	1171	The attribute '%s' of group '%s' on line %d has one or more values which are %s. This may lead to unexpected analysis results.	Error	The message is issued when attribute values defined are beyond the value/range per Liberty standard. Beyond the range could include, less than, less than equal to, greater than or greater than equal to. Valid values could be integers, 0,1 or n and float values. This may lead to unexpected results in analysis. To fix the issue, re-characterize the library with appropriate settings to have requisite values per the standard for the attributes for which this message is issued.
TECHLIB	1173	The '%s' attribute has not been defined for %s '%s' on line %d. This may lead to undesirable analysis results. To fix this issue, define the missing attribute in library.	Error	The message is issued when an expected attribute is not defined for a group. For example, directionattribute expected for the pingroup is not defined in the library. To fix the issue, define the missing attribute in the library or re-characterize the library with appropriate settings.

Valus User Guide

Valus Checks

LIB_SEMANTIC				
Message Prefix	ID	Message	Severity	Description
TECHLIB	1174	The attribute '%s' defined in group '%s' on line %d is not monotonically %s for values '%f' to '%f'. This may lead to undesirable analysis results.	Error	The message is issued when an attribute range does not increase or decrease monotonically as per the Liberty standard or convention. For example, instead of the expected range of monotonically increasing index_1: (1, 2, 3, 4, 5), the range defined in library is index_1: (1, 2, 3, 2, 3). To fix the issue, re-characterize the library with appropriate settings.
TECHLIB	1175	Template name '%s' used in group '%s' on line %d is not defined in the library. To fix this issue, define the missing template in the library.	Error	The message is issued when a template name which has not been defined in the library, has been specified in a table group. For example, the group 'cell_rise(abc)' given, but the template name 'abc' has not been defined in the library. To rectify the issue, correct the library to define the missing template definition.

Valus User Guide

Valus Checks

LIB_SEMANTIC				
Message Prefix	ID	Message	Severity	Description
TECHLIB	1176	The table size (%d) of the lookup table group '%s' on line %d is not as per the indices definition '%s'. This may lead to unexpected analysis results.	Error	The message is issued when the table size of the lookup table in the library is different from the expected size. The expected size is (size of index_1) x (size of index_2). For example, index_1: (1, 2); index_2: (3, 4); values (1, 2,3 ,4, 5, 6,7). The size of look up table here is 7, whereas the expected size of table in this case is (2x2). To fix the issue, re-characterize the library with appropriate settings.
TECHLIB	1179	The 'index_%d' is defined for '%s' group on line %d but its corresponding 'variable_%d' is not defined for template '%s' '%s' on line %d. This may lead to undesirable analysis results.	Error	This message is issued when an index is defined in the group but its corresponding variable name is not defined in the template. For index definition to be complete, its variable must be defined in the template. To fix the issue, re-characterize the library with appropriate settings such that all indices have defined variables in the template.
TECHLIB	1180	The 'index_%d' is not defined for '%s' group on line %d, but is required as per definition of template '%s' '%s' on line %d. This may lead to undesirable analysis results.	Error	This message is issued when an index is not defined for a group but is required as per the template definition. To fix the issue, re-characterize the library with appropriate settings such that all indices are defined in the group as per the template definition.

Valus User Guide

Valus Checks

LIB_SEMANTIC				
Message Prefix	ID	Message	Severity	Description
TECHLIB	1182	The higher level variable_%d is defined but variable_%d is not defined for template group '%s' '%s'. This template group will be ignored.	Error	This message is issued when a higher level variable is defined without the lower level variable definition. For example, variable_2 is defined but variable_1 definition is missing. To fix the issue, re-characterize the library with appropriate settings such that higher level variables are not defined without the lower level variable definitions.
TECHLIB	1183	The variable_%d and variable_%d have the same value %s for template group '%s' '%s' on line %d.	Error	This message is issued when two variables in a template have same value. Each variable should have unique value in the template. To fix the issue, re-characterize the library with appropriate settings such that all variables in a template have unique values.

Valus User Guide

Valus Checks

LIB_SEMANTIC				
Message Prefix	ID	Message	Severity	Description
TECHLIB	1184	The attribute '%s' is defined but the attribute '%s' is not defined in '%s' group on line %d.	Error	This message is issued when defining one attribute alone is not sufficient, but another attribute which is also mandatorily needed to be defined with that attribute, is missing. Not only either one but both attributes are necessary for analysis. For example, in one case where attribute 'sdf_cond' is defined, but 'when' attribute is missing from library or vice-versa. Another case, where the library does not specify the 'default_wire_load_selection' when wire load models are present. In both the cases, both attributes are required to be defined in the library. To fix the issue, re-characterize the library with appropriate settings, such that all mandatory dependent attributes are defined in the library.
TECHLIB	1185	The '%s' '%s' is defined in '%s' cell group but not defined in '%s' group.	Error	This message is issued when a cell object such as bus, bundle, pin etc., is inconsistent between the main and test cell defined in the library. To fix the issue, re-characterize the library with appropriate settings, such that cell objects are consistent between main cell and test cell definition

Valus User Guide

Valus Checks

LIB_SEMANTIC				
Message Prefix	ID	Message	Severity	Description
TECHLIB	1188	The cell '%s' on line %d has attribute pad_type defined, but it is not a pad cell.	Error	This message is issued when the attribute pad_type has been defined for a cell which is not a pad cell. Only pad cells are expected to have the attribute pad_type defined on them. To fix the issue, re-characterize the library with appropriate settings, such that the correct attributes are defined on the cell.
TECHLIB	1189	The cell '%s' on line %d has pad_cell attribute but has no 'is_pad' attribute set to true on any pin/pg_pin.	Error	This message is issued when the attribute is_pad is missing for all the pins or pg_pins of the cell which is defined as a pad cell. To fix the issue, re-characterize the library with appropriate settings, such that the correct attributes are defined on the pin and cell.
TECHLIB	1190	The cell '%s' is not a pad cell but has '%s' attribute specified for %s '%s'.	Error	This message is issued when pad attributes such as hysteresis, drive_current etc., have been defined for a pin of cell which is not a pad cell. Only pad pins are expected to have pad attributes defined on them. To fix the issue, re-characterize the library with appropriate settings, such that the correct attributes are defined on the pin/cell.

Valus User Guide

Valus Checks

LIB_SEMANTIC				
Message Prefix	ID	Message	Severity	Description
TECHLIB	1193	The %s '%s' on line %d is undefined in the library.	Error	This message is issued when any attribute is used but it is not defined in the library. To fix the issue, re-characterize the library with appropriate settings, such that there is no undefined attribute getting used in library.
TECHLIB	1194	The attribute 'bit_width' %d on line %d does not match with the bit range specified using attributes 'bit_from' %d and 'bit_to' %d.	Error	This message is issued when specified bit_width attribute does not match with the bus_width calculated from bit_from and bit_to attributes. To fix the issue, re-characterize the library with appropriate settings, such that there is no such conflict.
TECHLIB	1195	The bus pin '%s' on line %d, is out of range as per bus bit definition %d to %d.	Error	This message is issued when a bus pin is specified with width which is out of range as per the bus bit definition. For example, the pin X is specified with bus bit range 2 i.e. (0 to 1) but the bus pin is specified like X[3]. To fix the issue, re-characterize the library with appropriate settings, such that the correct range is defined.

Valus User Guide

Valus Checks

LIB_SEMANTIC				
Message Prefix	ID	Message	Severity	Description
TECHLIB	1196	The group '%s' with name '%s' on line %d is already defined on line %d. The first definition will be retained, rest will be ignored.	Error	This message is issued when an attribute is defined more than once. The same attribute cannot have multiple definition. First definition will be retained and rest will be ignored. To fix the issue, re-characterize the library with appropriate settings, such that the attributes are defined correctly.
TECHLIB	1197	The cell '%s' on line %d is neither a pad cell nor an auxiliary pad cell but has 'is_pad' attribute set to true on %s at line %d.	Error	This message is issued when the attribute pad_cell/ auxiliary_pad_cell is missing at cell level but has pin(s)/ pg_pin(s) with is_pad attribute. Pins should have is_pad attribute only if cell is a pad cell. To fix the issue, re-characterize the library with appropriate settings, such that the correct attributes are defined on pin/cell.
TECHLIB	1198	The 'index_%d' is defined but its corresponding 'variable_%d' is not defined for template '%s' '%s'. This may lead to undesirable analysis results.	Error	This message is issued when an index is defined but its corresponding variable name is not defined in the template. For index definition to be complete, its variable must be defined in the template. To fix the issue, re-characterize the library with appropriate settings such that all indices have defined variables in the template.

Valus User Guide

Valus Checks

LIB_SEMANTIC				
Message Prefix	ID	Message	Severity	Description
TECHLIB	1200	The attribute '%s' defined on line %d is not supported for '%s' group. It should be defined at '%s' level.	Error	The message is issued when an attribute defined for a group is not supported by that group and needs to be defined at a different parent level in the library. For example, if a library level attribute (such as time_unit) is defined at the cell or pin level definition. To fix the issue, re-characterize the library with appropriate settings, such that only attributes supported by a group definition are specified in that group.
TECHLIB	1201	The attribute '%s' defined on line %d is not supported for the '%s' group.	Error	The message is issued when an attribute defined for a group is not supported by that group and needs to be defined at a different parent level in the library. For example, if a library level attribute (such as time_unit) is defined at the cell or pin level definition. To fix the issue, re-characterize the library with appropriate settings, such that only attributes supported by a group definition are specified in that group.

Valus User Guide

Valus Checks

LIB_SEMANTIC				
Message Prefix	ID	Message	Severity	Description
TECHLIB	1203	The name of cell defined on line %d is missing. The definition will be ignored.	Error	The message is issued when the cell definition is missing the name string. In such cases, the cell definition will be completely ignored. To fix the issue, re-characterize the library with appropriate settings, such that the name of the cell definition is defined such that it can be referenced correctly.
TECHLIB	1204	Illegal value '%s' defined for attribute '%s'. The attribute will be ignored.	Error	The message is issued when non-legal and unacceptable values are specified for attributes. For example, per liberty guidelines, the acceptable values for attributes such as dont_touch and dont_use are only true/false. If the library has other values such as enum or integers specified, the attribute will not be honored. To fix the issue, re-characterize the library with appropriate settings, such that the attribute has legal value specified and can be honored.

Valus User Guide

Valus Checks

LIB_SEMANTIC				
Message Prefix	ID	Message	Severity	Description
TECHLIB	1206	The attribute '%s' defined on line %d has invalid syntax for floating number. It will be ignored.	Error	The message is issued when a library attribute having inconvertible floating point numbers such as single float is expected but multiple values are provided or unusable values are provided like nan/inf in float value or float array. For example, the attributes rise_capacitance and fall_capacitance are expected to have a single float value defined. To fix the issue, re-characterize the library with appropriate settings, such that the attribute takes a scalar float value.
TECHLIB	1207	The attribute wire_load_from_area defined on line %d of wire_load_selection group has max area value %f specified less than min area value %f.	Error	The message is issued when the min area value defined in the wire_load_from_area attribute of the wire_load_selection group, is less than that defined as the max area value. Min area should be lesser than the max area value. To fix the issue, re-characterize the library with appropriate settings, such that the area range is defined correctly.

Valus User Guide

Valus Checks

LIB_SEMANTIC				
Message Prefix	ID	Message	Severity	Description
TECHLIB	1208	The attribute '%s' of '%s' group '%s' has value '%f' which is '%s'.	Error	The message is issued when the attribute in a group has a value which is different (less than, less than equal to, greater than, greater than equal to defined values which could be integers or float values) than the expected value/range per the Liberty standard. For example, if max_area value defined in the wire_load_from_area attribute of the wire_load_selection group, is less than or equal to 0. Or min_area value is less than 0. To fix the issue, re-characterize the library with appropriate settings, such that the attribute values are defined per Liberty standard.
TECHLIB	1211	The %s '%s' is not a pad pin but has '%s' attribute specified on line %d.	Error	The message is issued when pad attributes such as hysteresis, drive_current etc., have been defined for a pin which is not a pad pin. Only pad pins are expected to have pad attributes defined on them. To fix the issue, re-characterize the library with appropriate settings, such that the correct attributes are defined on the pad pin

Valus User Guide

Valus Checks

LIB_SEMANTIC				
Message Prefix	ID	Message	Severity	Description
TECHLIB	1212	The pad attribute '%s' is defined on %s '%s' on line %d. It should be defined on '%s'.	Error	The message is issued when pad attributes such as hysteresis, drive_current are defined on incorrect pin based on direction. Per Liberty standard, the hysteresis attribute is valid only for input pad pins, whereas drive_current can only be defined for output pad pins. In cases where the library specifies it the differently, the message is issued. To fix the issue, re-characterize the library with appropriate settings, such that the correct attributes are defined with respect to the pad pin directions.
TECHLIB	1215	For cell '%s', pin name %s defined on line %d conflicts with sequential variable for '%s' group defined on line %d	Error	This message is issued when the pin name is defined same as the variable(s) specified for 'ff' and 'latch' group. To fix the issue, re-characterize the library with appropriate settings, such that the correct pin name is specified.
TECHLIB	1220	The '%s' attribute is not defined for the cell %s. This may lead to undesirable analysis results. To fix this issue, define the missing attribute in library.	Error	This message is issued when the attributes such as ecdsm_vtp and ecdsm_vtn are not defined for the cell in library with ecdsm version >= 1.2. To fix the issue, re-characterize the library with appropriate settings, such that these attributes are defined on the cell.

Valus User Guide

Valus Checks

LIB_SEMANTIC				
Message Prefix	ID	Message	Severity	Description
TECHLIB	1222	The 'index_%d' of type '%s' defined in '%s' group on line %d should have single float value. The group will be ignored.	Error	The message is issued when any index specified in the library is expected to have single float value but has more than one float value. To fix this issue, re-characterize the library such that all indices match the expected number of float values.
TECHLIB	1223	The attribute '%s' has value '%g' which is not '%s'.	Error	This message is issued when the attribute has float values beyond the defined range as per the Liberty standard.
TECHLIB	1225	For clock gating integrated cell '%s', output %s '%s', the function attribute has both '%s' and '%s' at line %d. This may lead to unexpected analysis results.	Error	This message is issued when pin function attribute is defined with both IQ and IQN pins for clock gating integrated cells. To fix the issue, re-characterize the library with appropriate settings.
TECHLIB	1231	The attribute fanout_length for wire_load group '%s' defined on line %d has %d arguments. It must have 2 or 5 arguments.	Error	This message is issued when the attribute fanout_length for wire_load group has incorrect number of arguments. It must have either 2 or 5 arguments. To fix the issue, re-characterize the library with appropriate settings, such that the correct value is set for the attribute.

Valus User Guide

Valus Checks

LIB_SEMANTIC				
Message Prefix	ID	Message	Severity	Description
TECHLIB	1234	The %s is not present for the %s '%s' on line '%d' for cell '%s'. This may cause incorrect voltage to be used during timing analysis	Error	This message is issued when the attribute related_power_pin or related_ground_pin is absent. For special cells (macro/level shifter/always on/retention/isolation/pad etc.), it is mandatory to specify the related_power_pin/related_ground_pin attribute for every pin definition. Absence of this attribute may cause incorrect voltage to be used during timing analysis. To fix the issue, re-characterize the library with appropriate settings, such that this information is correctly specified for every pin.
TECHLIB	1237	The related_pin '%s' specified for %s group on line '%d' is not defined for cell '%s'.	Error	This message is issued when an undefined pin is specified as related_pin. To fix the issue, re-characterize the library such that valid pin is used for related_pin.
TECHLIB	1253	'%s' group '%s' is undefined in library. This attribute will be ignored.	Error	This message is issued when input_voltage and/or output_voltage group(s) referred by these attributes at the pin level are not defined in the library. Attribute referring to undefined group will be ignored. To fix this issue, re-characterize the library with appropriate settings.

Valus User Guide

Valus Checks

LIB_SEMANTIC				
Message Prefix	ID	Message	Severity	Description
TECHLIB	1256	The %s is being ignored due to errors in this group. This group will be excluded for any further library checks. Refer to the previous messages issued for %s to find the details of the issues in this group.	Error	This message is issued in scenarios when any liberty group has problem. For example, if the template at library level and cell level do not match for number of indices or number of values in indices. It can also occur if indices are expected to have single value for some attributes per Liberty format but library has more than one value for that index. Another case can be when timing type is not specified/invalid and timing arc has sequential timing arcs. In such scenarios, the corresponding table data is ignored.
TECHLIB	1261	Found invalid/missing attribute value.	Error	This message is issued in scenarios when attribute value is invalid/missing. For example related_pin attribute is specified but its value is empty string or integer value is specified where string value is expected.
TECHLIB	1264	Pin index range [%d:%d] is out of range of bus %s in cell '%s	Error	This message is issued when the specified pin index range is out of range as per the bus definition in cell. To fix the issue, re-characterize the library with appropriate settings, such that the correct range is defined.

Valus User Guide

Valus Checks

LIB_SEMANTIC				
Message Prefix	ID	Message	Severity	Description
TECHLIB	1265	scalar pin %s specified as bus with index range [%d:%d] in boolean expression in cell %s. The boolean expression is ignored.	Error	This message is issued when the scalar pin is defined as bus bit in boolean expressions like function, when expression etc. To fix the issue, re-characterize the library with appropriate settings, such that the index range is specified for scaler pin.
TECHLIB	1298	Number of '%s' specified is %d in generated_clock group in cell '%s'. It must be odd number.	Error	This message is issued when the number of edges/shifts are specified as even number in generated_clock group. Number of edges/shifts must be an odd number. To fix the issue, re-characterize the library with appropriate settings, such that the generated clock is correctly defined.
TECHLIB	1299	Number of specified 'edges' (%d) is not equal to number of specified 'shifts' (%d) in generated_clock group in cell '%s'.	Error	This message is issued when there is a mismatch in the number of edges and shifts in generated_clock group. Number of edges must be equal to the number of shifts. To fix the issue, re-characterize the library with appropriate settings, such that the generated clock is correctly defined.

Valus User Guide

Valus Checks

LIB_SEMANTIC				
Message Prefix	ID	Message	Severity	Description
TECHLIB	1300	Invalid frequency %s factor '%d' specified in '%s' attribute in generated_clock group in cell '%s'. It must be a power of 2.	Warning	This message is issued when the invalid frequency multiplication factor or division factor is specified for the attribute multiplied_by/ divided_by in generated_clock group. The factor must be a power of two. To fix the issue, re-characterize the library with appropriate settings, such that the generated clock is correctly defined.
TECHLIB	1301	Invalid generated_clock definition in cell '%s'. Attribute(s) '%s' are not allowed together.	Error	This message is issued when the generated_clock group is defined incorrectly. The attribute divide_by can not be combined with any of these 'multiply_by', 'edges', 'shift' and the attribute 'multiply_by' can not be combined with any of these 'divide_by', 'edges', 'shift'. To fix the issue, re-characterize the library with appropriate settings, such that the generated clock is correctly defined.
TECHLIB	1302	First edge specified in '%s' is %d in generated_clock group in cell '%s'. It must be greater than equal to 1.	Error	This message is issued when the first edge is specified as less than 1 in generated_clock group. The first edge must be greater than or equal to 1. To fix the issue, re-characterize the library with appropriate settings, such that the correct value of first edge is specified.

Valus User Guide

Valus Checks

LIB_SEMANTIC				
Message Prefix	ID	Message	Severity	Description
TECHLIB	1303	Number of '%s' specified is %d in generated_clock group in cell '%s'. It must be greater than or equal to 3	Error	This message is issued when the number of edges specified is less than three in generated_clock group. The number of edges must be greater than or equal to 3. To fix the issue, re-characterize the library with appropriate settings, such that the correct value of number of edges is defined.
TECHLIB	1304	Invalid generated_clock definition in cell '%s'. Attribute 'shifts' is specified without attribute 'edges'.	Error	This message is issued when the shifts are specified but edges are not specified in generated_clock group. Both are required as the shifts are added to the edges specified in the edge list to generate the clock. To fix the issue, re-characterize the library with appropriate settings, such that the edges should be defined if shifts are specified.
TECHLIB	1305	The edges specified must be monotonically increasing in generated_clock group in cell '%s'.	Error	This message is issued when the edges specified in generated_clock group of cell do not monotonically increase. To fix the issue, re-characterize the library with appropriate settings, such that the edges defined are monotonic.

Valus User Guide

Valus Checks

LIB_SEMANTIC				
Message Prefix	ID	Message	Severity	Description
TECHLIB	1307	The %s group contains values which are not monotonically %s from %f to %f for %s %f and %s %f in group %s.	Error	This message is issued when ecdsm_waveform_set table values (PxQxR), for a fixed point of two of table indices do not monotonically increase/decrease across the third axis. For example, in the ecdsm_waveform_set table, for a fixed value of slew index and load index, the table values are not monotonically increasing across the values axis. To fix the issue, re-characterize the library with appropriate settings, such that table values are monotonic as per Liberty standard.
TECHLIB	1309	For point '%d' in index_%d (%s) in '%s' table on %s of cell '%s', values %f to %f are not monotonically %s across the index_%d (%s)	Error	This message is issued when delay table values (M x N), for a fixed point of index_1/ index_2 of table indices do not monotonically increase/decrease across the index_2/ index_1 axis. For example, in the cell_rise table, for a fixed value of slew index, the table values are not monotonically decreasing across the load axis or for fixed value of load index, the table values are not monotonically increasing across the slew axis. To fix the issue, re-characterize the library with appropriate settings, such that table values are monotonic as per Liberty standard.

Valus User Guide

Valus Checks

LIB_SEMANTIC				
Message Prefix	ID	Message	Severity	Description
TECHLIB	1319	The user-defined attribute '%s' is defined for group '%s' with type '%s'. This is an invalid definition.	Error	This message is issued when the define statement for user-defined attributes (related_spice_node/load_cap_rise/load_cap_fall) is not specified with appropriate group or type. The related_spice_node should be specified for ccsn_first_stage/ccsn_last_stage groups with type string and load_cap_rise/load_cap_fall should be specified for ccsn_first_stage group with type float. To fix the issue, re-characterize the library with appropriate settings, such that the required user-defined attributes are defined properly.
TECHLIB	1323	The define statement of user-defined %s is not valid. The definition contains %d parameters instead of %d parameters.	Error	The define statement should follow the syntax as: 'define(attr_name, group_name, attr_value_type);'. To fix the issue, re-characterize the library with appropriate settings, such that the required user-defined attributes are defined properly.

Valus User Guide

Valus Checks

LIB_SEMANTIC				
Message Prefix	ID	Message	Severity	Description
TECHLIB	1325	The specified parent group '%s' of user-defined %s '%s' is invalid. This definition will be ignored.	Error	The parent group of user-defined attribute or group should be either a standard group as per liberty or it should be a user-defined group which is defined using define_group() statement. To fix the issue, re-characterize the library with appropriate settings, such that the required user-defined attributes are defined properly.
TECHLIB	1331	The pg_pin '%s' with pg_type '%s' cannot be used in '%s' attribute for pin '%s' of the cell '%s'. This attribute is being ignored which may impact analysis accuracy.	Error	This message is issued when the pg_pin specified with related_power_pin/ related_ground_pin does not match with the corresponding pg_type e.g. pg_pin VDD with pg_type as primary_power is used in related_ground_pin. These attributes will be ignored which may impact analysis accuracy. To fix the issue, re-characterize the library with appropriate settings, such that the incompatible pg_pin is not used.

Valus User Guide

Valus Checks

LIB_SEMANTIC				
Message Prefix	ID	Message	Severity	Description
TECHLIB	1332	The ecdsm waveform associated with '%s' group of pin '%s' in cell '%s' will be ignored due to incorrect voltage values.	Error	This message is issued when the voltage values specified in ecdsm_waveform/ ecdsm_waveform_set is either outside of range [0.02, 0.098] or are too close (less than 0.01 difference between two consecutive values), then that point will be ignored and if after applying this criteria, there are less than 2 values left in voltage grid then the complete group in the rise_transition and fall_transition group will be ignored. To fix the issue, re-characterize the library with appropriate settings, such that the correct voltage values is specified
TECHLIB	1346	The attribute '%s' defined in group '%s' on line %d is not monotonically %s for values '%f' to '%f'. This may lead to undesirable analysis results. The attribute will be ignored.	Error	The message is issued when an attribute range does not increase or decrease monotonically as per the Liberty standard or convention. For example, instead of the expected range of monotonically increasing index_1: (1, 2, 3, 4, 5), the range defined in library is index_1: (1, 2, 3, 2, 3). To fix the issue, re-characterize the library with appropriate settings.

Valus User Guide

Valus Checks

LIB_SEMANTIC				
Message Prefix	ID	Message	Severity	Description
TECHLIB	1364	The attribute '%s' is defined but the attribute '%s' is not defined in '%s' group. The attribute '%s' will be ignored.	Error	This message is issued when defining one attribute alone is not sufficient, but another attribute which is also mandatorily needed to be defined with that attribute, is missing.
TECHLIB	1366	All combinations of %s in vector groups(s) of %s are not defined. The definition for %s vector group(s) is missing on %s and cell %s. The group will be ignored.	Error	This message is issued when any of the combination of input_net_transition and total_output_net_capacitance defined in vector group for the output_voltage_rise/ output_voltage_fall group in ccsn_first_stage/ ccsn_last_stage group or output_current_rise/ output_current_fall for timing group is missing. To fix the issue, re-characterize the library with appropriate settings, such that the vector groups are defined for all combination of input_net_transition and total_output_net_capacitance.

Valus User Guide

Valus Checks

LIB_SEMANTIC				
Message Prefix	ID	Message	Severity	Description
TECHLIB	1371	The '%s' pin '%s' defined for cell '%s' in library '%s' is either not defined or defined with different direction in the cell '%s' in library '%s'.	Error	This message is issued when either the pin is defined for cell in one library and not defined for the same cell in other library or pins are defined with different direction. For example: Pin(A) defined as input pin for cell in one library and is not defined or defined as output pin in other library. For such cases, tool will stop the design loading. To fix the issue, re-characterize the libraries with appropriate settings, such that the pins of same cell across libraries are consistent.
TECHLIB	1372	Scalar table specified for '%s' group defined on %s of cell '%s' is not supported. This group will be ignored.	Error	This message is issued when the scalar table is specified for eesm_capacitance group. Such eesm_capacitance group will be ignored. To fix the issue, re-characterize the library with appropriate settings, such that the tables are specified with correct dimension.

Valus User Guide

Valus Checks

LIB_SEMANTIC				
Message Prefix	ID	Message	Severity	Description
TECHLIB	1374	The normalized_driver_waveform %s is already defined on line %d. The first definition will be retained, rest will be ignored.	Error	This message is issued when the normalized_driver_waveform group is defined multiple times either with same driver_waveform_name or without driver_waveform_name at library level. The first definition will be retained and rest will be ignored. To fix the issue, re-characterize the library with appropriate settings, such that the normalized_driver_waveform groups defined at library level are unique.
TECHLIB	1383	Invalid pin '%s' is specified with the attribute related_output_pin for timing group on %s of cell '%s'. The pin specified with related_output_pin must be an output or inout pin.	Error	This message is issued when the pin specified with attribute related_output_pin is invalid. The direction of pin specified with related_output_pin must be either output or inout. To fix the issue, re-characterize the library with appropriate settings, such that the related_output_pin is correctly defined.

Valus User Guide

Valus Checks

LIB_SEMANTIC				
Message Prefix	ID	Message	Severity	Description
TECHLIB	1391	The test_cell group is defined for cell '%s' but its corresponding sequential group information is missing. The test_cell group will be ignored.	Error	This message is issued when the cell has test_cell group but its corresponding sequential group information is not defined. The behavior defined in test_cell group, must be already described by an ff, ff_bank, latch, latch_bank or statetable at cell level. To fix the issue, re-characterize the library with appropriate settings, such that the cell having test_cell group must have corresponding sequential group information.
TECHLIB	1396	The cell '%s' with ff/latch group has pin reference names but there is no pin with reference_input attribute. The ff/latch group will be ignored.	Error	This message is issued when the ff/latch group is defined with format that expects reference pin as reference_input in any pin but its not present. The ff groups will be ignored in such cases. To fix the issue, re-characterize the library with appropriate settings, such that the correct combination is used.

Valus User Guide

Valus Checks

LIB_SEMANTIC				
Message Prefix	ID	Message	Severity	Description
TECHLIB	1408	The ratio of attribute '%s' of group '%s' to the corresponding '%s' group specified on %s of cell '%s' has one or more values which are not in range [$\geq$ %g and $\leq$ %g]. This may lead to unexpected analysis results.	Warning	This message is issued when ratio of skewness and its corresponding std_dev is not in the pre-defined range. This may lead to unexpected results in analysis. To fix the issue, re-characterize the library with appropriate settings to have requisite values per the standard for the attributes for which this message is issued.
TECHLIB	1409	Attribute related_pin is missing for '%s' group specified on pin '%s' of cell '%s'. Attribute is mandatory if '%s' is specified on output pin.	Error	This message is issued when attribute related_pin is missing for max_trans/cap group. To fix the issue, re-characterize the library with appropriate settings, such that the max_trans/cap group is correctly defined.
TECHLIB	1410	For group '%s', 2-Dimensional table is specified on input pin '%s' of cell '%s'. 2-Dimensional table is not supported on input pin.	Error	This message is issued when 2-Dimensional table for group max_trans/cap is specified on input pin. To fix the issue, re-characterize the library with appropriate settings, such that the max_trans/cap tables are correctly defined.
TECHLIB	1428	The '%s' variable name '%s' is also used for pin '%s' defined on line %d in cell '%s'. The variable name must be unique name.	Error	This message is issued when variable name specified in ff/latch/ff_bank/latch_bank conflicts with pin/bus/bundle name. To fix the issue, please specify unique variable names.

Valus User Guide

Valus Checks

LIB_SEMANTIC				
Message Prefix	ID	Message	Severity	Description
TECHLIB	1430	The cell '%s' is being ignored due to errors in its definition. This library cell will not be loaded in this session. Refer to the previous messages issued for this cell to find the details of the issue.	Error	This message is issued when cell is defined incorrectly in library. Such cells will be ignored. Refer to previous messages issued for the cell. To fix the issue, re-characterize the library with appropriate settings, such that the cell is correctly defined in the library.
TECHLIB	1433	The pin '%s' defined for bundle '%s' in cell '%s' is not listed in 'members' attribute of bundle.	Error	This message is issued when pin defined as bundle pin but not specified in member attribute. To fix the issue, re-characterize the library with appropriate settings, such that the pins are defined correctly for bundle group.
TECHLIB	1434	The attribute '%s' defined in group '%s' has one or more values which are %s. This may lead to undesirable analysis results. The attribute will be ignored.	Error	The message is issued when attribute values defined are less than zero as per the Liberty standard or convention. For example, slew or cap index ranges cannot have negative values. To fix the issue, re-characterize the library with appropriate settings.
TECHLIB	1435	The dc_current values are not monotonically decreasing in the range '%.10g' to '%.10g' for '%s' '%f' for dc_current table defined in pin '%s' and cell '%s'. %s.	Warning	This message is issued when current values specified in dc_current table defined in CCSN/CCB group are not monotonically decreasing from VSS to VDD. This may impact delay calculation and glitch accuracy.

Valus User Guide

Valus Checks

LIB_SEMANTIC				
Message Prefix	ID	Message	Severity	Description
TECHLIB	1442	The timing arc defined for cell '%s' and for pin '%s', with related_pin '%s' and timing_type '%s' in library '%s' is not defined in the cell '%s' in library '%s'.	Warning	This message is issued when a timing arc is defined for cell in one library and not defined for the same cell in other library. This may impact accuracy in delay and glitch analysis. To fix the issue, re-characterize the libraries with appropriate settings, such that the timing arcs of same cell across libraries are consistent.
TECHLIB	1445	Attribute 'function' on output %s of cell '%s' is not defined in the library.	Warning	This message is issued when the attribute function is not defined on an output pin of cell in the library. To fix the issue, re-characterize the library with appropriate settings, such that the attribute is defined in the cell pin.
TECHLIB	1449	The value 'true' is set for complimentary attributes 'is_pass_gate' and 'is_inverting' for '%s' group on pin '%s' of cell '%s'. This can cause conflict for the tool. Either remove 'is_pass_gate' attribute or set the value for 'is_inverting' attribute as 'false'.	Error	A pass gate drives output by input, if the control is set to 'true'. On the contrary, is_inverting attribute specifies if the channel-connected block is inverting. Setting both attributes (is_pass_gate and is_inverting) as 'true' under the same group will cause a conflict. To avoid this issue, either remove 'is_pass_gate' attribute from the group or specify 'false' for 'is_inverting' attribute.

Valus User Guide

Valus Checks

LIB_SEMANTIC				
Message Prefix	ID	Message	Severity	Description
TECHLIB	1450	The attribute '%s' on pin '%s' of cell '%s' is not defined in the library, even though 'reference_input' attribute is defined.	Error	This message is issued when the attribute 'function' is not defined for a pin of cell in the library, even though 'reference_input' attribute is defined. To map, reference_input to FF/latch, function attribute on pin is needed. Such cells will not be loaded. To fix the issue, re-characterize the library with appropriate settings, such that the function is defined.
TECHLIB	1452	Unable to parse string because it is too long or an EOF has been encountered.	Error	This message is issued when there is when a very long string or EOF is encountered during library reading. To fix the issue, re-characterize the library with appropriate settings.
TECHLIB	1453	The pin '%s' is specified multiple times in the 'members' attribute of bundle '%s' for cell '%s'.	Error	This message is issued when the 'member' attribute of bundle group has multiple references of same pin. For example, members (D1, D1, D3, D4), such bundle definition is not supported.
TECHLIB	1454	Unable to process the sequential logic of cell '%s' because %s.	Error	This message is issued when the tool fails to process the sequential logic of a cell. This could be due to various reasons. For Example: If the function of an output pin being used in the logic is missing.

Valus User Guide

Valus Checks

LIB_SEMANTIC				
Message Prefix	ID	Message	Severity	Description
TECHLIB	1455	For point '%d' in index_%d (%s) in '%s' table on %s of cell '%s', values %f to %f are not monotonically %s across the index_%d (%s)	Warning	This message is issued when transition table values (M x N), for a fixed point of index_1/ index_2 of table indices do not monotonically increase/ decrease across the index_2/ index_1 axis. For example, in the rise_transition table, for a fixed value of slew index, the table values are not monotonically decreasing across the load axis or for fixed value of load index, the table values are not monotonically increasing across the slew axis. To fix the issue, re-characterize the library with appropriate settings, such that table values are monotonic as per Liberty standard.
TECHLIB	1456	The %s group contains values which are not monotonically %s from %f to %f for %s %f and %s %f in group %s.	Warning	This message is issued when transition table values (PxQxR), for fixed point of two of table indices do not monotonically increase/ decrease across the third axis. For example, in the rise_transition table, for a fixed value of load index and load index1, the table values are not monotonically increasing across the slew axis. To fix the issue, re-characterize the library with appropriate settings, such that table values are monotonic as per Liberty standard.

Valus User Guide

Valus Checks

LIB_SEMANTIC				
Message Prefix	ID	Message	Severity	Description
TECHLIB	1457	A %s arc from enable pin '%s' to %s pin '%s' is not present. The tlatch specification will be ignored and latch inferencing will not be done. Refer cell '%s'.(Line : %d)	Warning	This message is issued when the tlatch group is not consistent with the existing timing arc topology. The tlatch specification will be ignored and latch inferencing will not be done.
TECHLIB	1458	The existing timing arcs do not match with the edge_type:'%s' attribute in the tlatch group on pin '%s' and enable pin '%s'. The tlatch specification will be ignored and latch inferencing will not be done. Refer cell '%s'.(Line : %d)	Warning	This message is issued when the tlatch group is not consistent with the existing timing arc topology. The tlatch specification will be ignored and latch inferencing will not be done.
TECHLIB	1459	There are no combinational arcs in the fanout of data pin '%s' that can be considered as potentially transparent. The tlatch specification on pin '%s' and enable pin '%s' will be ignored and latch inferencing will not be done. Refer cell '%s'.(Line : %d)	Warning	This message is issued when the tlatch group is not consistent with the existing timing arc topology. The tlatch specification will be ignored and latch inferencing will not be done.
TECHLIB	1460	More than one tlatch group are defined on pin '%s' with same enable pin '%s' and same edge type '%s'. Tool accepts only two tlatch groups for same enable pin with different edge_type. Refer cell '%s'.	Warning	This message is issued when two tlatch groups defined on a pin with same enable pin also have same edge_type. To fix this change the edge_type of one tlatch group. Also ensure that there are not more than 2 tlatch groups with the same enable pin.

Valus User Guide

Valus Checks

LIB_SEMANTIC				
Message Prefix	ID	Message	Severity	Description
TECHLIB	1462	The master_pin attribute is missing in the generated_clock group in cell '%s'. This generated_clock group will be ignored.	Error	This message is issued when the master_pin attribute is missing in generated_clock group. To fix the issue, re-characterize the library by adding the missing data in order to define the generated clock group correctly.
TECHLIB	1466	The when condition '%s' for timing arc defined on pin '%s' of cell '%s' with timing_sense '%s' and related_pin '%s' will be ignored. Refer to the previous message 'TECHLIB-%d' issued for this timing arc.	Error	This message is issued when the pin specified in when condition for specified timing_sense and related_pin attributes is not valid for the given timing arc. Tool will ignore the when attribute.
TECHLIB	1481	The index_%d value is either not specified or has invalid value in user defined group table '%s'%s". Table will be ignored.	Error	This message is issued when an index is either not specified in the user-defined group table or has an invalid value. To fix the issue, re-characterize the library with appropriate settings, such that the index is specified correctly.
TECHLIB	1483	The library attribute '%s' is defined for %s pin %s but missing for %s pin %s of cell %s. For consistency, this attribute should be defined for all pins of the bus/bundle group.	Warning	This message is issued when a library attribute timing/internal_power/ccb/receiver capacitance is defined for few pins of a bus/bundle but missing for other pins of the same bus/bundle. For library cell definition consistency, this attribute should be defined for all the pins of a bus group.

Valus User Guide

Valus Checks

LIB_SEMANTIC				
Message Prefix	ID	Message	Severity	Description
TECHLIB	9023	Pin/bus/bundle '%s' definition not found in cell '%s'.	Error	Pin/bus/bundle (<PinName>) definition not found in cell (<CellName>). The indicated cell <CellName> does not have description for the pin/bus/bundle <PinName>. Check if <PinName> has been used correctly. If yes, add the necessary description, otherwise modify the cell by deleting the reference(s) to <PinName>.
TECHLIB	9024	Pin '%s' is invalid for bus '%s' in cell '%s'.	Error	This message is issued when there is a mismatch in pin name and bus name or pin index is out of range as per bus index range defined in a cell. To fix the issue, re-characterize the library with appropriate settings, such that the pin name is matched with bus name.
TECHLIB	9025	Pin name '%s' in bus '%s' of cell '%s' does not match library bus_naming_style '%s'.	Error	Pin name (<PinName>) in bus (<PinBusName>) of cell (<CellName>) does not match library bus_naming_style (<bus_naming_style>). Bus pin name must match the bus naming style of library.

Valus User Guide

Valus Checks

LIB_SEMANTIC				
Message Prefix	ID	Message	Severity	Description
TECHLIB	9026	Look-up table template '%s' definition not found.	Error	Look-up table template (<TemplateName>) definition not found. Undefined look-up table template is referred in the timing description. Check the correctness of <TemplateName>. In case it has been used correctly, add definition for it at the library level, otherwise delete reference(s) to this template.
TECHLIB	9031	'%s' absent in pin/bundle/bus '%s' of cell '%s'.	Error	(<AttributeName>) absent in pin/bundle/bus (<PinBusBundleName>) of cell (<CellName>). The specified attribute <AttributeName> is absent in the indicated pin/bundle bus/bundle <PinBusBundleName> of the cell <CellName>. This attribute is essential for translation. Rectify the library by specifying the missing information.
TECHLIB	9032	Definition of type group for bus_type '%s' is absent for cell '%s' in library. Library reading will be aborted.	Error	This message is issued when the type group definition is missing at library level for specified bus_type. Library reading will be aborted. To fix the issue, re-characterize the library with appropriate settings, such that the type is defined for bus_type.

Valus User Guide

Valus Checks

LIB_SEMANTIC				
Message Prefix	ID	Message	Severity	Description
TECHLIB	9034	Conflicting value for direction found in pin '%s' of bundle/ bus '%s' in cell '%s'.	Error	This message is issued when the direction of a pin is not matched with the direction specified in its parent bus/ bundle. To fix the issue, re-characterize the library with appropriate settings, such that the correct direction is defined.
TECHLIB	9040	Improper definition of related pin '%s'.	Error	To resolve the problem, check the definition of related pin.
TECHLIB	9048	Inconsistency in the expression '%s' in cell '%s'; %s.	Error	Inconsistency in the expression '<Expr>' in cell (<CellName>); Extra parenthesis. There is a mismatch in parenthesis in the expression <Expr> in cell <CellName>. Correct the expression.
TECHLIB	9056	'%s' not found in the library. Assuming '%s	Error	This message is issued when the attribute leakage_power_unit is not defined in the library. It defaults to 1nW. To fix the issue, re-characterize the library with appropriate settings, such that the leakage_power_unit is defined.
TECHLIB	9058	Incorrect '%s' unit specified in the library. Assuming '%s	Error	This message is issued when incorrect unit is specified in the library. To fix the issue, re-characterize the library with appropriate settings, such that the correct unit is specified.

Valus User Guide

Valus Checks

LIB_SEMANTIC				
Message Prefix	ID	Message	Severity	Description
TECHLIB	9063	Invalid delay model specified for the library '%s'. Delay model other than table_lookup/cmos2/generic_cmos is not supported.	Error	This message is issued when invalid delay model is specified. To fix the issue, re-characterize the library with appropriate settings, such that the correct delay model is specified.
TECHLIB	9073	The number of member pins in the bundles '%s' and '%s' is not the same.	Error	The number of member pins in the bundles (<ReferencedBundleName>) and (<OriginalBundleName>) is not the same. Since one of the bundles is defined as a function to the other, the number of member pins in both should be the same.
TECHLIB	9074	Voltage Group '%s' definition not found in library.	Error	This message is issued when the undefined voltage group is used in the library. To fix the issue, re-characterize the library with appropriate settings, such that the voltage group is defined.
TECHLIB	9096	Pulling resistance specified for a non pull_up/pull_down pin '%s' in cell '%s'.	Error	Pulling resistance specified for a non pull_up/pull_down pin (<PinName>) in cell (<CellName>). Pulling resistance can only be specified for pull_up/pull_down devices. Specify the device as pull_up/pull_down by using driver_type attribute to rectify the problem.

Valus User Guide

Valus Checks

LIB_SEMANTIC				
Message Prefix	ID	Message	Severity	Description
TECHLIB	9097	No pulling resistance unit specified for pulling resistance of pull_up/ pull_down devices in cell '%s'.	Error	This message is issued when the pulling resistance is specified without unit. Valid unit values are 1ohm, 10ohm, 100ohm, and 1kohm. To fix the issue, re-characterize the library with appropriate settings, such that the unit of pulling resistance is defined.
TECHLIB	9102	Incorrectly specified slew measurement points. Reported slews falling beyond the range (0.0 - 100.0).	Error	Incorrectly specified slew measurement points. Reported slews falling beyond the range (0.0 - 100.0). To resolve the issue, check/ modify the definitions of the slew measurement thresholds and the slew_derate_from_library construct such that the reported slew range falls in between (0 - 100). To calculate reported slew use, (measured slew upper - measured slew lower) slew derate factor

Valus User Guide

Valus Checks

LIB_SEMANTIC				
Message Prefix	ID	Message	Severity	Description
TECHLIB	9103	The slew_derate_from_library and Input slew measurement points produce reported slews which fall beyond the range (0.0 - 100.0).	Error	This warning is issued if reported slew threshold is computed to be beyond the range (0.0-100.0). The slew derate from library is the factor which defines the relationship between the reported and measured slew thresholds for the library. The formula is as follows - $\text{delta reported} = \text{delta measured} / \text{slew_derate_from_library}$ 'delta' signifies the difference between 'slew_upper' and 'slew_lower'. If delta reported comes out to be greater than hundred with the measure thresholds and slew derate specified, then the reported slews will fall beyond 0-100 range - which is not possible. In such case the above warning is issued. To avoid this warning, make sure proper values of slew_derate_from_library and Input slew measurement points are specified in the timing library.
TECHLIB	9105	Slew_derate_from_library specified as $0 \leq x > 1$ in the library where x is the value of slew_derate.	Error	Slew_derate_from_library specified as $0 \leq x > 1$ in the library where x is the value of slew_derate. To rectify the issue change slew derating factor (slew_derate_from_library) inside the library.

Valus User Guide

Valus Checks

LIB_SEMANTIC				
Message Prefix	ID	Message	Severity	Description
TECHLIB	9118	The statetable for the %s %s is syntactically incorrect. This may be because of %s.	Error	This message is issued when the statetable is incorrectly defined. This might be due to incorrectly placed commas/ colons. To fix the issue, re-characterize the library with appropriate settings, such that the statetable is correctly defined.
TECHLIB	9121	Mismatch in bit_from/bit_to values while overwriting the bus_type for bus %s in cell %s.	Error	This message is issued when there is mismatch in the attributes bit_from/bit_to values while overwriting the bus_type for bus. To fix the issue, re-characterize the library with appropriate settings, such that the attributes bit_from/bit_to are defined correctly.
TECHLIB	9123	Slew threshold difference '%s' is less than 1%.	Error	Slew threshold difference (<Rise>/<Fall>) is less than 1%. This library might not be usable with some of the downstream tools.
TECHLIB	9125	Missing axis name in template definition '%s'.	Error	Missing axis name in template definition <Template name>. Number of template variables in a template should correspond with the number of Index statements.
TECHLIB	9131	Unnamed '%s' found in input technology library. This is incorrect liberty Syntax.	Error	The library does not have pin/cell/library name defined in the library. The input technology library should have pin/cell/library name defined. Check the library correctness.

Valus User Guide

Valus Checks

LIB_SEMANTIC				
Message Prefix	ID	Message	Severity	Description
TECHLIB	9138	The 'default_operation_condition' should be defined after the 'operation_condition' group. Check the library for correctness.	Error	The attribute 'default_operation_condition' should be defined in the library after the operation_condition' group. This is done to ensure the validity of the properties signified by these attributes.
TECHLIB	9144	The statetable for the %s %s is syntactically incorrect. This may be because the number of %s columns (%d) is %s than %d	Error	This message is issued when there is a mismatch in number of entries in statetable and number of entries in the table header for next state/current state/inputs. To fix the issue, re-characterize the library with appropriate settings, such that the statetable is defined correctly.
TECHLIB	9147	The 'default_operating_condition' '%s' is not defined in the library. This will be ignored and nominal operating condition will be used for analysis. To avoid this error, you need to add the missing operating condition definition in the library and then re-run.	Error	This error is issued when the default_operating_condition definition is missing in the library. In such cases, the nominal operating condition is used to proceed with the analysis. However, this might lead to unexpected analysis results and may impact signoff quality. To rectify the error, either re-characterize the library or add the definition of the missing operating condition in the library.

Valus User Guide

Valus Checks

LIB_SEMANTIC				
Message Prefix	ID	Message	Severity	Description
TECHLIB	9149	For cell '%s', the bundle group '%s' (on line %d) does not have the mandatory attribute '%s' present.	Error	This message is issued when mandatory attribute in library is missing. The attribute mentioned is an essential attribute for a bundle group. The library loading cannot proceed without this. To fix the issue, re-characterize the library with appropriate settings, such that all mandatory attributes are defined in the library.
TECHLIB	9156	Syntax error: library header is missing or this file may not be a liberty file.	Error	Syntax error: library header is missing. The input tech file does not contain the library header. Correct the library to contain a valid library header.
TECHLIB	9162	Illegal comments in a statetable %s of the cell %s	Error	This message is issued when some illegal symbols are found in the statetable definition. These are more likely to be unclosed or runaway nested comments.
TECHLIB	9168	Unknown value '%s' encountered for Liberty simple attribute '%s'. Ignoring	Error	Unknown value <value> encountered for Liberty simple attribute <name>. Ignoring. Software does not support the specified value <value> for the simple attribute <name> at present.

Valus User Guide

Valus Checks

LIB_SEMANTIC				
Message Prefix	ID	Message	Severity	Description
TECHLIB	282	Table values less than expected in user defined group '%s'%s". Table index points being truncated	Warning	This message is issued when the number of table values specified in user defined group are less than product of dimensions of index values. To fix the issue, re-characterize the library with appropriate settings, such that the table values are specified correctly.
TECHLIB	302	No function defined for cell '%s'. The cell will only be used for analysis.	Warning	This message is issued when the function of the cell is not defined. This cell will not be used for optimization. To fix the issue, re-characterize the library with appropriate settings, such that the function should be specified.
TECHLIB	311	Can't derive the asynchronous type of input pin '%s' from the function of output pin '%s' in sequential cell '%s	Warning	The asynchronous timing type (preset or clear) of an input-to-output timing arc in a sequential cell is derived by checking if the input is in the support set of the corresponding asynchronous function of that output. If the asynchronous function of the output is not specified, the asynchronous type cannot be determined and therefore the timing type of that timing arc will be set to combinational by default

Valus User Guide

Valus Checks

LIB_SEMANTIC				
Message Prefix	ID	Message	Severity	Description
TECHLIB	313	Property '%s' can not be set on pin '%s' of cell '%s'	Warning	Some attributes are specific to certain pin types (input/output/inout). The reported property may not be applicable to the given pin based on its pin type.
TECHLIB	318	Derating table '%s' already defined, ignored	Warning	Currently the old derating table is not overridden by the new one
TECHLIB	319	Template '%s' already defined, ignored	Warning	Currently the old template is not overridden by the new one
TECHLIB	327	Cell '%s' has more than %d input pins. The cell will only be used for analysis.	Warning	This message is issued when a cell has more than 24 input pins. Such cells are not used during optimization and are used only for analysis.
TECHLIB	357	Test cell '%s' uses '%s' as state variable whereas scan cell uses '%s' in library '%s'; '%s' is used as state variable in test cell	Warning	This message is issued when the state variables defined in the library are not same in the test cell and scan cell. To fix the issue, re-characterize the library with appropriate settings, such that the state variables are defined correctly.
TECHLIB	368	Unable to find the ff/latch '%s' functionality pin for cell '%s'	Warning	This message is issued when the ff/latch functionality pin (IQ/IQn) for cell is undefined. To fix the issue, re-characterize the library with appropriate settings, such that the function of a cell is defined.

Valus User Guide

Valus Checks

LIB_SEMANTIC				
Message Prefix	ID	Message	Severity	Description
TECHLIB	390	No delay threshold points for %s transition specified in library '%s', assuming delay thresholds %.0f%-%.0f%	Warning	This warning is encountered if the following constructs are missing from the library. input_threshold_pct_fall input_threshold_pct_rise output_threshold_pct_fall output_threshold_pct_rise These constructs in the timing library define how the delay of the characterized cells has been deduced. For example a input_threshold_pct_rise of 30 and output_threshold_pct_rise of 60 means the delay value in the tables are the time difference between the input rising to 30% of its values and output rising to 70% of its value. If these are missing, then they are assumed to be 50% for both input and output and rise and fall. To avoid this warning make sure that the timing libraries have the correct values of input and output rise/fall thresholds specified.

Valus User Guide

Valus Checks

LIB_SEMANTIC				
Message Prefix	ID	Message	Severity	Description
TECHLIB	391	No slew threshold points for %s transition specified in library '%s', assuming slew thresholds %.0f%-%.0f% (measured) %.0f%-%.0f% (reported).	Warning	This warning is encountered if the following constructs are missing from the library. slew_lower_threshold_pct_fall slew_upper_threshold_pct_fall slew_lower_threshold_pct_rise slew_upper_threshold_pct_rise These constructs in the timing libraries are the slew thresholds which define how slew values are calculated. For example a slew_lower_threshold_pct_rise of 20 and slew_upper_threshold_pct_rise of 80 means that the time values present in the library were written as if 'measured' at 20% of the signal and 80% of the signal, and the slew is represented as their difference. If these are missing, they are assumed to be 40% -60% (measured) 10%-90% (reported). The measured thresholds are the ones at which timing is actually measured from spice. These can be then stretched to reported thresholds linearly, to generate the data in the timing library. To avoid this warning make sure that the timing libraries have the correct slew threshold values specified.

Valus User Guide

Valus Checks

LIB_SEMANTIC				
Message Prefix	ID	Message	Severity	Description
TECHLIB	395	Library defines an invalid %s threshold for %s transition of %.0f%. Default value of %.0f% is used instead.	Warning	Delay and slew thresholds of 0 or 100% is physically impossible. Rising exponential waveform will never reach 100%. Falling exponential waveform will never reach 0%
TECHLIB	399	No %s threshold for %s transition specified, assuming default value of %.0f%	Warning	The mentioned threshold value is missing from the library and the default value is used to compensate for the missing information. This message occurs only if partial threshold information is specified. For example, delay threshold for rising transition is specified while delay threshold for falling transition is not.
TECHLIB	400	State table for cell '%s' is ignored because it contains multiple references to input map. The input_map attribute can be defined more than once for multi-bit sequential cells only.	Warning	Examples of multiple instances of a sequential device in a single cell include master-slave devices, shift registers or counters. Only simple register/latch banks with explicit bundle specification are supported
TECHLIB	406	Ignoring the specified 'steady_state_current_tristate' table in the non three_state pin/bus/bundle %s of cell %s. The above specification applies only to three_state pins.	Warning	The steady_state_current_tristate can only be specified for three state pins, buses or bundles. If it is specified for a non three-state pin, its value is ignored.

Valus User Guide

Valus Checks

LIB_SEMANTIC				
Message Prefix	ID	Message	Severity	Description
TECHLIB	407	The specified value is less than 0 for %s steady_state_resistance in pin %s of cell %s. Values must be positive float numbers.	Warning	The steady_state_resistance can not take negative values. All such negative values are neglected.
TECHLIB	409	Invalid axis specified for 'iv_lut_template' %s. It must be function of output voltage (iv_output_voltage). Ignoring this template.	Warning	The iv_lut_template must be a single dimensional table. The only axis is the output voltage axis, specified by iv_output_voltage
TECHLIB	411	The lookup table template %s specified in the library is a Scalar lookup template. All of the axis in the template contain only single co-ordinate points, which would not be useful during table lookup.	Warning	A scalar lookup template is one which has only one axis point in each of its axis. Such a lookup template is useless for table lookup.
TECHLIB	417	Multiple definitions of template %s found in the library. Preserving the latest definition.	Warning	If a template is defined multiple number of times at the library level, only the latest definition is preserved.
TECHLIB	418	Number of data points in the table %s in pin %s in cell %s does not match the number of axis points specified in the template.	Warning	The number of data points should preferably match the number of axis points in a template. If this is not the case, table lookups might not be accurate.
TECHLIB	423	Unable to infer polarity for pin %s of cell %s. Positive polarity assumed.	Warning	Couldn't find two different delay values in the table for pair of input slew and output load.

Valus User Guide

Valus Checks

LIB_SEMANTIC				
Message Prefix	ID	Message	Severity	Description
TECHLIB	435	Nominal Voltage for library %s is specified to be 0. This may cause problems with delay calculation	Warning	Nominal voltage of a library defines the voltage at which the library was characterized. A zero value is unrealistic for this attribute.
TECHLIB	436	Attribute 'fanout_load' on '%s' %s of cell '%s' is not defined in the library.	Warning	When a library is loaded, software checks for the presence of fanout_load attribute on input pins of all cells in library automatically. This message is issued when any of the mentioned attribute is missing.
TECHLIB	452	The %s capacitances %s table group %s%s%s for %s transition in pin '%s' of cell '%s' are not in the range of min cap value	Warning	For correct calculation of delay value the capacitances value should be more than or equal to min cap value.
TECHLIB	701	Invalid axis specified for 'lut_template' %s. This axis type should be input_voltage/ output_voltage. Ignoring this template.	Warning	This message is issued when the axis type specified for the lut_template for dc_current group is invalid. The valid axes are input_voltage/ output_voltage. Such template will be ignored. To fix the issue, re-characterize the library with appropriate settings, such that the valid axis type is defined.

Valus User Guide

Valus Checks

LIB_SEMANTIC				
Message Prefix	ID	Message	Severity	Description
TECHLIB	703	Setting '%s' as the default value for the attribute '%s' for the pin '%s' for cell '%s	'Warning	This message is issued when the pg_pin is not defined for the attributes related_power_pin or related_ground_pin of cell. In such cases the first pg_pin group having pg_type primary_power/primary_ground will be considered. To fix the issue, re-characterize the library with appropriate settings, such that the pg_pin is defined.
TECHLIB	712	For the cell '%s', the lowerbound value '%f' is greater than upperbound value '%f' for the attribute '%s'. This is not permitted. The attribute is being ignored.	Warning	This message is issued when the lowerbound value of attributes input_voltage_range/output_voltage_range is greater than the upperbound value of these attributes at cell level. To fix the issue, re-characterize the library with appropriate settings, such that the lowerbound/upperbound values are specified correctly.
TECHLIB	713	For the pin '%s' of cell '%s', the lowerbound value '%f' is greater than upperbound value '%f' for the attribute '%s'. The attribute is being ignored	Warning	This message is issued when the lowerbound value of input_voltage_range and output_voltage_range is greater than upperbound value for the pin of cell. To fix the issue, re-characterize the library with appropriate settings, such that the lowerbound value should be specified lower than the upperbound value.

Valus User Guide

Valus Checks

LIB_SEMANTIC				
Message Prefix	ID	Message	Severity	Description
TECHLIB	716	Number of primary outputs specified as the bundle members do not match with the number of bits in the sequential bank for cell '%s'. The bundle definition needs to be checked.	Warning	This message is issued when there is a mismatch in bundle width and number of bits in the sequential bank. To fix the issue, re-characterize the library with appropriate settings, such that the number of bits should match with bundle width.
TECHLIB	828	The 'pulse_generator' and 'pulse_generator_polarity' attributes are obsolete, as specified over the pin '%s' of the cell '%s'. The 'pulse_clock' attribute should be used instead.	Warning	This message is issued when the attributes pulse_generator and pulse_generator_polarity are specified on the pins of the cell. The pulse_clock attribute should be defined instead of pulse_generator and pulse_generator_polarity, as they are discontinued now. To fix the issue, re-characterize the library with appropriate settings, such that the pulse_clock should be defined.
TECHLIB	905	Found a function attribute specified on pin '%s' with 'interface_timing' attribute set to true on cell '%s'. As per the liberty standard, a function attribute should not be specified for a cell which has interface_timing attribute set to true. This function attribute will be ignored.	Warning	When 'interface_timing: true', the cell is regarded as a complex black box cell instead of a regular standard cell and 'function' attribute should not be specified for a black box cell. Re-characterize the library such that the correct attribute is set on the cell.

Valus User Guide

Valus Checks

LIB_SEMANTIC				
Message Prefix	ID	Message	Severity	Description
TECHLIB	1001	Multiple normalized driver waveform groups without a driver_waveform_name were found. The latest definition would overwrite the earlier ones.	Warning	This message is issued when the normalized driver waveform group is defined multiple times but driver_waveform_name is absent. The latest definition will be retained and rest will be ignored.
TECHLIB	1002	Mismatch found in the number of index points for '%s' in the normalized_driver_waveform and its template. The normalized_driver_waveform data at the group level will be used.	Warning	This message is issued when the number of index points are different in the normalized_driver_waveform group and its template at the library level. Data specified in the template will be ignored and data specified in the normalized_driver_waveform group will be used.
TECHLIB	1004	The normalized_driver_waveform group is specified with only one axis. This may result in accuracy loss during analysis.	Warning	This message is issued when the normalized_driver_waveform is specified with only one axis i.e. either input_net_transition or normalized_voltage only is specified as axis. This may cause inaccuracy in timing analysis. To fix the issue, re-characterize the library with appropriate settings, such that the normalized_driver_waveform should be specified with both axes.

Valus User Guide

Valus Checks

LIB_SEMANTIC				
Message Prefix	ID	Message	Severity	Description
TECHLIB	1103	The number of pins in input_map for the pin '%s' of cell '%s' doesn't match the number of inputs given in the state table. This may cause issues in function inference from state table.	Warning	This message is issued when there is a discrepancy between the number of inputs given in the state table and number of pins in input_map for the pin of the cell. This may cause issues in function inference from state table. To fix the issue, re-characterize the library with appropriate settings, such that the number of inputs in the state table is equal to the number of pins in input_map.
TECHLIB	1107	The phase %d is not supported for retention pin for cell %s, pin %s. The retention pin would be ignored.	Warning	Unrecognized phase/retention_disable_value for state retention power gating pin/retention pin. The cell cannot be used for SRPG synthesis
TECHLIB	1109	Nominal Condition '%s' is not defined in timing library. Assuming default value %.3f. Setting this attribute is recommended as it can impact timing and SI results.	Warning	This message is issued when the nominal condition PVT is not defined in the timing library and thus it assumes the default values. The default value of Process is 1, default value of voltage is 5 and default value of temperature is 25. The nominal condition attributes should be set appropriately.

Valus User Guide

Valus Checks

LIB_SEMANTIC				
Message Prefix	ID	Message	Severity	Description
TECHLIB	1117	Power supply group at line %d has no default_power_rail attribute defined.	Warning	This message is issued when the attribute default_power_rail is not present for power supply group. To fix the issue, re-characterize the library with appropriate settings, such that this attribute is specified for power supply group.
TECHLIB	1120	Input map for bus/bundle %s for cell %s is not defined on all pins.	Warning	This message is issued when the input map for bus/bundle is not defined on all pins of a bus/bundle. To fix the issue, re-characterize the library with appropriate settings, such that the input map for bus/bundle should be defined on all pins at cell level.
TECHLIB	1142	No 'values' attribute found for normalized driver waveform group at line number %d. This is a mandatory attribute for normalized waveform group. This group will be ignored.	Warning	The normalized driver waveform group should have mandatory attribute 'values'. If this attribute is not present the characterized library is incorrect and the library should be re-characterized.
TECHLIB	1161	The library level attribute %s on line %d is defined after at least one cell definition. The attribute will be ignored.	Warning	This message is issued when any library level simple or complex attribute is defined after the cell definition in library file. Since such attributes are applicable for all cells in design, they must be specified before the cell definition.

Valus User Guide

Valus Checks

LIB_SEMANTIC				
Message Prefix	ID	Message	Severity	Description
TECHLIB	1172	The '%s' attribute has been defined more than once for group '%s' on line %d. The attribute should be defined only once.	Warning	The message is issued when multiple values has been defined within a group. For example, index_1 in lu_table_template group is defined more than once. Each attribute should be defined only once per group. To fix the issue, re-characterize the library with appropriate settings.
TECHLIB	1177	'index_%d' defined in '%s' group should have at least '%d' float values. This may lead to undesirable analysis results.	Warning	The message is issued when any index specified in the library has less than the minimum number of float values expected. This may impact accuracy in delay and glitch analysis. To fix this issue, re-characterize the library such that all indices match the expected number of float values.
TECHLIB	1181	The 'variable_%d' is defined but 'index_%d' is not defined for %s '%s' on line %d.	Warning	This message is issued when a variable is defined in the template but its corresponding index is not defined. All variables should have defined indices in the template. To fix the issue, re-characterize the library with appropriate settings such that all indices exist corresponding to the variables in the template definition.

Valus User Guide

Valus Checks

LIB_SEMANTIC				
Message Prefix	ID	Message	Severity	Description
TECHLIB	1187	The cell '%s' on line %d has both pad cell and auxiliary pad cell attributes defined.	Warning	This message is issued when both pad_cell and auxiliary_pad_cell attributes are defined for a cell. The cell can be classified either as a pad cell or an auxiliary pad cell, but not both. To fix the issue, re-characterize the library with appropriate settings, such that the cell is defined with the correct attribute.
TECHLIB	1192	The attribute '%s' has been defined for %s %s '%s' on line %d.	Warning	This message is issued when the attributes function/ state_function/ Clock_gate_out_pin are defined incorrectly on pins with certain direction i.e. function attribute cannot be defined for input pins, state_function cannot be defined for input and internal pins and clock_gate_out_pin cannot be set true for input and internal pins. To fix the issue, re-characterize the library with appropriate settings, such that the attributes function/ state_function/ Clock_gate_out_pin should be defined on correct pins.

Valus User Guide

Valus Checks

LIB_SEMANTIC				
Message Prefix	ID	Message	Severity	Description
TECHLIB	1199	The delay_model attribute specified on line %d should be the first attribute in the library.	Warning	This message is issued when delay_model attribute is specified in library but it is not the first attribute in library. As per Liberty standards, delay_model attribute must be first attribute if technology attribute is not specified in library.
TECHLIB	1202	The %s statement for '%s' on line %d is first defined on line %d. The first definition will be retained.	Warning	The message is issued when more than one definitions of the define_group attribute are defined in the library. To fix the issue, re-characterize the library with appropriate settings, such that only one definition per define_group is present in the library.
TECHLIB	1205	Boolean value '%s' defined for attribute %s on line %d. The acceptable value is true/false. However, the value and the attribute will be honored.	Warning	The message is issued when boolean values are specified for attributes. For example, per liberty guidelines the acceptable values for attributes such as dont_touch and dont_use are only true/false. However, if the library has boolean values specified, the tool honors the val and the attribute, along with this message. To fix the issue, re-characterize the library with appropriate settings, such that the attribute has values specified per the required syntax.

Valus User Guide

Valus Checks

LIB_SEMANTIC				
Message Prefix	ID	Message	Severity	Description
TECHLIB	1209	The area range %f-%f specified for wire_load_from_area attribute defined on line %d of wire_load_selection group '%s' overlaps with the range %f-%f of the wire_load_from_area attribute defined on line %d.	Warning	The message is issued when area range defined in one wire_load_from_area attribute of the wire_load_selection group, overlaps the area range defined in the another wire_load_from_area attribute. The range of area defined should be mutually exclusive. To fix the issue, re-characterize the library with appropriate settings, such that the area range is defined correctly.
TECHLIB	1210	The %s attribute(s) is not defined for %s group on line %d, even though %s attributes(s) are defined.	Warning	The message is issued when the ff/ff_bank attributes such as preset, clear, clear_preset_var1 and clear_preset_var2 are not all defined in the library when either of the attributes are defined. For example, in case one, if preset and clear are not defined when clear_preset_var1 or clear_preset_var2 are defined or in case two, if when both clear and preset are defined, but clear_preset_var1 and/or clear_preset_var2 are not defined. To fix the issue, re-characterize the library with appropriate settings, such that the attributes are defined per the requirement.

Valus User Guide

Valus Checks

LIB_SEMANTIC				
Message Prefix	ID	Message	Severity	Description
TECHLIB	1226	For the cell '%s', inout %s '%s' on line %d has missing three_state function.	Warning	This message is issued when the three_state function is not specified for the inout pin of the cell. To fix the issue, re-characterize the library with appropriate settings, such that the three_state function should be specified.
TECHLIB	1228	The pin name '%s' defined on line %d is part of the list of reserved words for sequential cells. However, the pin name is honored.	Warning	This message is issued when a sequential cell has pin names that are part of the reserved words for sequential cells. List of the reserved words is following: clocked_on_also, data_in, enable, enable_also, preset, clear, force_00, force_01, force_10, force_11. To fix the issue, re-characterize the library with appropriate settings, such that the correct pin name is specified.
TECHLIB	1238	The %s group on line %d can have only %d parameters. The extra parameters will be ignored.	Warning	This message is issued when the extra parameters are defined for the ff/latch and ff_bank/latch_bank. For ff/latch parameter limit is 2 and ff_bank/latch_bank parameter limit is 3, rest will be ignored. To fix the issue, re-characterize the library with appropriate settings, such that the parameter should be defined within the limit.

Valus User Guide

Valus Checks

LIB_SEMANTIC				
Message Prefix	ID	Message	Severity	Description
TECHLIB	1252	The '%s' attribute is defined for '%s %s %s' of cell '%s'. This attribute will be ignored.	Warning	The input_voltage attribute can be defined only for input or inout pins, bus or bundle and output_voltage attribute can be defined only for output or inout pins, bus or bundle. Incorrectly applied attributes will be ignored during timing analysis. To fix this issue, re-characterize the library with appropriate settings.
TECHLIB	1254	The '%s - %s' table of the '%s' group on the '%s' %s '%s' does not have a template that uses '%s' as a variable. Ensure that the templates used are in accordance with the direction of the pin.	Warning	This message is issued when the input_net_transition and total_output_net_capacitance tables defined with incorrect direction of the pin for the power/rise_power/fall_power groups. For 1-d tables input pin should have input_net_transition. For 2-d tables input and output pins should be defined with both input_net_transition and total_output_net_capacitance tables and inout pins can be defined with any of the attribute. To fix the issue, re-characterize the library with appropriate settings, such that the templates used are defined in accordance with the direction of pin.

Valus User Guide

Valus Checks

LIB_SEMANTIC				
Message Prefix	ID	Message	Severity	Description
TECHLIB	1267	Pin '%s' does not exist in cell '%s' or it is the part of the function string mapped to a latch/ff element, which has been ignored due to the cell having 'interface_timing' set to true.	Warning	The message is issued when the cell is a black box cell but it contains function attribute or ff group.
TECHLIB	1269	Pin '%s' does not exist in cell '%s'. The functions expression will be ignored.	Warning	This message is issued when the pin is not defined in the cell and referred in the function. Function statement will be ignored. This may impact accuracy in delay and glitch analysis. To fix the issue, re-characterize the library with appropriate settings, such that the pin is defined in the cell.
TECHLIB	1270	The '%s' attribute/group has been defined more than once. Only the %s occurrence of the attribute will be retained.	Warning	The message is issued when there are multiple definitions for an attribute/group. Each attribute/group should be defined only once. To fix this issue, re-characterize the library with the appropriate settings.
TECHLIB	1272	The '%s' %s has been defined more than once on %s. Only the %s occurrence of the %s will be retained.	Warning	The message is issued when there are multiple definitions for an attribute/group. Each attribute/group should be defined only once. To fix this issue, re-characterize the library with the appropriate settings.

Valus User Guide

Valus Checks

LIB_SEMANTIC				
Message Prefix	ID	Message	Severity	Description
TECHLIB	1277	The %s '%s' has been defined for %s %s '%s'. '%s' cannot be defined at this level and is being ignored.	Warning	This message is issued when any attribute/group is specified at incorrect level/context. Such attributes/groups are ignored. This may impact accuracy in delay and glitch analysis. To fix the issue, recharacterize the library such that attributes/groups are specified correctly.
TECHLIB	1278	The mandatory %s '%s' is missing for %s %s '%s'.	Warning	This message is issued when any mandatory attribute/groups is missing in library. This can cause issues later in tool applications.

Valus User Guide

Valus Checks

LIB_SEMANTIC				
Message Prefix	ID	Message	Severity	Description
TECHLIB	1286	%s data has been defined on one or more cells in the library, but is missing for cell '%s'.	Warning	This message is issued when the data for CCS timing/ ECSM/LVF/FIT data is defined for one or more cell but missing on other cell in the same library. The valid slew groups are rise_transition, fall_transition, retain_rise_slew, retain_fall_slew, ocv_sigma_rise_transition, ocv_sigma_fall_transition, ocv_sigma_cell_rise, ocv_sigma_cell_fall, ocv_sigma_rise_constraint and ocv_sigma_fall_constraint. To fix the issue, re-characterize the library with appropriate settings, such that the CCS timing/ECSM waveform/LVF/ FIT data is defined corresponding to all cell within the same library.

Valus User Guide

Valus Checks

LIB_SEMANTIC				
Message Prefix	ID	Message	Severity	Description
TECHLIB	1287	For point '%d' in index_%d (%s) in '%s' table on %s of cell '%s', values %f to %f are not monotonically %s across the index_%d (%s)	Warning	This message is issued when em_max_toggle_rate table values (M x N), for a fixed point of index_1/index_2 of table indices do not monotonically increase/decrease across the index_2/index_1 axis. For example, in the em_max_toggle_rate table, for a fixed value of slew index, the table values are not monotonically decreasing across the load axis or for fixed value of load index, the table values are not monotonically increasing across the slew axis. To fix the issue, re-characterize the library with appropriate settings, such that table values are monotonic as per liberty standard.
TECHLIB	1290	The cell '%s' is a phase-locked loop cell but '%s' attribute is not defined on any pin of this cell.	Warning	This message is issued when attributes is_pll_reference_pin/ is_pll_feedback_pin/ is_pll_output_pin, are not defined for a pin of cell which is a phase-locked loop cell. These attributes are mandatory for phase-locked loop cell. To fix the issue, re-characterize the library with appropriate settings, such that the correct attributes are defined on the pin/cell.

Valus User Guide

Valus Checks

LIB_SEMANTIC				
Message Prefix	ID	Message	Severity	Description
TECHLIB	1291	The '%s' attribute has been defined on more than one pin for cell '%s'. First definition at line %d would be retained and rest will be ignored.	Warning	This message is issued when the attributes is_pll_reference_pin/ is_pll_feedback_pin are defined on more than one pin for a phase-locked loop cell. The same attribute cannot have multiple definition. First definition will be retained and rest will be ignored. To fix the issue, re-characterize the library with appropriate settings, such that the attributes are defined correctly.
TECHLIB	1292	The cell '%s' is not a phase-locked loop cell but has '%s' attribute specified for %s.	Warning	This message is issued when attributes is_pll_reference_pin/ is_pll_feedback_pin/ is_pll_output_pin , have been defined for a pin of cell which is not a phase-locked loop (PLL) cell. Only phase-locked loop pins are expected to have phase-locked loop attributes defined on them. To fix the issue, re-characterize the library with appropriate settings, such that the correct attributes are defined on the pin/cell.
TECHLIB	1295	Voltage (%0.3f) specified in nom_voltage attribute does not match the voltage (%0.3f) specified in default_operating_conditions (%s) group.	Warning	The nominal voltage and the voltage specified in the default operating condition of the library must be identical.

Valus User Guide

Valus Checks

LIB_SEMANTIC				
Message Prefix	ID	Message	Severity	Description
TECHLIB	1308	The user defined group '%s' defined on parent group '%s' has missing 'define_group' statement at library level. However, group data will be honored.	Warning	This message is issued when certain user defined groups are used but not defined at library level using define_group statements. In such cases, Cadence tools will accept the data but other tools may not do so. It is recommended that library is re-characterized such that undefined user-defined groups are not used.
TECHLIB	1310	The %s group contains values which are not monotonically %s from %f to %f for %s %f and %s %f in group %s.	Warning	This message is issued when delay table values (PxQxR), for fixed point of two of table indices do not monotonically increase/decrease across the third axis. For example, in the cell_rise table, for a fixed value of load index and load index1, the table values are not monotonically increasing across the slew axis. To fix the issue, re-characterize the library with appropriate settings, such that table values are monotonic as per Liberty standard.

Valus User Guide

Valus Checks

LIB_SEMANTIC				
Message Prefix	ID	Message	Severity	Description
TECHLIB	1311	For point '%d' in index_%d (%s) in '%s' table on %s of cell '%s', values %f to %f are not monotonically %s across the index_%d (%s)	Warning	This message is issued when transition table values (M x N), for a fixed point of index_1/ index_2 of table indices do not monotonically increase/ decrease across the index_2/ index_1 axis. For example, in the rise_transition table, for a fixed value of slew index, the table values are not monotonically decreasing across the load axis or for fixed value of load index, the table values are not monotonically increasing across the slew axis. To fix the issue, re-characterize the library with appropriate settings, such that table values are monotonic as per Liberty standard.
TECHLIB	1312	The %s group contains values which are not monotonically %s from %f to %f for %s %f and %s %f in group %s.	Warning	This message is issued when transition table values (PxQxR), for fixed point of two of table indices do not monotonically increase/ decrease across the third axis. For example, in the rise_transition table, for a fixed value of load index and load index1, the table values are not monotonically increasing across the slew axis. To fix the issue, re-characterize the library with appropriate settings, such that table values are monotonic as per Liberty standard.

Valus User Guide

Valus Checks

LIB_SEMANTIC				
Message Prefix	ID	Message	Severity	Description
TECHLIB	1320	The user-defined attribute '%s' is not present in any of the '%s' group. This attribute is required for Tempus if SPICE correlation of ROP glitch needs to be performed. The missing attribute does not affect SI delay or glitch analysis.	Warning	This message is issued when the user-defined attribute related_spice_node attribute is not present in library. To fix the issue, re-characterize the library with appropriate settings, such that the required user-defined attributes are defined and used properly.
TECHLIB	1321	The user-defined attribute '%s' is not specified in the group '%s' on %s of cell '%s'.	Warning	This message is issued when the related_spice_node/ load_cap_rise/load_cap_fall attribute is defined in library but is not specified on corresponding group ccsn_first_stage/ ccsn_last_stage. To fix the issue, re-characterize the library with appropriate settings, such that the required user-defined attributes are specified properly.
TECHLIB	1324	The specified type '%s' of user-defined attribute '%s' is invalid. It will be taken as string as default.	Warning	The type of user-defined attribute should one of these: integer, boolean, string and float. To fix the issue, re-characterize the library with appropriate settings, such that the required user-defined attributes are defined properly.
TECHLIB	1327	The 'retention_pin' attribute on the pin '%s' in cell '%s' is honored, even though the 'retention_cell' attribute is missing at cell level.	Warning	It is advisable to have the retention_cell attribute set on a cell having a retention pin. The retention_cell simple attribute identifies a retention cell.

Valus User Guide

Valus Checks

LIB_SEMANTIC				
Message Prefix	ID	Message	Severity	Description
TECHLIB	1329	The attribute '%s' on the %s in cell '%s' is missing, even though the cell has multiple supply voltages.	Warning	This message is issued when the attributes related_power_pin or related_ground_pin is not specified on the pin in cell and cell is marked as MSV cell. These are mandatory attributes for MSV cell. To fix the issue, re-characterize the library with appropriate settings, such that the related_power_pin or related_ground_pin is specified on the pin in cell.
TECHLIB	1330	The attribute '%s' on the %s in cell '%s' is missing, it will be considered as standard cell.	Warning	This message is issued when the attributes related_power_pin or related_ground_pin is not specified on the pin in cell and cell has multiple pg pins but only one pg pin is used as related_power_pin. Such cell will be considered as standard cell. To fix the issue, re-characterize the library with appropriate settings, such that the related_power_pin or related_ground_pin is specified on the pin in cell

Valus User Guide

Valus Checks

LIB_SEMANTIC				
Message Prefix	ID	Message	Severity	Description
TECHLIB	1365	The %s vector group for %s has a duplicate definition (line '%d') on %s and cell '%s'. The duplicate definition will be ignored.	Warning	This message is issued when the vector group specified in output_voltage_rise/ output_voltage_fall group for ccsn_first_stage/ ccsn_last_stage group or output_current_rise/ output_current_fall for timing group is defined more than once with the same value of 'total_output_net_capacitance' and 'input_net_transition'. Multiple definition of vector group with same values of 'total_output_net_capacitance' and 'input_net_transition' is not allowed. To fix the issue, re-characterize the library with appropriate settings, such that the vector group defined is unique.
TECHLIB	1397	The cell '%s' doesn't have ff/ latch group with pin reference names but there is pin with reference_input attribute.	Warning	This message is issued when the cell has pin with reference_input attribute but it doesn't have any ff/latch group with reference pin names. This may impact accuracy in delay and glitch analysis. To fix the issue, re-characterize the library with appropriate settings, such that the correct combination is used.

Valus User Guide

Valus Checks

LIB_SEMANTIC				
Message Prefix	ID	Message	Severity	Description
TECHLIB	1421	Attribute 'max_capacitance' on '%s' %s of cell '%s' is not defined in the library.	Warning	This message is issued when the attribute max_capacitance is not defined on output/inout pin of cell in the library. To fix the issue, re-characterize the library with appropriate settings, such that the attributes are defined in the cell pin.
TECHLIB	1422	Attribute 'max_transition' on '%s' %s of cell '%s' is not defined in the library.	Warning	This message is issued when the attribute max_transition is not defined on any pin of cell in the library. To fix the issue, re-characterize the library with appropriate settings, such that the attributes are defined in the cell pin.
TECHLIB	1423	Attribute 'max_fanout' on '%s' %s of cell '%s' is not defined in the library.	Warning	This message is issued when the attribute max_fanout is not defined on output/inout pin of cell in the library. To fix the issue, re-characterize the library with appropriate settings, such that the attributes are defined in the cell pin.

Valus User Guide

Valus Checks

LIB_SEMANTIC				
Message Prefix	ID	Message	Severity	Description
TECHLIB	9018	Nominal Condition '%s' is not defined in timing library. Assuming default value '%f'. Setting this attribute is recommended as it can impact timing and SI results.	Warning	This message is issued when the attribute '<NomCondName>' is not defined in timing library. This attribute is a must requirement in case the user uses the set_op_cond/setOpCond or -opcond command with delay corners in MMMC. This value would be used to derate the library values to the operating conditions of the design. To avoid this warning, make sure that '<NomCondName>' is specified in the timing library.
TECHLIB	9064	Missing delay_model statement in the library '%s'. %s delay model is assumed.	Warning	The delay_model statement is missing in the library. Assuming default model

Valus User Guide

Valus Checks

LIB_SEMANTIC				
Message Prefix	ID	Message	Severity	Description
TECHLIB	9108	'%s' not specified in the library, using .lib default of '%s'.	Warning	This warning is encountered if the construct (<ThresholdConstruct>) is missing from the library. This construct in the timing library defines how the delay of the characterized cells has been deduced. For example a input_threshold_pct_rise of 30 and output_threshold_pct_rise of 60 means that the delay value in the tables are the time difference between the input rising to 30% of its values and output rising to 70% of its value. If this is missing, then a value of (<PercentageValue>) is assumed. To avoid this warning make sure that the timing libraries have the correct values of (<ThresholdConstruct>) specified.
TECHLIB	9122	Multiple statetables found in cell %s. Only the last statetable definition will be preserved.	Warning	Multiple statetables found in cell <Cell>. Only the last statetable definition will be preserved. A cell having multiple statetables is considered to be an incorrect style of modelling. Thus only the latest statetable definition is preserved.
TECHLIB	9127	Multiple '%s' were detected at (Line %d). Using the last value.	Warning	The last value of the multiple definitions would be used.

Valus User Guide

Valus Checks

LIB_SEMANTIC				
Message Prefix	ID	Message	Severity	Description
TECHLIB	9128	Lookup tables are only supported in 'table_lookup' libraries. Table seen at line %d. Verify that the delay_model for this library is table_lookup.	Warning	Table can be seen when delay model type is defined as table_lookup. The library needs to be reviewed.
TECHLIB	9134	Multiple unit definitions detected (Line %d). A library can only have one definition for capacitive_load_units. First definition would be retained.	Warning	The first value of the multiple unit definitions would be used.
TECHLIB	9139	'%s' not found in the library. Assuming '%s'	'Warning	The attribute leakage_power_unit is found missing in the library. Assuming it to be 1nW. To avoid this assumption, specify (leakage_power_unit) in the library.
TECHLIB	9142	Mismatch between number of values(%d) defined in template '%s' and number of values(%d) characterized in table, on line no. %d.	Warning	If any library template has n*m entries, then its corresponding table entries values should be n*m.

Valus User Guide

Valus Checks

LIB_SEMANTIC				
Message Prefix	ID	Message	Severity	Description
TECHLIB	9145	Cell '%s' has interface_timing attribute set to true and it also has statetable defined. The statetable will be ignored.	Warning	The library cell does not require to have a statetable defined if the cell definition has interface_timing attribute set to true. In the case both interface_timing attribute and statetable are defined for a cell, the state table will be ignored and interface_timing attribute would be honored. In this case, to enable tool honor the statetable, either set the interface_timing attribute to false in the cell definition or remove the attribute from cell definition
TECHLIB	9153	Duplicate definition for attribute '%s' encountered. The last definition will be retained.	Warning	Duplicating definition for attribute encountered. The last definition will be retained.
TECHLIB	9161	Empty Group encountered; Ignoring.	Warning	This message is issued when the group is defined without any attributes. To fix the issue, re-characterize the library with appropriate settings, such that the defined group is not empty.
TECHLIB	9167	Found complex attribute in user defined group. As per liberty specification, complex attributes are not allowed in user defined groups. The attribute will be ignored.	Warning	Found complex attribute in user defined group. As per liberty specification, complex attributes are not allowed in user defined groups. The attribute will be ignored.

Valus User Guide

Valus Checks

LIB_SEMANTIC				
Message Prefix	ID	Message	Severity	Description
TECHLIB	9170	Multiple definitions of the fanout length '%d' for the wire_load group. The first definition would be retained	Warning	This message is issued when same fanout length is defined more than once with different values in wire_load group. The same fanout length cannot have multiple definition. First definition will be retained and rest will be ignored.
TECHLIB	517	Both ff/latch and statetable are present in cell '%s'. Only ff/latch will be used to model sequential cell behavior and statetable will be ignored	Info	This message is issued when the cell is specified with both ff/latch and statetable then ff/latch will be used to model sequential cell behavior and stateable will be ignored.
TECHLIB	520	The slew_derate_from_library is not specified in the library. Default value of 1 is used.	Info	If slew_derate_from_library is not specified, then based on the liberty specifications, the slew_derate is default to one. This means that the slew threshold specified in the library is used for both measured and reported slew threshold
TECHLIB	1178	For 'index_%d', the number of float values in '%s' '%s' (%d), on line %d and '%s' (%d) group on line %d, do not match. Float values in '%s' group will be used.	Info	The message is issued when number of float values for an index in the template is different than the float values for the same index in the group. Index defined at the group level will be used. To avoid the issue, re-characterize the library to have same number of float values in both template and group.

Valus User Guide

Valus Checks

LIB_SEMANTIC				
Message Prefix	ID	Message	Severity	Description
TECHLIB	1326	Assuming the pin '%s' in cell '%s' to have '%s' with the default value '%s'.	Info	As per Liberty Standard, the default value for restore_edge_type attribute is leading.

LIB_MISC

The LIB_MISC checks are performed during the validation of various parameters of a library. This category includes checks related to file read/write permission, incorrect command line options, invalid library format, and so on.

LIB_MISC				
Message Prefix	ID	Message	Severity	Description
VALUS	40	No write permission in the directory. Check the write permission and run the tool.	Error	This message is issued when the software cannot open a file for writing. It could be the unix write permission issue.
VALUS	1032	The set_message -id string '%s' is incorrectly specified. Check the tool help for usage of command.	Error	This message is issued when command set_message is incorrectly specified. The example format is 'set_message -id TECHLIB-436 -limit 10'.
TECHLIB	246	Error encountered while reading cell '%s' in library '%s'	Error	This message is issued when the error is already issued while reading a cell in library. Refer the log file for the failure reason.
TECHLIB	281	Attempt to read .lib library '%s' failed	Error	This message is issued when the error is already issued while reading a library. Refer the log file for the failure reason.
TECHLIB	622	Could not read %s. Check the file path and read permissions.	Error	This message is issued when software cannot open a file to read. It could be the Unix read permission issue.

Valus User Guide

Valus Checks

LIB_MISC				
Message Prefix	ID	Message	Severity	Description
TECHLIB	1141	Cell '%s' redefined in library. The first cell definition will be retained, rest will be ignored.	Error	Cell re-definitions in a library is not allowed. In case of multiple definitions for the same cell name, only the first definition is retained and rest are ignored. In order to retain all cell definitions, the cell names must be unique. Modify the cell definition in library by changing name of cells such that all the cells are distinct
TECHLIB	1145	Message with id '%d' not found.	Error	This message is issued when valid message id is not specified.
TECHLIB	1230	Failed to parse '%s' expression for %s group on line number %d. %s	Error	This message is issued when software encounters an error in parsing the boolean expressions like next_state/ function/when/when_start/ when_end/sdf_cond etc.. To rectify the issue, fix the error issued by correcting the Boolean expression definition.
TECHLIB	1242	The library format in the file '%s' is not supported by software. Check the format.	Error	The software is not able to recognize the library format of file. Liberty timing libraries or side-file files must be in either text-readable format or Cadence compiled .ldb format.
TECHLIB	1276	Cannot find timing library file '%s' specified with '%s'. Check unix permissions.	Error	This message is issued when the specified library is not found

Valus User Guide

Valus Checks

LIB_MISC				
Message Prefix	ID	Message	Severity	Description
TECHLIB	1370	Identified inconsistency between function and timing sense for timing arc on pin '%s' of cell '%s'	Error	This message is issued when the timing sense defined for timing arc is not consistent with the function of pin. For example, in case of inverter, which has function '!A' then arc should have a negative_unate timing sense. To fix the issue, re-characterize the library with appropriate settings, such that the timing_sense and function are consistent.
TECHLIB	1419	The expression '%s' specified for attribute %s for %s is invalid. %s. The attribute will be ignored.	Error	This message is issued when software encounters an error in parsing the boolean expressions like power_down_function etc.. To rectify the issue, fix the error issued by correcting the Boolean expression definition.
TECHLIB	1472	The library attribute '%s' preexists in library '%s', with a value of '%f'. Preexisting library attributes cannot be overwritten by user-level annotations. The attribute value has not been changed.	Error	This message is issued when the sigma constraint folding factor being annotated to the specified library using command 'set_socv_constraint_config', preexists in the specified library. The preexisting sigma constraint folding factors cannot be redefined, thus, attribute value will not be changed.

Valus User Guide

Valus Checks

LIB_MISC				
Message Prefix	ID	Message	Severity	Description
TECHLIB	9164	The group name is larger than %d characters. Such large names would cause problems in downstream tools. Rename the group to remove this error.	Error	The group name cannot be larger than the limit specified. The downstream tools might not be able to handle such large names. Rename the group to meet the size limit.
TECHLIB	1297	LDB generation could not complete. Refer to previously issued error messages in the session.	Warning	This message is issued when library has some serious errors which prevent LDB file generation. To fix the issue, re-characterize the library to fix the error messages issued in the session.
TECHLIB	1420	The receiver capacitance group specified for '%s' input transition is invalid for '%s' timing arc defined for %s in cell '%s'.	Warning	This message is issued when a timing arc had receiver capacitance group specified which is not invalid. The validity check is done on the basis of valid input pin transitions for the given timing group.
TECHLIB	927	Message <%s> has exceeded the message display limit of '%d'. Use 'set_message -no_limit -id list_of_msgIDs' to reset the message limit.	Info	This message is issued when the message has exceeded the message display limit which defaults to 20 and can be changed with the help of command 'set_message -no_limit -id list_of_msgIDs'. For example: set_message -id TECHLIB-436 -limit 5.

Valus User Guide

Valus Checks

LIB_UI

The LIB_UI checks are performed during the validation of various parameters of a library.

LIB_UI				
Message Prefix	ID	Message	Severity	Description
VALUS	1042	Invalid message ID '%s'. Specify only one message ID at a time. For example, '%s VALUS-1005	Error	This message is issued when incorrect or more than one message id is specified in detail help.
VALUS	1043	write_lib will not write out '%s' library as there are no modifications done.	Error	This message is issued when the library is not modified and user is trying to write library using command write_lib.
VALUS	1044	The input library '%s' is not in the correct format to modify. The LDB or encrypted libraries are not supported.	Error	This message is issued when the library cannot be modified due to its format. The LDB or encrypted libraries are not supported. To fix the issue, specify the library in correct format.
VALUS	1045	The input library '%s' is not an ETM/PTM library. The library other than ETM/PTM are not allowed to modify.	Error	This message is issued when the library cannot be modified as it is not an ETM/PTM library. The library other than ETM/PTM are not allowed to modify. To fix the issue, specify the correct library.

Valus User Guide

Valus Checks

LIB_UI				
Message Prefix	ID	Message	Severity	Description
VALUS	1047	Invalid name '%s' specified with '%s' option.	Error	This message is issued when the name of library/cell/pin/timing_arc/pg_pin specified is invalid. For example, the library name specified may not match the name of the library being updated. To fix the issue, specify the correct name. This message is also issued when pin name specified in format fromPin_toPinwhen group_name is timing_arc, does not exists in library or when the pin/pg_pin name specified is not present in the library.
VALUS	1048	Attribute will be %s in all occurrence of group '%s' specified with option '-group' with command '%s'. Specify name of pin in option '-name' to add the group for specific pin.	Warning	The attribute specified in command 'add_attribute/delete_attribute' will be added to or deleted from all occurrence of group defined with option '-group'. Specify the name of pin in option '-name' to add/delete the group for specific pin.
VALUS	1049	Invalid value '%s' for '-value' option.	Error	The message is issue when the value specified for '-value' option of 'add_attribute' or 'add_power_info' command
VALUS	1050	The ETM library '%s' consists more than one cell. Rerun the command with correct ETM library.	Error	This message is issued when the ETM library cannot be modified as it is having more than one cell. ETM library can have only one cell. To fix the issue, specify the correct library.

Valus User Guide

Valus Checks

LIB_UI				
Message Prefix	ID	Message	Severity	Description
VALUS	1051	The %s '%s' specified for option '%s' with command '%s' does not exist in library and is not valid.	Error	This message is issued when incorrect pin name or pin name with improper format is passed to '-to' or '-from' or '-ref' option of 'add_timing_info' or 'add_power_info' or 'add_noise_info' or 'add_attribute' or 'copy_attribute' command
VALUS	1052	Command '%s' cannot be executed as no library is read in the session.	Error	This message is issued when no library is read in the current session. To fix the issue, read the library which needs to be modified using command 'read_libs'.
VALUS	1053	Command 'update_library' is mandatory in case multiple libraries are read in the session to specify which library needs to be modified.	Error	This message is issued when multiple libraries are read in the current session and update_library command is not specified. To fix the issue, run update_library command and specify the library name which needs to be modified.

Valus User Guide

Valus Checks

LIB_UI				
Message Prefix	ID	Message	Severity	Description
VALUS	1054	Options '%s' and '%s' are mutually exclusive for command '%s'	Error	This message is issued to notify that options '-rise_transition -fall_transition -cell_rise -cell_fall' and '-rise_constraint -fall_constraint' are mutually exclusive for command 'add_timing_info'. The options '-ref' and '-from' are mutually exclusive for command 'add_noise_info'. The options '-ref' and '-propagated_noise_*' are mutually exclusive for command 'add_noise_info'.
VALUS	1055	The attribute '%s' defined on pin '%s' which is %s pin. The attribute '%s' cannot be added on this pin.	Error	This message is issued when the attribute is specified on incorrect pin. The correct pin type for the attributes are: max_transition can be defined on input or output pin but not on an internal pin, min_pulse_width_low/ min_pulse_width_high can be defined on input/internal pin, min_period can be defined on clock pin, max_capacitance can be defined on output or inout pin but not on an internal pin and capacitance can be defined on input, output, inout, or internal pin. To fix the issue, specify the attribute correctly.

Valus User Guide

Valus Checks

LIB_UI				
Message Prefix	ID	Message	Severity	Description
VALUS	1056	The '%s' is already present for '%s' and would be over-written.	Warning	This message is issued when attribute to be added/copied with command add_attribute/copy_attribute is already present and would be over-written.
VALUS	1058	'%s' pin specified for option '%s' is invalid for '%s' arc. The correct pin type for option '%s' is '%s'.	Error	The correct pin type for combinational/trigger arcs is output/bidi pin and for constraint arcs is input pin or inout pin.
VALUS	1059	None of the valid attributes i.e. '%s', have been specified with timing type '%s'.	Error	This message is issued when the none of the valid attributes are not given with the specified timing_type.
VALUS	1060	Mode definition specified with command option '%s -mode' does not exist in the library. Specify valid mode condition.	Error	This message is issued when mode definition specified with command 'add_timing_info/add_power_info -mode' is not present in the library. Mode definition already deinfed in library can be specified with this option.
VALUS	1062	The '%s' is specified for incorrect group '%s' with command add_attribute.	Error	This message is issued when the attribute is specified for incorrect group with command add_attribute. The valid attributes for group 'pin' are minimum_pulse_width_low, minimum_pulse_width_high, min_period, max_transition, max_capacitance and capacitance and for groups 'cell' is 'cell_leakage_power'. To fix the issue, specify the correct group name.

Valus User Guide

Valus Checks

LIB_UI				
Message Prefix	ID	Message	Severity	Description
VALUS	1064	The attribute '%s' is already present on '%s' and would be over-written.	Warning	This message is issued when attribute to be added/copied with command add_attribute/ copy_attribute is already present and would be over-written.
VALUS	1065	The group '%s' cannot be specified with command '%s'.	Error	This message is issued when incorrect group is specified for the given command. Valid groups for command 'add_attribute/ delete_attribute': library, cell, pin and timing_arc, for command 'copy_attribute' : pin and timing_arc.
VALUS	1066	The attribute '%s' with '%s' group has not been specified using the 'define_attribute' command and thus, cannot be copied/added/deleted.	Error	This message is issued when an attribute is not defined by the user, is attempted to be copied/added/deleted, using the copy_attribute/ add_attribute/delete_attribute command. To fix this issue, attribute should be predefined before adding/copying.
VALUS	1067	The '%s' attribute with '%s' group is already defined using 'define_attribute' command.	Error	This message is issued when the attribute with same group_name is already defined using define_attribute command. For example, attribute is defined as 'define_attribute -name abc -group pin -type int' thus cannot be reefined as 'define_attribute -name abc -group pin -type float'.

Valus User Guide

Valus Checks

LIB_UI				
Message Prefix	ID	Message	Severity	Description
VALUS	1068	The attribute '%s' specified with command 'delete_attribute' to delete is not present in the library.	Error	This message is issued when the attributes defined with command 'delete_attribute' to delete is not present in the library.
VALUS	1069	'%s' is the valid and mandatory attribute for power_arcs, to be specified on '%s' pin.	Error	This message is issued when a power arc is being added on an output/inout pin without a related pin.
VALUS	1070	The option '-name' is mandatory in case of adding/deleting attributes on %s group.	Error	This message is issued when an attribute is added/deleted using add_attribute/delete_attribute command on timings arcs/operating_conditions/generated_clock/pg_pin groups without specifying the corresponding pin/group using -name option.
VALUS	1071	The pin names specified with '-to' and '-from' option differ, in the command '%s'. These must be the same for arc with timing_type '%s'.	Error	This message is issued, when different pin names are specified with '-to' and '-from' option with command add_timing_info. These must be the same for arcs with timing_type 'minimum_period' and 'min_pulse_width'.
VALUS	1072	The attribute name '%s' specified to be copied using 'copy_attribute' command, does not exist on the reference pin '%s' specified with the option '-reference'.	Error	This message is issued, when the attribute to be copied is not present on the reference pin specified with the option '-reference' while using 'copy_attribute' command.

Valus User Guide

Valus Checks

LIB_UI				
Message Prefix	ID	Message	Severity	Description
VALUS	1073	The attribute '%s' cannot be %s, as no timing arc exists from pin '%s' to pin '%s'.	Error	This message is issued, when the attribute cannot be added/ deleted as the timing arc between the from and to pins specified, does not exist.
VALUS	1074	The attribute '%s' is not supported with '%s' power arcs.	Error	This message is issued, when the attribute cannot be added with the power arc type specified with the command 'add_power_info'.
VALUS	1075	The '%s' power arc can not be added at the '%s' level.	Error	This message is issued, when a power group is added at an incorrect level. 'internal_power' can only be added at pin level and 'leakage_power' can only be added at cell level.
VALUS	1076	The option '%s' is mandatory while adding an internal power arc.	Error	This message is issued, when an internal power group is added without specifying the to pin through '-to' option, using command 'add_power_info'.
VALUS	1077	The attribute '%s' is specified with negative value with command option 'add_attribute -value'.	Error	This message is issued when the value specified for attribute with command option add_attribute -value option is negative. Attributes area, default_max_transition, default_max_capacitance and fanout_load cannot have negative value.

Valus User Guide

Valus Checks

LIB_UI				
Message Prefix	ID	Message	Severity	Description
VALUS	1078	The value '%s' provided with '%s' option of command '%s' is invalid.	Error	This message is issued when the value specified with '-name' option of command add_attribute/delete_attribute is invalid. The value to be provided with this option should be in the syntax 'toPin_fromPin'.
VALUS	1079	The voltage_map for voltage_name '%s' does not exist in the library '%s', thus cannot be added.	Error	This message is issued when the voltage_name specified with '-value' option of command add_attribute is invalid. The voltage_name provided should have a corresponding voltage_map in the library.
VALUS	1080	The attribute '%s' is not valid attribute for internal power on '%s' pin. Remove the option and rerun the 'add_power_info' command.	Error	This message is issued when the option 'related_pin' is specified while adding an internal power arc on an 'input' pin, which is not valid. This option should be removed and command can be rerun.
VALUS	1081	Can not add 'default_operating_condition' attribute to library as the operating condition '%s' does not exist in library '%s'.	Error	This message is issued when the undefined operating_condition is specified with '-value' option of command add_attribute is invalid. The operating condition should exist in the library for it to make default_operating_condition.

Valus User Guide

Valus Checks

LIB_UI				
Message Prefix	ID	Message	Severity	Description
VALUS	1082	Arguments given to 'voltage_map' in 'add_attribute' command must be exactly two, having string name and its float value, thus '%s' given with '-value' option is invalid and cannot be added.	Error	This message is issued when the voltage_map specified with '-value' option of command add_attribute is invalid. The voltage_map must have exactly two arguments.
VALUS	1083	The '%s' group is specified for incorrect parent group '%s' with command add_group.	Error	This message is issued when group is specified for incorrect parent group with command add_group. The valid groups for parent group 'cell' are 'pg_pin' and 'generated_clock' and for parent group 'library' is 'operating_conditions'. To fix the issue, specify the correct parent group name.
VALUS	1084	The '%s' name specified with group '%s' already exists and cannot be added.	Error	This message is issued when the name specified for group already exists in the library. The group name specified should be unique.
VALUS	1085	Command option '%s' is specified but its corresponding option '%s' is missing. Height and width tables for a particular region (low, high, below_low, above_high) must be specified in pairs.	Error	This message is issued if propagated_noise_width_low/high is defined but its corresponding propagated_noise_height_low/high command option is missing.
VALUS	1086	The library '%s' specified for option '%s' with command '%s' does not exist.	Error	This message is issued if the library specified by user does not exist.

Valus User Guide

Valus Checks

LIB_UI				
Message Prefix	ID	Message	Severity	Description
VALUS	1087	The library specified for option '%s' with command '%s' does not have CCSN data to be copied.	Error	This message is issued if the library specified by user does not have CCSN data to be copied. This can be triggered under any of two scenarios. (1) While copying default CCSN data from pin/arc level and CCSN data is not available on ref_pin. (2) While copying CCSN data from pin/arc level with when_condition specified with the command and CCSN data is not available for that specified when_condition.
VALUS	1088	Pin '%s' specified with command option '%s' is invalid. The valid pin type is '%s'.	Error	This message is issued when 'ref_pin/to_pin' is not 'input/inout' pin while adding ccsn_first_stage data or 'to_pin/from_pin' is not 'output/inout' while adding noise attributes at arc level.
VALUS	1089	Mode definition name '%s' specified with command option 'add_mode_definition' already exists. Only unique mode_definition can be added.	Error	This message is issued when the 'mode_definition' name specified by the user with command 'add_mode_definition', already exists on the ETM/PTM cell. Only unique mde_definition can be added.

Valus User Guide

Valus Checks

LIB_UI				
Message Prefix	ID	Message	Severity	Description
VALUS	1090	Table '%s' defined with command option 'add_power_info' but table(s) '%s' are not defined. Arc cannot be added. Both tables 'rise_power' and 'fall_power' are mandatory if any one is specified.	Error	This message is issued when all the required tables are not defined while adding the arc using command 'add_power_info'. For example if rise_power is specified, then fall_power must also be specified, and if fall_power is specified, then rise_power must also be specified.
VALUS	1091	The option '%s' cannot be used with command 'add_timing_info' for type '%s'.	Error	This message is issued, when min_delay_arc value is set with command 'add_timing_info' on arcs other than combinational and rising edge/falling edge. This option can be used with combinational delay arcs and rising edge/falling edge only.
VALUS	1092	Either the option 'power' or both 'rise_power' and 'fall_power' must be specified while using the command 'add_power_info'.	Error	This message is issued, when none of power, rise_power or fall_power value is specified with command 'add_power_info'. Either the option 'power' or both 'rise_power' and 'fall_power' must be specified while using the command 'add_power_info'.

Valus User Guide

Valus Checks

LIB_UI				
Message Prefix	ID	Message	Severity	Description
VALUS	1093	The ccsn_first_stage group with the specified when_condition could not be found, while using command 'add_noise_info'.	Error	This message is issued, when the ccsn_first_stage to be copied from a reference library, cannot be found with the specified when_condition, while using command 'add_noise_info'. To avoid this issue, the correct when condition needs to be specified.
VALUS	1094	The options 'to_pin' and 'from_pin' are mandatory while adding propagated noise data to a pin. The options 'to_pin' and 'ref_pin' are mandatory when copying ccsn_first_stage group to a pin from a reference library pin and the option 'to_ref_pin' is also mandatory when copying ccsn_first_stage group to a pin from a reference library arc, while using command 'add_noise_info'.	Error	This message is issued, when an incorrect combination of options is specified while using command 'add_noise_info'. The options 'to_pin' and 'from_pin' are mandatory while adding propagated noise data to a pin. The options 'to_pin' and 'ref_pin' are mandatory when copying ccsn_first_stage group to a pin from a reference library pin and the option 'to_ref_pin' is also mandatory when copying ccsn_first_stage group to a pin from a reference library arc.
VALUS	1095	The option 'input_ccb' is only valid with options '-ref_pin' and '-to_pin'.	Error	This message is issued, when an incorrect combination of options is specified while using command 'add_noise_info'. The options 'to_pin' and 'ref_pin' are mandatory and valid while copying input_ccb group to a pin from a reference library pin.

Data Integrity Checks

The data integrity checks are performed to check data sanity of liberty attributes or groups, wherever applicable. These checks are performed to ensure valid values or range, data monotonicity, and mandatory attributes/groups.

These checks include:

- [Positive Checks](#) on page 238
- [Range Checks](#) on page 241
- [Monotonic Increase Checks](#) on page 244
- [Mandatory Checks](#) on page 245
- [Mandatory Ignore Checks](#) on page 246
- [Count Checks](#) on page 247
- [Enum Value Checks](#) on page 252

Valus User Guide

Valus Checks

Positive Checks

The positive value checks issue the following message:

TECHLIB-1223: The attribute '%s' on line %d has value '%f' which is not '%s'.

The given below attributes must be positive for a defined group in a library:

Group Name	Attribute Name
bundle	capacitance, fall_capacitance, fanout_load, max_capacitance, rise_capacitance, rise_current_slope_before_threshold
bus	capacitance, fall_capacitance, fanout_load, max_capacitance, rise_capacitance, rise_current_slope_before_threshold
fall_power	index_1, index_2, index_3
internal_power	index_1, index_2, index_3
library	default_cell_leakage_power, default_fanout_load, default_max_capacitance, default_wire_load_area, default_wire_load_capacitance, default_wire_load_resistance
min_pulse_width	constraint_high, constraint_low
pin	capacitance, fall_capacitance, fanout_load, max_capacitance, rise_capacitance, rise_current_slope_before_threshold
rise_power	index_1, index_2, index_3
wire_load	capacitance, fall_capacitance, rise_capacitance
wire_load_table	fanout_area, fanout_capacitance, fanout_length, fanout_resistance
hyperbolic_noise_low	area_coefficient, height_coefficient, width_coefficient
hyperbolic_noise_high	area_coefficient, height_coefficient, width_coefficient

Valus User Guide

Valus Checks

Group Name	Attribute Name
hyperbolic_noise_below_low	area_coefficient, height_coefficient, width_coefficient
hyperbolic_noise_above_high	area_coefficient, height_coefficient, width_coefficient
noise_lut_template	index_1, index_2
noise_immunity_high	index_1, index_2, values
noise_immunity_low	index_1, index_2, values
noise_immunity_below_low	index_1, index_2, values
noise_immunity_above_high	index_1, index_2, values
propagation_lut_template	index_1, index_2, index_3
propagated_noise_height_low	index_1, index_2, index_3, values
propagated_noise_height_high	index_1, index_2, index_3, values
propagated_noise_height_above_high	index_1, index_2, index_3, values
propagated_noise_height_below_low	index_1, index_2, index_3, values
propagated_noise_width_high	index_1, index_2, index_3, values
propagated_noise_width_above_high	index_1, index_2, index_3, values
propagated_noise_width_below_low	index_1, index_2, index_3, values
propagated_noise_width_low	index_1, index_2, index_3, values
ccsn_first_stage	millier_cap_fall, millier_cap_rise
ccsn_last_stage	millier_cap_fall, millier_cap_rise
ocv_sigma_rise_transition	values
ocv_sigma_fall_transition	values
ocv_sigma_cell_rise	values
ocv_sigma_cell_fall	values
ocv_sigma_rise_constraint	values
ocv_sigma_fall_constraint	values
timing	timing_sense
vector	reference_time

Valus User Guide

Valus Checks

Group Name	Attribute Name
gate_leakage	input_high_value
ocv_std_dev_fall_constraint	values
ocv_std_dev_rise_constraint	values
ocv_std_dev_rise_transition	values
ocv_std_dev_fall_transition	values
ocv_std_dev_cell_rise	values
ocv_std_dev_cell_fall	values
input_ccb	millier_cap_rise
input_ccb	millier_cap_fall
output_ccb	millier_cap_rise
output_ccb	millier_cap_fall
ocv_sigma_retaining_rise	values
Begin_Group	values
Begin_Group	values
ocv_sigma_retain_fall_slew	values

Valus User Guide

Valus Checks

Range Checks

The range checks issue the following message:

TECHLIB-1223: The attribute '%s' on line %d has value '%f' which is not '%s'.

The given below attributes must be within the specified range for a defined group in a library:

Group Name	Attribute Name	Range_Value
bundle	input_threshold_pct_fall	>=0.0 && <=100.0
bundle	input_threshold_pct_rise	>=0.0 && <=100.0
bundle	output_threshold_pct_fall	>=0.0 && <=100.0
bundle	output_threshold_pct_rise	>=0.0 && <=100.0
bundle	slew_lower_threshold_pct_fall	>=0.0 && <=100.0
bundle	slew_lower_threshold_pct_rise	>=0.0 && <=100.0
bundle	slew_upper_threshold_pct_fall	>=0.0 && <=100.0
bundle	slew_upper_threshold_pct_rise	>=0.0 && <=100.0
bus	input_threshold_pct_fall	>=0.0 && <=100.0
bus	input_threshold_pct_rise	>=0.0 && <=100.0
bus	output_threshold_pct_fall	>=0.0 && <=100.0
bus	output_threshold_pct_rise	>=0.0 && <=100.0
bus	slew_lower_threshold_pct_fall	>=0.0 && <=100.0
bus	slew_lower_threshold_pct_rise	>=0.0 && <=100.0
bus	slew_upper_threshold_pct_fall	>=0.0 && <=100.0
bus	slew_upper_threshold_pct_rise	>=0.0 && <=100.0
library	input_threshold_pct_fall	>=0.0 && <=100.0
library	input_threshold_pct_rise	>=0.0 && <=100.0
library	k_temp_nochange_fall	> -100 && < 100
library	k_temp_skew_rise	> -100 && < 100
library	k_temp_skew_fall	> -100 && < 100
library	output_threshold_pct_fall	>=0.0 && <=100.0

Valus User Guide

Valus Checks

Group Name	Attribute Name	Range_Value
library	output_threshold_pct_rise	>=0.0 && <=100.0
library	slew_lower_threshold_pct_fall	>=0.0 && <=100.0
library	slew_lower_threshold_pct_rise	>=0.0 && <=100.0
library	slew_upper_threshold_pct_fall	>=0.0 && <=100.0
library	slew_upper_threshold_pct_rise	>=0.0 && <=100.0
library	slew_derate_from_library	>=0.0 && <=1.0
library	setup_constraint_sigma_folding_factor	> -100 && < 100
library	hold_constraint_sigma_folding_factor	> -100 && < 100
library	recovery_constraint_sigma_folding_factor	> -100 && < 100
library	removal_constraint_sigma_folding_factor	> -100 && < 100
library	non_seq_setup_constraint_sigma_folding_factor	> -100 && < 100
library	non_seq_hold_constraint_sigma_folding_factor	> -100 && < 100
library	nochange_setup_constraint_sigma_folding_factor	> -100 && < 100
library	nochange_hold_constraint_sigma_folding_factor	> -100 && < 100
library	min_period_constraint_sigma_folding_factor	> -100 && < 100
library	min_pulse_width_constraint_sigma_folding_factor	> -100 && < 100
library	skew_constraint_sigma_folding_factor	> -100 && < 100
library	default_constraint_sigma_folding_factor	> -100 && < 100
pin	input_threshold_pct_fall	>=0.0 && <=100.0
pin	input_threshold_pct_rise	>=0.0 && <=100.0
pin	output_threshold_pct_fall	>=0.0 && <=100.0
pin	output_threshold_pct_rise	>=0.0 && <=100.0

Valus User Guide

Valus Checks

Group Name	Attribute Name	Range_Value
pin	slew_lower_threshold_pct_fall	>=0.0 && <=100.0
pin	slew_lower_threshold_pct_rise	>=0.0 && <=100.0
pin	slew_upper_threshold_pct_fall	>=0.0 && <=100.0
pin	slew_upper_threshold_pct_rise	>=0.0 && <=100.0

Monotonic Increase Checks

The following attributes should be monotonically increasing for the defined group in library.

TECHLIB-1174: The attribute '%s' defined in group '%s' on line %d is not monotonically %s for values '%f' to '%f'. This may lead to undesirable analysis results.

The given below attributes should be monotonically increasing for a defined group in a library.

Attribute Name	Group Name
index_1	ca_edge_table_template, em_lut_template, internal_power, iv_lut_template, lu_table_template, maxcap_lut_template, maxtrans_lut_template, noise_immunity_above_high, noise_immunity_below_low, noise_immunity_high, noise_immunity_low, noise_lut_template, ocv_table_template, output_current_template, pg_current_template, propagation_lut_template
index_2	ca_edge_table_template, em_lut_template, internal_power, lu_table_template, maxcap_lut_template, maxtrans_lut_template, noise_immunity_above_high, noise_immunity_below_low, noise_immunity_high, noise_immunity_low, noise_lut_template, ocv_table_template, output_current_template, pg_current_template, power_lut_template, propagation_lut_template
index_3	internal_power, lu_table_template, output_current_template, pg_current_template, power_lut_template, propagation_lut_template
index_4	lu_table_template, pg_current_template
curve_x	base_curves

Mandatory Checks

The mandatory value checks issue the following message:

TECHLIB-1173: The '%s' attribute has not been defined for %s '%s' on line %d. This may lead to undesirable analysis results. To fix this issue, define the missing attribute in library.

Given below is a list of mandatory attributes in a library for the corresponding specified groups.

Group Name	Attribute Name
ocv_derate_factors	derate_type, path_type, rf_type
ff_bank	clocked_on, next_state
ff	clocked_on, next_state
compact_lut_template	index_1, index_2
bus	bus_type
library	pulling_resistance_unit
memory_read	address
memory_write	address
minimum_period	constraint
power_lut_template	index_1
type	base_type
ecsm_waveform	index_1
ecsm_waveform_set	index_1
ecsm_waveform_set	values
sensitization	pin_names
sensitization	vector

Mandatory Ignore Checks

The mandatory ignore value check issues the following message:

TECHLIB-1173: The '%s' attribute has not been defined for %s '%s' on line %d. This may lead to undesirable analysis results. To fix this issue, define the missing attribute in the library.

This is followed by the message below:

TECHLIB-1256: The '%s' is being ignored due to errors in this group. This group will be excluded for any further library checks. Refer to the previous messages issued for %s to find the details of the issues in this group.

Given below is a list of mandatory ignore attributes in a library for the corresponding specified group:

Group Name	Attribute Name
lut_values	values
ecsm_waveform	values
ecsm_capacitance	values
receiver_capacitance_rise	segment
receiver_capacitance_rise	values
receiver_capacitance_fall	segment
receiver_capacitance_fall	values

Count Checks

The count checks issue the following message:

TECHLIB-1172: The '%s' attribute has been defined more than once for group '%s' on line %d. The attribute should be defined only once.

Given below is a list of attributes that are allowed to be defined only once in a library for the specified group.

Attribute Name	Group Name
index_1_count	Begin_Group, Begin_Group, ca_edge_table_template, cell_fall, cell_rise, dc_current, ecdsm_capacitance_set, ecdsm_waveform, ecdsm_waveform_set, em_lut_template, em_max_toggle_rate, fall_constraint, fall_power, fall_propagation, fall_transition, lle_delta_cell_fall, lle_delta_cell_rise, lle_delta_fall_constraint, lle_delta_fall_transition, lle_delta_leakage_power, lle_delta_rise_constraint, lle_delta_rise_transition, lu_table_template, lut_values, maxcap_lut_template, maxtrans_lut_template, normalized_driver_waveform, ocv_cell_fall_mean_shift, ocv_cell_fall_skewness, ocv_cell_fall_std_dev, ocv_cell_rise_mean_shift, ocv_cell_rise_skewness, ocv_cell_rise_std_dev, ocv_derate_factors, ocv_fall_constraint_mean_shift, ocv_fall_constraint_skewness, ocv_fall_constraint_std_dev, ocv_fall_transition_mean_shift, ocv_fall_transition_skewness, ocv_fall_transition_std_dev, ocv_mean_shift_cell_fall, ocv_mean_shift_cell_rise, ocv_mean_shift_fall_constraint, ocv_mean_shift_fall_transition, ocv_mean_shift_rise_constraint, ocv_mean_shift_rise_transition, ocv_rise_constraint_mean_shift, ocv_rise_constraint_skewness, ocv_rise_constraint_std_dev, ocv_rise_transition_mean_shift, ocv_rise_transition_skewness, ocv_rise_transition_std_dev, ocv_sigma_cell_fall, ocv_sigma_cell_rise, ocv_sigma_fall_constraint, ocv_sigma_fall_transition, ocv_sigma_retain_fall_slew, ocv_sigma_retaining_rise, ocv_sigma_rise_constraint, ocv_sigma_rise_transition, ocv_skewness_cell_fall, ocv_skewness_cell_rise, ocv_skewness_fall_constraint, ocv_skewness_fall_transition, ocv_skewness_rise_constraint, ocv_skewness_rise_transition, ocv_std_dev_cell_fall,

Valus User Guide

Valus Checks

Attribute Name	Group Name
	ocv_std_dev_cell_rise, ocv_std_dev_fall_constraint, ocv_std_dev_fall_transition, ocv_std_dev_rise_constraint, ocv_std_dev_rise_transition, ocv_table_template, output_current_template, pg_current_template, power, receiver_capacitance1_fall, receiver_capacitance1_rise, receiver_capacitance2_fall, receiver_capacitance2_rise, receiver_capacitance_fall, receiver_capacitance_rise, recovery_supply_derate_lut_template, recovery_supply_derate_table, retain_fall_slew, retain_rise_slew, retaining_fall, retaining_rise, rise_constraint, rise_power, rise_transition, stress_lut_template, stress_mapping_lut_template, stress_mapping_table, stress_table, vector
index_2	Begin_Group, Begin_Group, ca_edge_table_template, cell_fall, cell_rise, dc_current, ecdm_capacitance_set, em_lut_template, em_max_toggle_rate, fall_constraint, fall_power, fall_propagation, fall_transition, lle_delta_cell_fall, lle_delta_cell_rise, lle_delta_fall_constraint, lle_delta_fall_transition, lle_delta_rise_constraint, lle_delta_rise_transition, lu_table_template, maxcap_lut_template, maxtrans_lut_template, normalized_driver_waveform, ocv_cell_fall_mean_shift, ocv_cell_fall_skewness, ocv_cell_fall_std_dev, ocv_cell_rise_mean_shift, ocv_cell_rise_skewness, ocv_cell_rise_std_dev, ocv_derate_factors, ocv_fall_constraint_mean_shift, ocv_fall_constraint_skewness, ocv_fall_constraint_std_dev, ocv_fall_transition_mean_shift, ocv_fall_transition_skewness, ocv_fall_transition_std_dev, ocv_mean_shift_cell_fall, ocv_mean_shift_cell_rise, ocv_mean_shift_fall_constraint, ocv_mean_shift_fall_transition, ocv_mean_shift_rise_constraint, ocv_mean_shift_rise_transition, ocv_rise_constraint_mean_shift, ocv_rise_constraint_skewness, ocv_rise_constraint_std_dev, ocv_rise_transition_mean_shift, ocv_rise_transition_skewness, ocv_rise_transition_std_dev, ocv_sigma_cell_fall, ocv_sigma_cell_rise, ocv_sigma_fall_constraint,

Valus User Guide

Valus Checks

Attribute Name	Group Name
	ocv_sigma_fall_transition, ocv_sigma_retain_fall_slew, ocv_sigma_retaining_rise, ocv_sigma_rise_constraint, ocv_sigma_rise_transition, ocv_skewness_cell_fall, ocv_skewness_cell_rise, ocv_skewness_fall_constraint, ocv_skewness_fall_transition, ocv_skewness_rise_constraint, ocv_skewness_rise_transition, ocv_std_dev_cell_fall, ocv_std_dev_cell_rise, ocv_std_dev_fall_constraint, ocv_std_dev_fall_transition, ocv_std_dev_rise_constraint, ocv_std_dev_rise_transition, ocv_table_template, output_current_template, pg_current_template, power, power_lut_template, receiver_capacitance1_fall, receiver_capacitance1_rise, receiver_capacitance2_fall, receiver_capacitance2_rise, receiver_capacitance_fall, receiver_capacitance_rise, retain_fall_slew, retain_rise_slew, retaining_fall, retaining_rise, rise_constraint, rise_power, rise_transition, stress_lut_template, stress_table, vector
index_3	Begin_Group, Begin_Group, cell_fall, cell_rise, ecsm_capacitance_set, fall_constraint, fall_power, fall_propagation, fall_transition, lu_table_template, ocv_cell_fall_mean_shift, ocv_cell_fall_skewness, ocv_cell_fall_std_dev, ocv_cell_rise_mean_shift, ocv_cell_rise_skewness, ocv_cell_rise_std_dev, ocv_fall_constraint_mean_shift, ocv_fall_constraint_skewness, ocv_fall_constraint_std_dev, ocv_fall_transition_mean_shift, ocv_fall_transition_skewness, ocv_fall_transition_std_dev, ocv_mean_shift_cell_fall, ocv_mean_shift_cell_rise, ocv_mean_shift_fall_constraint, ocv_mean_shift_fall_transition, ocv_mean_shift_rise_constraint, ocv_mean_shift_rise_transition, ocv_rise_constraint_mean_shift, ocv_rise_constraint_skewness, ocv_rise_constraint_std_dev, ocv_rise_transition_mean_shift, ocv_rise_transition_skewness, ocv_rise_transition_std_dev, ocv_sigma_cell_fall, ocv_sigma_cell_rise, ocv_sigma_fall_constraint, ocv_sigma_fall_transition, ocv_sigma_retain_fall_slew, ocv_sigma_retaining_rise, ocv_sigma_rise_constraint, ocv_sigma_rise_transition, ocv_skewness_cell_fall, ocv_skewness_cell_rise, ocv_skewness_fall_constraint, ocv_skewness_fall_transition, ocv_skewness_rise_constraint, ocv_skewness_rise_transition, ocv_std_dev_cell_fall,

Valus User Guide

Valus Checks

Attribute Name	Group Name
	ocv_std_dev_cell_rise, ocv_std_dev_fall_constraint, ocv_std_dev_fall_transition, ocv_std_dev_rise_constraint, ocv_std_dev_rise_transition, output_current_template, pg_current_template, power, power_lut_template, retain_fall_slew, retain_rise_slew, retaining_fall, retaining_rise, rise_constraint, rise_power, rise_transition, stress_lut_template, stress_table, vector
index_4	lu_table_template, pg_current_template, vector
values	Begin_Group, Begin_Group, ca_edge_table_template, ca_edge_table_template, cell_fall, cell_rise, dc_current, ecdsm_capacitance_set, ecdsm_waveform, ecdsm_waveform_set, em_max_toggle_rate, fall_constraint, fall_power, fall_propagation, fall_transition, lle_delta_cell_fall, lle_delta_cell_rise, lle_delta_fall_constraint, lle_delta_fall_transition, lle_delta_leakage_power, lle_delta_rise_constraint, lle_delta_rise_transition, lut_values, normalized_driver_waveform, ocv_cell_fall_mean_shift, ocv_cell_fall_skewness, ocv_cell_fall_std_dev, ocv_cell_rise_mean_shift, ocv_cell_rise_skewness, ocv_cell_rise_std_dev, ocv_derate_factors, ocv_fall_constraint_mean_shift, ocv_fall_constraint_skewness, ocv_fall_constraint_std_dev, ocv_fall_transition_mean_shift, ocv_fall_transition_skewness, ocv_fall_transition_std_dev, ocv_mean_shift_cell_fall, ocv_mean_shift_cell_rise, ocv_mean_shift_fall_constraint, ocv_mean_shift_fall_transition, ocv_mean_shift_rise_constraint, ocv_mean_shift_rise_transition, ocv_rise_constraint_mean_shift, ocv_rise_constraint_skewness, ocv_rise_constraint_std_dev, ocv_rise_transition_mean_shift, ocv_rise_transition_skewness, ocv_rise_transition_std_dev, ocv_sigma_cell_fall, ocv_sigma_cell_rise, ocv_sigma_fall_constraint, ocv_sigma_fall_transition, ocv_sigma_retain_fall_slew, ocv_sigma_retaining_rise, ocv_sigma_rise_constraint, ocv_sigma_rise_transition, ocv_skewness_cell_fall,

Valus User Guide

Valus Checks

Attribute Name	Group Name
	ocv_skewness_cell_rise, ocv_skewness_fall_constraint, ocv_skewness_fall_transition, ocv_skewness_rise_constraint, ocv_skewness_rise_transition, ocv_std_dev_cell_fall, ocv_std_dev_cell_rise, ocv_std_dev_fall_constraint, ocv_std_dev_fall_transition, ocv_std_dev_rise_constraint, ocv_std_dev_rise_transition, power, receiver_capacitance1_fall, receiver_capacitance1_rise, receiver_capacitance2_fall, receiver_capacitance2_rise, receiver_capacitance_fall, receiver_capacitance_rise, recovery_supply_derate_table, retain_fall_slew, retain_rise_slew, retaining_fall, retaining_rise, rise_constraint, rise_power, rise_transition, stress_mapping_table, stress_table, vector
edges	generated_clock
shifts	generated_clock
receiver_capacitance_rise_threshold_pct	library
receiver_capacitance_fall_threshold_pct	library
active_input_ccb	receiver_capacitance
pin_names	sensitization
variation_type	ca_edge_delay_derate
derate_type	ca_edge_delay_derate
rf_type	ca_edge_delay_derate
path_type	ca_edge_delay_derate
variation_type	ca_edge_check_derate
check_type	ca_edge_check_derate
rf_type	ca_edge_check_derate

Enum Value Checks

The following message is issued for enum value checks:

TECHLIB-1204: Illegal value '%s' defined for attribute %s. The attribute will be ignored.

The given below attributes must be defined with the valid values for a defined group in a library.

Group Name	Attribute Name	Valid Values
bundle	direction	inout, input, internal, output
bundle	dont_fault	sa0, sa01, sa1
bundle	nextstate_type	clear, data, load, preset, scan_enable, scan_in
bundle	prefer_tied	0, 1
bundle	pin_func_type	active_falling, active_high, active_low, active_rising, clock_enable
bundle	signal_type	test_clock, test_scan_clock, test_scan_clock_a, test_scan_clock_b, test_scan_enable, test_scan_enable_inverted, test_scan_in, test_scan_in_inverted, test_scan_out, test_scan_out_inverted
bundle	slew_control	high, low, medium, none
bundle	map_to_logic	0, 1
bundle	antenna_diode_type	ground, power, power_and_ground
bundle	save_action	F, H, L, R
bundle	restore_action	F, H, L, R
bundle	restore_edge_type	edge_trigger, leading, trailing
bus	direction	inout, input, internal, output
bus	dont_fault	sa0, sa01, sa1
bus	nextstate_type	clear, data, load, preset, scan_enable, scan_in
bus	prefer_tied	0, 1

Valus User Guide

Valus Checks

Group Name	Attribute Name	Valid Values
bus	pin_func_type	active_falling, active_high, active_low, active_rising, clock_enable
bus	slew_control	high, low, medium, none
bus	map_to_logic	0, 1
bus	signal_type	test_clock, test_scan_clock, test_scan_clock_a, test_scan_clock_b, test_scan_enable, test_scan_enable_inverted, test_scan_in, test_scan_in_inverted, test_scan_out, test_scan_out_inverted
bus	antenna_diode_type	ground, power, power_and_ground
bus	save_action	F, H, L, R
bus	restore_action	F, H, L, R
bus	restore_edge_type	edge_trigger, leading, trailing
cell	dont_fault	sa0, sa01, sa1
cell	timing_model_type	abstracted, extracted, qtm
cell	level_shifter_type	HL, HL_LH, LH
cell	switch_cell_type	coarse_grain, fine_grain
cell	power_cell_type	macro, stdcell
cell	antenna_diode_type	ground, power, power_and_ground

Valus User Guide

Valus Checks

Group Name	Attribute Name	Valid Values
cell	builtin_clock_gating_integrated_cell	latch_negedge, latch_negedge_postcontrol, latch_negedge_postcontrol_obs, latch_negedge_precontrol, latch_negedge_precontrol_obs, latch_posedge, latch_posedge_postcontrol, latch_posedge_postcontrol_obs, latch_posedge_precontrol, latch_posedge_precontrol_obs, none_negedge, none_negedge_control, none_negedge_control_obs, none_posedge, none_posedge_control, none_posedge_control_obs
ocv_sigma_rise_transition	sigma_type	early, early_and_late, late
ocv_sigma_fall_transition	sigma_type	early, early_and_late, late
ocv_sigma_cell_rise	sigma_type	early, early_and_late, late
ocv_sigma_cell_fall	sigma_type	early, early_and_late, late
ff_bank	clear_preset_var1	H, L, N, T, X
ff_bank	clear_preset_var2	H, L, N, T, X
ff	clear_preset_var1	H, L, N, T, X
ff	clear_preset_var2	H, L, N, T, X
latch_bank	clear_preset_var1	H, L, N, T, X
latch_bank	clear_preset_var2	H, L, N, T, X
latch	clear_preset_var1	H, L, N, T, X
latch	clear_preset_var2	H, L, N, T, X
library	current_unit	100mA, 100ma, 100uA, 100ua, 10mA, 10ma, 10uA, 10ua, 1A, 1a, 1mA, 1ma, 1uA, 1ua
library	default_wire_load_mode	enclosed, segmented, top

Valus User Guide

Valus Checks

Group Name	Attribute Name	Valid Values
library	delay_model	cmos2, generic_cmos, lsi_cmde, piecewise_cmos, polynomial, table_lookup
library	in_place_swap_mode	match_footprint, no_swapping
library	leakage_power_unit	100mW, 100nW, 100nw, 100pW, 100pw, 100uW, 100uw, 10mW, 10nW, 10nw, 10pW, 10pw, 10uW, 10uw, 1mW, 1mw, 1nW, 1nw, 1pW, 1pw, 1uW, 1uw
library	piece_type	piece_length, piece_pin_cap, piece_total_cap, piece_wire_cap
library	power_model	polynomial, table_lookup
library	power_unit	100mw, 100uw, 10mw, 10uw, 1mw, 1uw, 1w
library	distance_unit	1mm, 1um
library	time_unit	100ps, 10ps, 1ns, 1ps
library	voltage_unit	100mV, 10mV, 1V, 1mV
lu_table_template	variable_1	connect_delay, constrained_pin_transition, input_net_transition, input_noise_height, input_noise_width, input_voltage, normalized_voltage, output_net_length, output_net_pin_cap, output_net_wire_cap, output_pin_transition, output_voltage, parameter_value_delta, pg_voltage, pg_voltage_difference, related_out_output_net_length, related_out_output_net_pin_cap, related_out_output_net_wire_cap, related_out_total_output_net_capacitance, related_pin_transition, time, total_output_net_capacitance

Valus User Guide

Valus Checks

Group Name	Attribute Name	Valid Values
lu_table_template	variable_2	connect_delay, constrained_pin_transition, input_net_transition, input_noise_height, input_noise_width, input_voltage, normalized_voltage, output_net_length, output_net_pin_cap, output_net_wire_cap, output_pin_transition, output_voltage, parameter_value_delta, related_out_output_net_length, related_out_output_net_pin_cap, related_out_output_net_wire_cap, related_out_total_output_net_capacitance, related_pin_transition, time, total_output_net_capacitance
lu_table_template	variable_3	constrained_pin_transition, input_net_transition, input_noise_height, input_noise_width, output_net_length, output_net_pin_cap, output_net_wire_cap, parameter_value_delta, related_out_output_net_length, related_out_output_net_pin_cap, related_out_output_net_wire_cap, related_out_total_output_net_capacitance, related_pin_transition, time, total_output_net_capacitance

Valus User Guide

Valus Checks

Group Name	Attribute Name	Valid Values
lu_table_template	variable_4	constrained_pin_transition, input_net_transition, input_noise_height, input_noise_width, output_net_length, output_net_pin_cap, output_net_wire_cap, parameter_value_delta, related_out_output_net_length, related_out_output_net_pin_cap, related_out_output_net_wire_cap, related_out_total_output_net_capacitance, related_pin_transition, time, total_output_net_capacitance
memory	type	ram, rom
operating_conditions	tree_type	balanced_tree, best_case_tree, worst_case_tree
pin	dont_fault	sa0, sa01, sa1
pin	direction	inout, input, internal, output
pin	nextstate_type	clear, data, load, preset, scan_enable, scan_in
pin	prefer_tied	0, 1
pin	pin_func_type	active_falling, active_high, active_low, active_rising, clock_enable
pin	signal_type	test_clock, test_scan_clock, test_scan_clock_a, test_scan_clock_b, test_scan_enable, test_scan_enable_inverted, test_scan_in, test_scan_in_inverted, test_scan_out, test_scan_out_inverted
pin	slew_control	high, low, medium, none
pin	map_to_logic	0, 1
pin	pulse_clock	fall_triggered_high_pulse, fall_triggered_low_pulse, rise_triggered_high_pulse, rise_triggered_low_pulse

Valus User Guide

Valus Checks

Group Name	Attribute Name	Valid Values
pin	antenna_diode_type	ground, power, power_and_ground
pin	save_action	F, H, L, R
pin	restore_action	F, H, L, R
pin	restore_edge_type	edge_trigger, leading, trailing
power_lut_template	variable_1	equal_or_opposite_output_net_capacitance, input_transition_time, total_output2_net_capacitance, total_output_net_capacitance
power_lut_template	variable_2	equal_or_opposite_output_net_capacitance, input_transition_time, total_output2_net_capacitance, total_output_net_capacitance
power_lut_template	variable_3	equal_or_opposite_output_net_capacitance, input_transition_time, total_output2_net_capacitance, total_output_net_capacitance
em_max_toggle_rate	current_type	average, peak, rms
scaled_cell	dont_fault	sa0, sa01, sa1
timing	timing_sense	negative_unate, non_unate, positive_unate

Valus User Guide

Valus Checks

Group Name	Attribute Name	Valid Values
timing	timing_type	clear, combinational, combinational_fall, combinational_rise, falling_edge, hold_falling, hold_rising, max_clock_tree_path, min_clock_tree_path, min_pulse_width, minimum_period, nochange_high_high, nochange_high_low, nochange_low_high, nochange_low_low, non_seq_hold_falling, non_seq_hold_rising, non_seq_setup_falling, non_seq_setup_rising, preset, recovery_falling, recovery_rising, removal_falling, removal_rising, rising_edge, setup_falling, setup_rising, skew_falling, skew_rising, three_state_disable, three_state_disable_fall, three_state_disable_rise, three_state_enable, three_state_enable_fall, three_state_enable_rise
timing	sdf_edges	both_edges, end_edge, noedge, start_edge
type	data_type	bit
tlatch	edge_type	falling, rising
iv_lut_template	variable_1	iv_output_voltage
noise_lut_template	variable_1	input_noise_width, total_output_capacitance, total_output_net_capacitance
noise_lut_template	variable_2	input_noise_width, total_output_capacitance, total_output_net_capacitance
em_lut_template	variable_1	input_transition_time, total_output_net_capacitance

Valus User Guide

Valus Checks

Group Name	Attribute Name	Valid Values
em_lut_template	variable_2	input_transition_time, total_output_net_capacitance
maxcap_lut_template	variable_1	frequency
maxcap_lut_template	variable_2	input_transition_time
maxtrans_lut_template	variable_1	frequency
maxtrans_lut_template	variable_2	input_transition_time
ocv_table_template	variable_1	path_depth, path_distance
ocv_table_template	variable_2	path_depth, path_distance
pg_current_template	variable_1	input_net_transition, time, total_output_net_capacitance
pg_current_template	variable_2	input_net_transition, time, total_output_net_capacitance
pg_current_template	variable_3	input_net_transition, time, total_output_net_capacitance
pg_current_template	variable_4	input_net_transition, time, total_output_net_capacitance
propagation_lut_template	variable_1	input_noise_height, input_noise_width, total_output_net_capacitance
propagation_lut_template	variable_2	input_noise_height, input_noise_width, total_output_net_capacitance
propagation_lut_template	variable_3	input_noise_height, input_noise_width, total_output_net_capacitance
ecsm_power	ecsm_power_type	fall, rise
pg_pin	pg_type	backup_ground, backup_power, deepnwell, deeppwell, internal_ground, internal_power, nwell, primary_ground, primary_power, pwell
pg_pin	direction	inout, input, internal, output
pg_pin	physical_connection	device_layer, routing_pin
output_current_template	variable_1	input_net_transition, total_output_net_capacitance

Valus User Guide

Valus Checks

Group Name	Attribute Name	Valid Values
output_current_template	variable_2	input_net_transition, total_output_net_capacitance
output_current_template	variable_3	time
ccsn_first_stage	stage_type	both, pull_down, pull_up
ccsn_last_stage	stage_type	both, pull_down, pull_up
base_curves	base_curve_type	ccs_half_curve, ccs_timing_half_curve
compact_lut_template	variable_1	input_net_transition, total_output_net_capacitance
compact_lut_template	variable_2	curve_parameters, input_net_transition, total_output_net_capacitance
compact_lut_template	variable_3	curve_parameters
input_ccb	stage_type	both, pull_down, pull_up
output_ccb	stage_type	both, pull_down, pull_up
ocv_sigma_retaining_rise	sigma_type	early, early_and_late, late
Begin_Group	sigma_type	early, early_and_late, late
Begin_Group	sigma_type	early, early_and_late, late
ocv_sigma_retain_fall_slew	sigma_type	early, early_and_late, late
lle_delta_cell_rise	related_params	u0, vth
lle_delta_cell_fall	related_params	u0, vth
lle_delta_rise_transition	related_params	u0, vth
lle_delta_fall_transition	related_params	u0, vth
lle_delta_rise_constraint	related_params	u0, vth
lle_delta_fall_constraint	related_params	u0, vth
char_config	internal_power_calculation	exclude_switching_on_rise, exclude_switching_on_rise_and_fall, include_switching
lle_delta_leakage_power	related_params	u0, vth
ca_edge_table_template	variable_1	left_site_distance, right_site_distance

Valus User Guide

Valus Checks

Group Name	Attribute Name	Valid Values
ca_edge_table_template	variable_2	left_site_distance, right_site_distance
stress_lut_template	variable_1	effective_age, temperature, voltage
stress_lut_template	variable_2	effective_age, temperature, voltage
stress_lut_template	variable_3	effective_age, temperature, voltage
age_param	device_type	NMOS, PMOS
stress_mapping_lut_template	variable_1	input_param
recovery_supply_derate_lut_template	variable_1	duty_cycle